GE Medical Systems

gehealthcare.com

Technical Publication

Direction 2378261-100
Revision 15

GE Medical Systems
GRE Software Installation Procedures

The information in this document applies to the GOC3 and GOC4 GRE Consoles, as used on the following CT Scanners:

- LightSpeed 1.X (QX/i)
- LightSpeed 2.X (Plus, QX/i)
- LightSpeed 3.X (Ultra & Plus) & HiSpeed QX/i
- LightSpeed 4.X (LightSpeed 16)
- LightSpeed 5.X (Pro16 100kW and Pro16 80kW)
- LightSpeed 5.X (LS16, Ultra, & Plus)
- LightSpeed 5.X (RT)

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。
GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触
电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

Call 1-800-548-3366 and use option 8.

Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm

Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage that may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.

ATTENTION Danger d’Explosion

Il y a danger d’explosion s’il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Rev	Date	Reason for change
0	06/17/03	Initial Release.
1	06/25/03	Revised Irix to Linux conversion procedure in Chapter 2
2	9/24/03	• Additions required for CSA Protocol Retention issues in the field • Incorporated M3 LFC for GRE Console information • Converted **System State Conversion** chapter to an appendix • Added new appendix: **LightSpeed 16 M3 LFC Procedure** • Updates to Lightspeed Pro^{16} chapter, including addition of Hpower ME-6 Patch A instructions.
3	10/06/03	• Included HiSpeed QX/i information to chapter 2 • Added new chapter: **LFC Procedures - LightSpeed16, Ultra & Plus M3**
4	10/31/03	• Updates for CSA Protocol Retention issues in field. • Updated product name titles for better clarification to users. • Updates to Preferences and Hardware screens. • General updates for software load steps. • Added chapter: **LFC Procedures - LightSpeed 5.X (RT M3)** • Added new appendix: **Example config/INFO Output**
5	11/17/03	Updates to LightSpeed 5.X (LS^{16}, Ultra & Plus) chapter.
6	12/3/03	• Changed "**Preface**" to "**Introduction**"; added Section 1 - Software Compatibility. • Added new chapters for LS Ultra/Plus M4, LS 16 M4, LS Pro^{16} (100kW) M3.1, and LS Pro^{16} (80kW) M3. • Added Appendix D for DARC OS Load Pop-Up Boxes • Updated load procedures for Network printer selection. • Updated LS RT chapter for OS 1.7.10 changes.
7	12/18/03	• Introduction: Updated "Section 1.0 Software Compatibility". • Chapter 1: Updated "Section 2.1.2 Software Versions". • Added a new chapter for LS 5.X (LS^{16}, Ultra & Plus 500.15M3).
8	2/20/04	• Updated chapters LS Ultra/Plus M4, LS 16 M4, LS Pro^{16} (100kW) M3.1, LS Pro^{16} (80kW) M3, LS 5.X (LS^{16}, Ultra & Plus 500.15M3), and LS RT for minor corrections, to clarify software CD-ROM names, update System Settings screen, and add graphics for HP PC DVD and DARC CD drive locations. • Moved section 3.1.3-3.1.7 information to new Appendix E.
9	3/19/04	Added a Notice for the Install Options procedure in the LightSpeed 5.X (LS16, Ultra & Plus) chapters.
10	5/3/04	• Corrected LS3.X and LS4.X Apps software version number. • Removed obsolete chapter LS5.X Pro16 100KW M3 and Appendix LS4.X M3 • Added chapter GRE Common Applications 04MK13.7 and Appendix Hardware Parameters Selection. • Minor updates to all other Appendices.
11	6/2/04	• Renamed Chapter 1 from 04MW13.7 to 04MW13.X; added LS3.X (8 slice), LS 4.X, LS5.X (100kW, 80kW, RT) and HiSpeed QX/i products. • Chapter 4 (LS16 M4) CSA Protocol retention: added M3.1 systems.

Rev	Date	Reason for change
12	8/10/04	• Updated first Warning in Section 3.3.2 for Chapters 1-10. • Corrected Option Install section for LS5.X (CTCge91590).
13	10/5/04	Updated Chapters 2, 4, 5, 6, 8, & 9 for PRQ 13021744 (clarify CSA Protocol retention).
14	1/24/05	• Updated Introduction and Chapter 1 for 04MW44.6 software. • Added "gettubeusagedata" script to Chapter 1, Section 3.8 and Appendix A (FCTge02242) • Added netmask value note to Chapters 1-10 (PQR 13026918) • Added "fix_irix_sd" script to Chapter 1, Section 3.4.
15	3/24/05	Added P/Ns for Release 44.9.

Table of Contents

Introduction

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
Software Compatibility

This section identifies the CT products and software versions included in this document. To find the software installation procedure that applies to your system, do the following:

1.) Open a shell.
2.) Type: **`check_config`** **ENTER**.
3.) The "check_config" command displays a number of output lines, including "Product" and "sw_version" (software version). Refer to the Table A below to determine which software installation procedure to use.

SYSTEM	MODEL	"PRODUCT"	"SW_VERSION"	CHAPTER TO USE
LightSpeed 5.X	All	--	04MW44.9, 04MW44.6 *See Note below.*	Chapter 1
LS 1.X, 2.X, 3.X, 4.X, 5.X (Pro16, RT); HiSpeed QX/i	--	--	04MW13.X *See Note below.*	Chapter 1
LightSpeed 3.X	Plus/Ultra	LightSpeed QX/i	308L.8_H3.3M4	Chapter 2
			307L.2_H3.3GREM3	Chapter 3
HiSpeed QX/i	QX/i	HiSpeed QX/i	103.4.HSQXIM4	Chapter 2
LightSpeed 4.X	H16	LightSpeed 16	405L.8_H4.2M4	Chapter 4
			404L.2_H4.2GREM3.1	Chapter 5
LightSpeed 5.X	HP100	LightSpeed PRO	502HP100.2_H5.0M3.1	Chapter 6
	HP80	LightSpeed PRO	502HP80.0_H5.0M3	Chapter 7
	HP60	LightSpeed 16	500.15_HP60_5.1_M3	Chapter 8
			500.13_HP60_5.1_M3	Chapter 9
	RT	LightSpeed RT	rel 501WB.2_H5.2M3	Chapter 10

Table A Software Compatibility

Note: The GRE Common Applications software supports multiple hardware configurations via a single software set. The new software version string has the following format:

<software_label>.<G>_<T>_<D>_<R>

The "software label" format is: *YYLWxx.ZZ*

Where: YY = Year. For example `04`.
L = Location. M - Milwaukee, H - Hino, B - Beijing.
Wxx = Fiscal Week. For example, `W15`.
ZZ = Revision number for each build/reroll.

Refer to Table B for an explanation of the *<G>_<T>_<D>_<R>* format.

G = GANTRY TYPE	T = TUBE TYPE	D = DAS TYPE	R = RECON HW TYPE
H1 = LightSpeed 1.X	P = Performix	S4 = SDAS4	P = Pegasus
H2 = LightSpeed 2.X, 3.X, 4.X	O = OEM	M4 = MDAS4	G = GRE
HP = LightSpeed 5.X	H = Hercules	M8 = MDAS8	
WB = LightSpeed 5.x RT		M16 = MDAS16	
QX = HighSpeed QXi		G16 = GDAS16	

Table B Common Applications Sotware Naming Conventions

The variables G, T, D, and R are generated dynamically based on the type of hardware configuration this software is installed on. The information for the dynamic part of the string is obtained from the INFO files which are located in /usr/g/config directory.

An example string based on the above format is as follows:

```
04MW13.7.H2_P_M4_G
```

The break down is as follows: This is 2004 software release at Milwaukee in Fiscal Week 13 with a revision number "7". The hardware configuration is "H2" gantry with "Performix" tube, "MDAS4", and "GRE" recon hardware.

Section 2.0
Safety & Hazard Information

2.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS:

DANGER
EXCESSIVE VOLTAGE
CRUSH POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING
ROTATING EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMAND MOTION.**

CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

NOTICE
Equipment Damage Possible

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk.They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

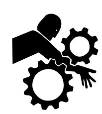
It is important that the reader not ignore hazard statements in this document.

2.2 Graphical Representation

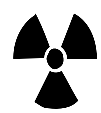
Important information is preceded by an exclamation point contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL

MECHANICAL

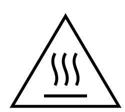


RADIATION

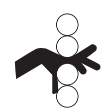


LASER

HEAT



PINCH

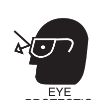


Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY

TAG AND LOCK OUT

WEAR EYE PROTECTION



Section 3.0
Publication Conventions

3.1 General Paragraph and Character Styles

Paragraph prefixes (such as Example, Comment and Note) are used to identify important but non-safety related information. Text styles and shading are also applied to text within each paragraph modified by the specific prefix. Examples follow:

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

3.2 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example:
Fixed Output

```
This paragraph denotes computer screen fixed output. Its output  is fixed
in the sense that it does not vary from application to application. It is
the most commonly used style used to indicate filenames, paths and text.
```

Example:
Variable Output

```
This paragraph denotes computer screen output that varies from
application to application. Variable output is sometimes placed between
greater-than and less-than operators. For example: <variable_ouput>
```

Example:
Fixed Input

`This paragraph denotes computer input that does not vary from application to application.`

Example:
Variable Input

`This paragraph denotes computer input that can vary from application to application. Variable input is sometimes placed between greater-than and less-than operators. For example: <variable_input> (the operators are dropped prior to input - exceptions are noted in the text).`

3.3 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example:
Hard Keys

A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example:
Soft Keys

Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Chapter 1
LFC Procedure – GRE Common Applications Version 04MW13.X, 04MW44.6 & 04MW44.9

Section 1.0
Scope

This LFC (Load from Cold) procedure presently supports the following systems using the version 04MW13.X GRE/Xtream Applications Disk #1 & Disk #2 software CD-ROMs:

- LightSpeed 1.X (QX/i)
- LightSpeed 2.X (Plus, QX/i)
- HiSpeed QXi
- LightSpeed 3.X (Ultra, Plus, QX/i) (required for sites using Wideview option)
- LightSpeed 4.X (16)
- LightSpeed 5.X Pro^{16} (100kW)
- LightSpeed 5.X Pro^{16} (80kW)
- LightSpeed 5.X RT

This LFC (Load from Cold) procedure presently supports the following systems using the version 04MW44.6 or 04MW44.9 GRE/Xtream Applications Disk #1 & Disk #2 software CD-ROMs:

- LightSpeed 5.X Pro^{16} (100kW)
- LightSpeed 5.X Pro^{16} (80kW)
- LightSpeed 5.x LS^{16}, Ultra, Plus (a.k.a "HP60")
- LightSpeed 5.X RT

Note: If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0
Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of four (4) CDROMs. See Section 2.1.2.

- Console Operating System (OS) software - loaded on HP Host Computer and DARC Node.
- Host Applications software (2 CD set) - loaded on HP Host Computer.
- DARC Applications software - loaded on HP Host Computer.

Note: The Xtream Serial-Over-Lan Service Software is not used during the LFC unless instructed to via a popup message box "SOL Service - ...run the Serial Over LAN Service CD...".

2.1.2 Software Version 04MW13.x

2.1.2.1 For LightSpeed 1.X, 2.X, HS QXi, 3.X, 4.X, and 5.X (100kW, 80kW, RT)

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **5114320, 5114321 – GRE/Xtream Applications Disk #1, Disk #2**
  Version 04MW13.7
- **5114322 – GRE/Xtream DARC Applications**
  Version 04MW13.7
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.1.2.2 For HiSpeed QX/i

- **5115543 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **5115544, 5115545 – HiSpeed QX/i/Xtream M4.1 Applications Disk #1, Disk #2**
  Version 04MW13.6
- **5115546 – HiSpeed QX/i/Xtream M4.1 DARC Applications**
  Version 04MW13.6
- **5115552 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.1.3 Software Version 04MW44.6 (For LightSpeed 5.X)

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **5129592, 5129593 – GRE/XtreamApplications Disk #1, Disk #2**
  Version 04MW44.6
- **5129595 – GRE/Xtream DARC Applications**
  Version 04MW13.7
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.1.4 Software Version 04MW44.9 (For LightSpeed 5.X)

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
- **5135962, 5135963 – GRE/XtreamApplications Disk #1, Disk #2**
- **5135964 – GRE/Xtream DARC Applications**
- **2399297 – Xtream Serial-Over-Lan Service Software**

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: `cat /usr/g/config/INFO` ENTER

c.) Record screen information in Section 4.0. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) Perform Save System State; save Protocols if applicable. See Section 3.2.
2.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 GRE SW Load Checklist

Note: This section is a checklist only. The actual procedures start in Section 3.2.

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration in Section 4.0.
  Open a shell and type: **`cat /usr/g/config/INFO`** **ENTER**
- ❑ Install the Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – YES to autorun. After Disk1 load, Insert Disk 2, select OK
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Start CT Applications by opening a shell and typing: **`st`** **ENTER**.
- ❑ Configure the user's system access.
- ❑ Restore System State. (IRIX to Linux upgrade sites: Restore MOD System State.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Set the time and date.
- ❑ LightSpeed 5.x RT systems: Enable Extended CT Number Range.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **`#bigguy`**
- ctuser: **`4$apps`**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) If reloading software, select: UTILITIES. If upgrading from earlier version software, select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.
2.) Remove the GRE Operator Console's front cover.
3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

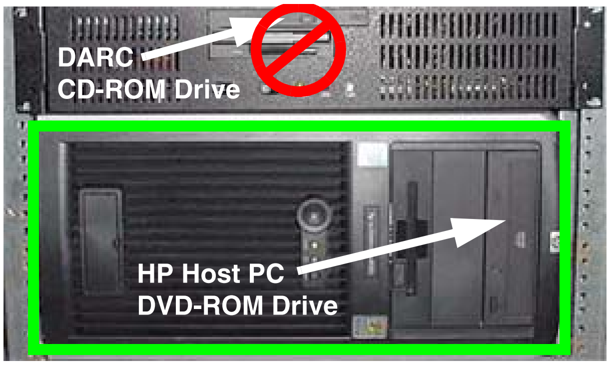


Figure 1-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console..

Figure 1-2 Shutdown Icon

The following options appear:

Logout User (this appears only if **HIPAA Present** was set to ON during the Load process)
Restart
Shutdown

5.) Select RESTART then OK to shut down and restart the system, or if you powered OFF the console, wait 30 seconds to allow the disk drive to settle then power **ON** the Console.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the `boot:` prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload. The **workaround** procedure is:

a.) Power cycle the HP Host Computer.

b.) While system is powering up, push and hold the HP Host Computer eject button and remove the disk.

c.) At the login prompt type:

`ctuser:` **`root`** **ENTER**

`password:` **`install`** **ENTER**

8.) The HP Host Computer reboots. **Do not insert the GRE/Xtream Applications Disk #1** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt, insert the **GRE/Xtream Applications Disk #1** CDROM into the DVD-ROM drive on the HP Host PC.

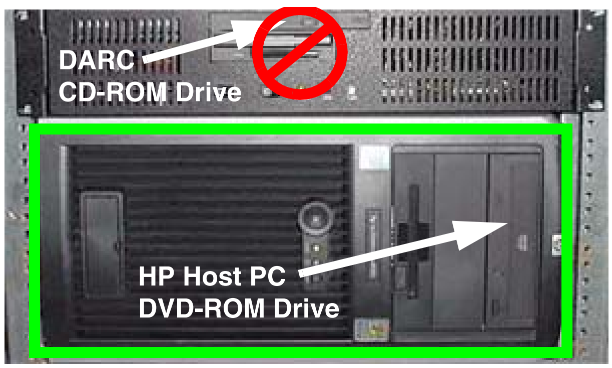


Figure 1-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear after two minutes, type the following then wait approximately one minute for the pop-up box to appear.

`mount /mnt/cdrom1` <u>ENTER</u>

`/mnt/cdrom1/autorun` <u>ENTER</u>,

3.) Click <u>YES</u>. After 2-3 minutes, a pink pop-up box appears with "`Install of cd 1 completed successfully! Insert disk 2 to continue applications load.`"

4.) Remove Applications Disk #1 and insert the **GRE/Xtream Applications Disk #2** CDROM into the DVD-ROM drive on the HP Host PC.

5.) Wait till the green LED on the HP Host PC DVD-ROM goes OFF, then click YES. After 1-2 minutes, an Installation Utility window appears.

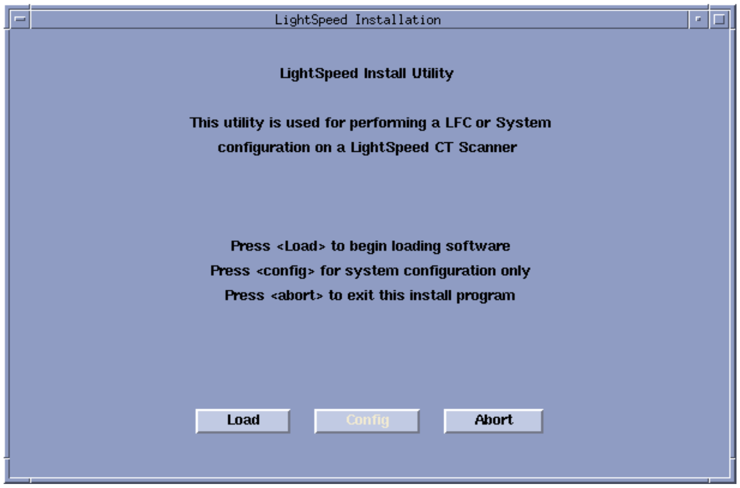


Figure 1-4 Lightspeed Install Utility window

6.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

7.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

8.) Click the SYSTEM button. See Figure 1-5 for an example.

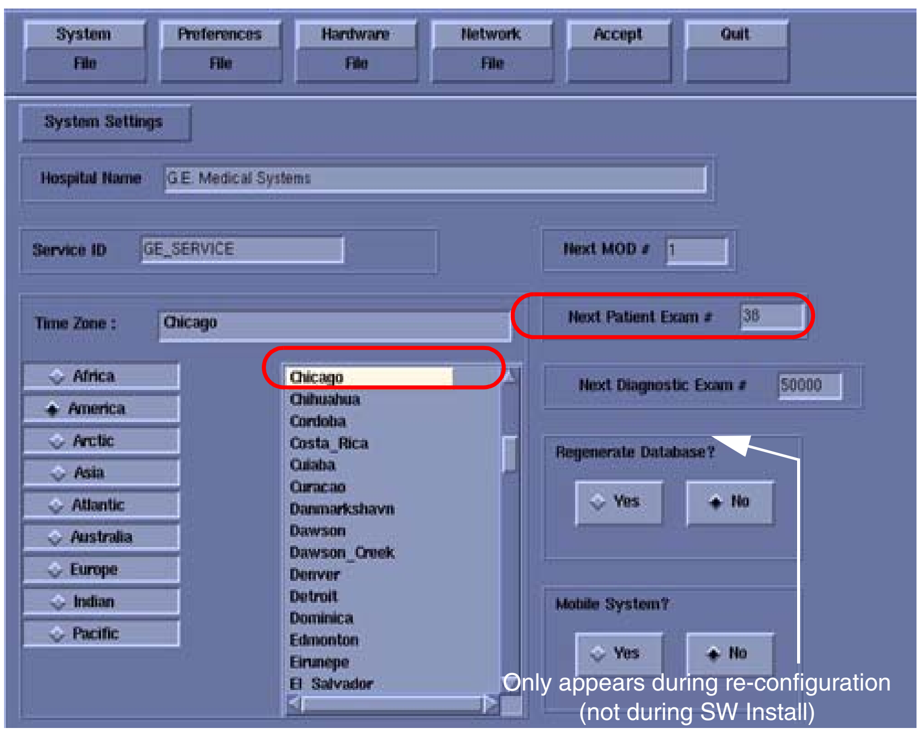


Figure 1-5 System Settings screen (Example)

9.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

10.) Verify that the proper **Time Zone** is selected (example: Chicago).

11.) Click the PREFERENCES button. See Figure 1-6 for an example.

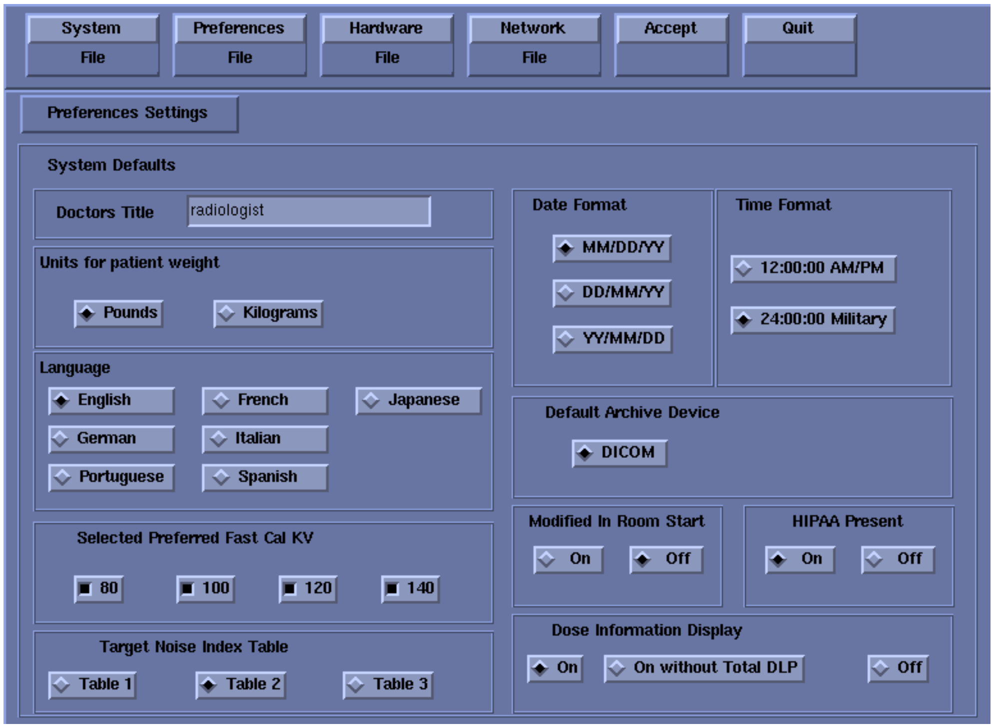


Figure 1-6 Preferences screen (Example)

12.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

13.) Select **Language**: (ENGLISH or OTHER). See Notice in Section 3.12.

14.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

15.) Select **Target Noise Index Table**: default is TABLE 2.

16.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

17.) Verify proper **Time Format** is selected (Example: 24:00:00 MILITARY).

18.) Default **Archive Device**: add your device here (Example: DICOM.)

19.) Verify **Modified in Room Start** is set to OFF.

20.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

21.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFO file System State.

22.) Click the HARDWARE button. See Figure 1-7 for an example Hardware Settings screen (the

Hardware Settings screen is different for each system).

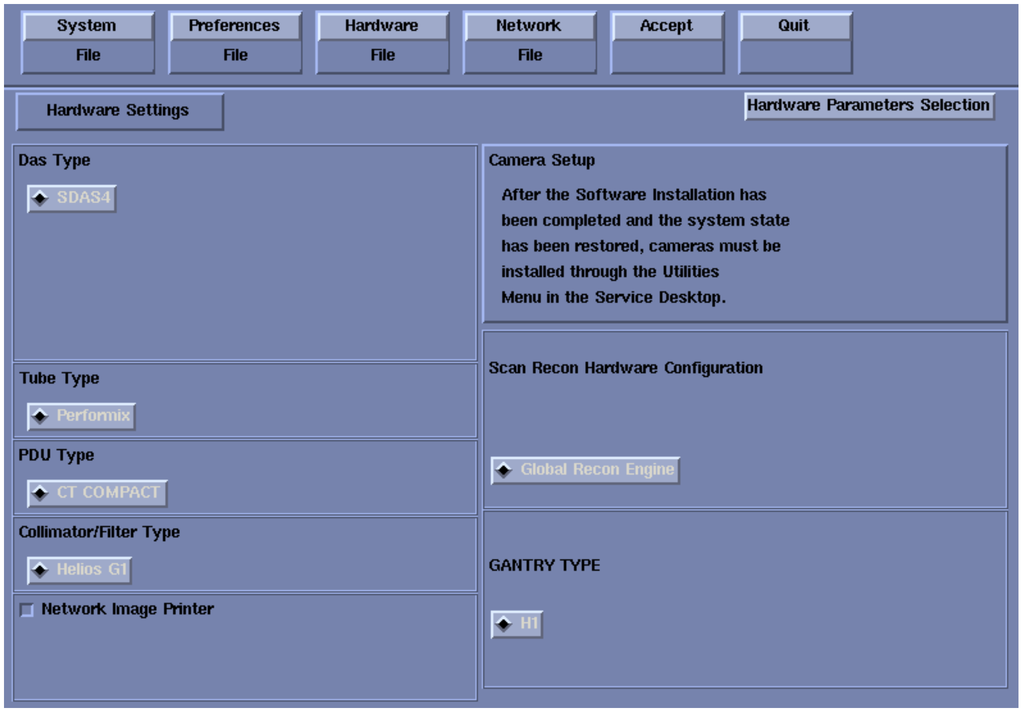


Figure 1-7 Hardware Settings Screen (Example - LightSpeed 1.X SDAS shown)

23.) Refer to Table 1-1 to verify: DAS Type, Tube Type, PDU Type, and Gantry Type. If not correct, click the HARDWARE PARAMETERS SELECTION button then select the correct system configuration from the displayed list. For additional information, refer to Appendix B.

SYSTEM	DAS	TUBE	PDU	GANTRY
LightSpeed 1.X (QX/i)	SDAS4	Performix	CT COMPACT	H1
LightSpeed 2.X (Plus, QX/i)	SDAS4	Performix	CT COMPACT	H2
HiSpeed QX/i	MDAS4	Performix	CT COMPACT or NG PDU	HiSpeed QX/i
LightSpeed 3.X (Plus, QX/i)	MDAS4	Performix	CT COMPACT or NG PDU	H2
LightSpeed 3.X (Ultra)	MDAS8	Performix	CT COMPACT or NG PDU	H2
LightSpeed 4.X (H16)	MDAS16	Performix	CT COMPACT	H2
LightSpeed 5.X Pro^{16} (100kW) (HP100)	GDAS16 or MDAS16	Hercules	NG PDU	HP
LightSpeed 5.X Pro^{16} (80kW) (HP80)	GDAS16 or MDAS16	OEM	NG PDU	HP
LightSpeed 5.x LS^{16}	GDAS16 or MDAS16	Performix	NG PDU	HP
LightSpeed 5.x Ultra	GDAS8	Performix	NG PDU	HP
LightSpeed 5.x Plus	GDAS4	Performix	NG PDU	HP
LightSpeed 5.x RT	MDAS4	OEM	NG PDU	WB

Table 1-1 DAS, Tube, PDU, and Gantry Type Selections

NOTICE Scanning with the wrong Tube Type will damage the tube! Before continuing, verify the Tube Type is correct per Table 1-1.

24.) Verify Collimator/Filter Type: HELIOS G1.

25.) Verify Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

27.) Click the NETWORK button. See Figure 1-8 for an example; your site will be different).

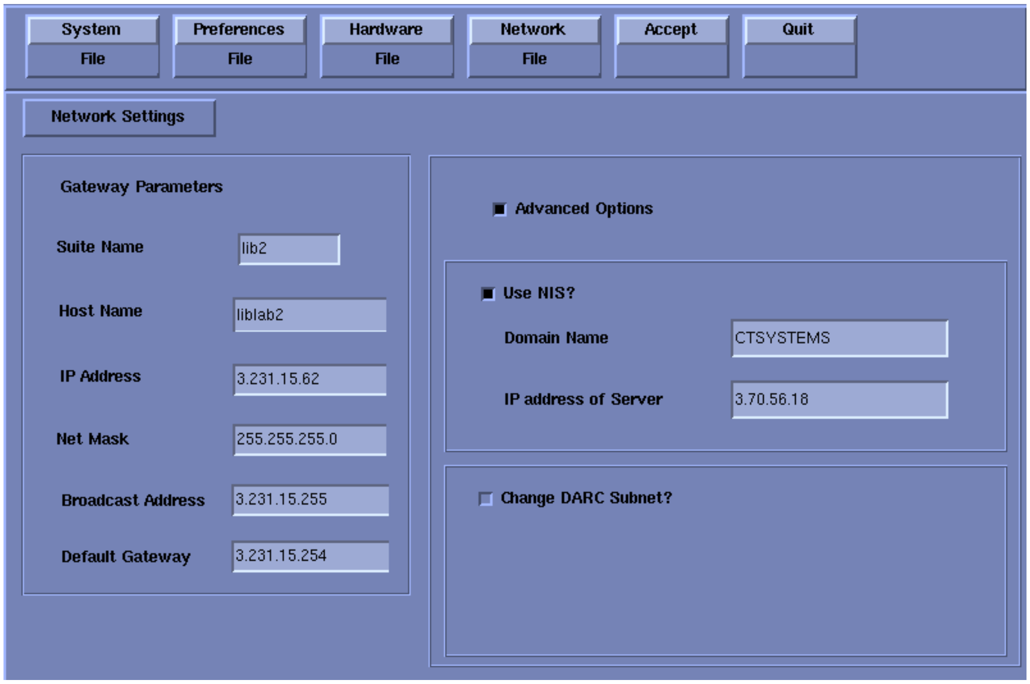


Figure 1-8 Network Settings screen (Example)

28.) Select the **Suite Name**: add your name here (Example: ***lib2***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

29.) Select the Host Name: add your host name here (Example: ***liblab2***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

30.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

31.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

32.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

33.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

34.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

35.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**. (This option might not appear on your system; it is not shown in Figure 1-8.)

36.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

a.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

b.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

Note: Make sure the **NUM LOCK** button on the keyboard is not active. If it is, deselect it to enable the **ACCEPT** button in the next step.

37.) Click the **ACCEPT** button at the right corner of the User Interface.

38.) An Install INFO pop-up box is displayed. See Figure 1-9 for a typical Install INFO message (the Install INFO message is different for each system configuration).

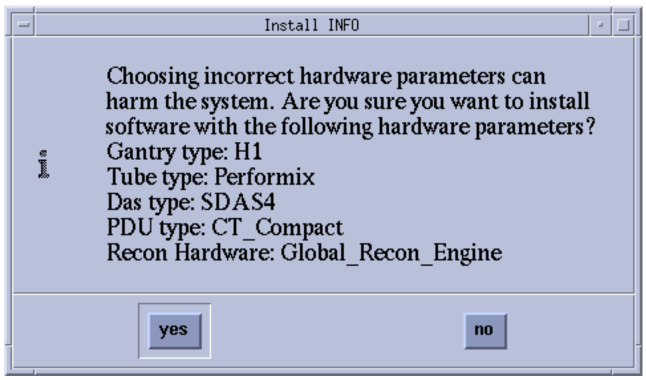


Figure 1-9 Hardware Confirmation Screen (Example - LightSpeed 1.X SDAS Shown)

39.) If the information is correct, click the YES button.

Note: If the information is not correct, click the NO button. This returns you to the Hardware screen. Click the HARDWARE PARAMETERS SELECTION button and select the correct system configuration from the displayed list. For additional information, refer to Appendix B. Click the **ACCEPT** button again, followed by the YES button on the Install INFO pop-up to accept the new hardware configuration.

40.) The following pop-up message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the **GRE/Xtream Applications Disk #2** from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~7 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop-up box appears, stating `CT Software Auto-Start Disabled`. In this box, click OK.

44.) Press the button on the DVD ROM drive to remove the **GRE/Xtream Applications Disk #2** CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following:

`ctuser:` **`su -`** <u>**ENTER**</u>

`password:` **`#bigguy`** <u>**ENTER**</u>

`root:` **`cd /usr/g/scripts`** <u>**ENTER**</u>

`root:` **`./start_darcOS`** <u>**ENTER**</u>

An Attention window appears. See Figure 1-10.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 1-10 Insert OS CD Pop-Up Window

3.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

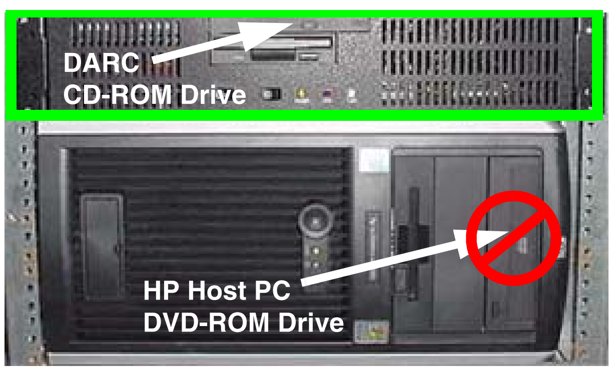


Figure 1-11 DARC CD-ROM Drive Location

NOTICE **Make sure the Operating System CD-ROM is securely loaded on the CD tray. If not pressed securely into place, the CD-ROM will slide into the DARC Node. This requires dismantling the DARC Node to retrieve the CD-ROM.**

4.) Click on <u>**OK**</u>. A sh window displays initial information, repetitive dpccli> commands, then the message below:

```
The OS load on the DARC Node is started.  This takes about 9-11
minutes.
```

```
##.## % done.
```

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Refer to Appendix D for additional information on these error pop-up windows. Also, if any problems arise, verify the correct CD is installed.

5.) When the DARC OS load has completed, a pop-up message is displayed. See Figure 1-12.

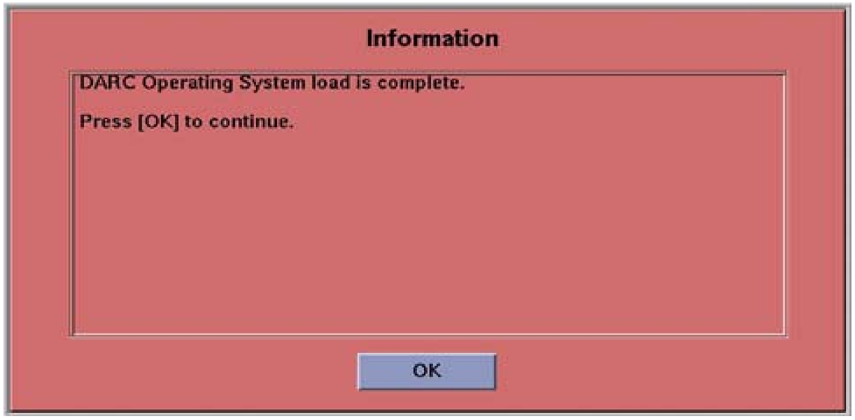


Figure 1-12 DARC OS Load Complete Pop-up Window

6.) Click on **OK**.

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch . Manually eject the CD after approximately 2 minutes.

8.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

9.) This completes the OS load on the DARC.

10.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully loaded. The DARC Applications Software CD-ROM will then be used and copied onto the HP Host Computer and then loaded onto the DARC Node. If any problems arise, verify that the correct CD is installed.

1.) Insert the **GRE/Xtrea DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

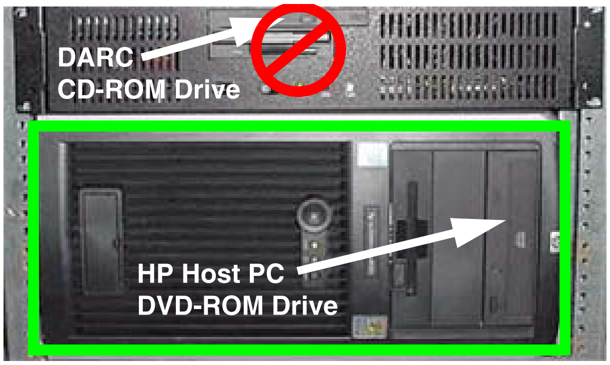


Figure 1-13 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su -`** **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

`root:` **`rsh darc`** **ENTER**

`[root@localhost root]` (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

`[root@localhost root]` **`exit`** **ENTER**

4.) If login was successful, continue with DARC Apps load:

`root:` **`cd /`** **ENTER**

`root:` **`mount /mnt/cdrom`** **ENTER**

`root:` **`cd /mnt/cdrom`** **ENTER**

`root:` **`./copy_rpms`** **ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

`root:` **`cd /`** **ENTER**

`root:` **`umount /mnt/cdrom`** **ENTER**

`root:` **`eject`** **ENTER**

6.) Remove the DARC Applications CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

`root:` **`cd /usr/g/scripts`** **ENTER**

`root:` **`./start_darc_load`** **ENTER**

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **`su -`** **ENTER**

3.) Enter the root password: **`#bigguy`** **ENTER**

⚠ NOTICE

There can be an IP address conflict between LightSpeed 5.x and LightSpeed 1.X (QX/i) systems resulting in the QX/i shutting down when the LightSpeed 5.X system is rebooted.

For 04MW13.X software: Call the OnLine Center for instructions.

For 04MW44.6 software: To correct this potential problem, execute the following command:

`fix_irix_sd -m` **ENTER**. (This command re-configures the conflicting ethernet port.)

4.) Type: **`sync`** **ENTER**

5.) Type: **`sync`** **ENTER**

6.) Type: **`reboot`** **ENTER**

A message appears: `The system is going down for reboot NOW!.`

7.) After approximately 4–5 minutes a pink pop up window appears with the message: `CT Software Auto-Start Disabled.`Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: **`st`** ENTER or **`startup`** ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER (this appears only if **HIPAA Present** was set to ON during the Load process) and then OK, to bring up the login/admin screen.

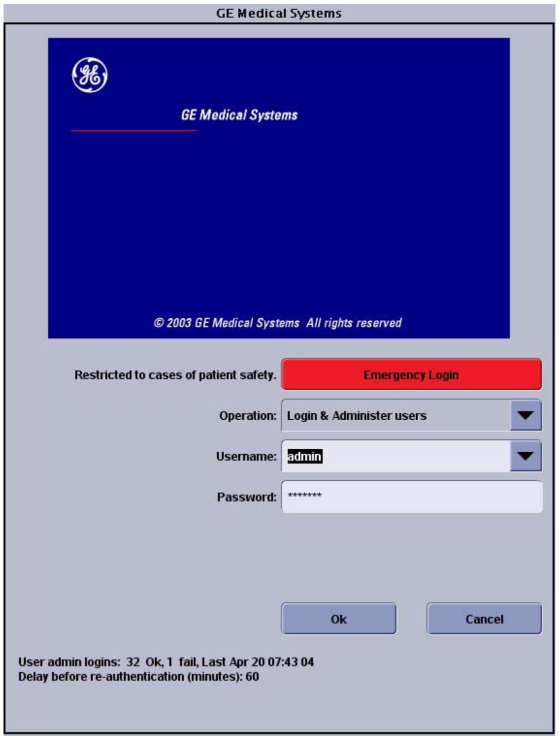


Figure 1-14 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select <u>OK</u>.

5.) On the left hand of the screens you see tabs for admin functions. Select the <u>USER, GROUPS PERMISSIONS</u> tab.

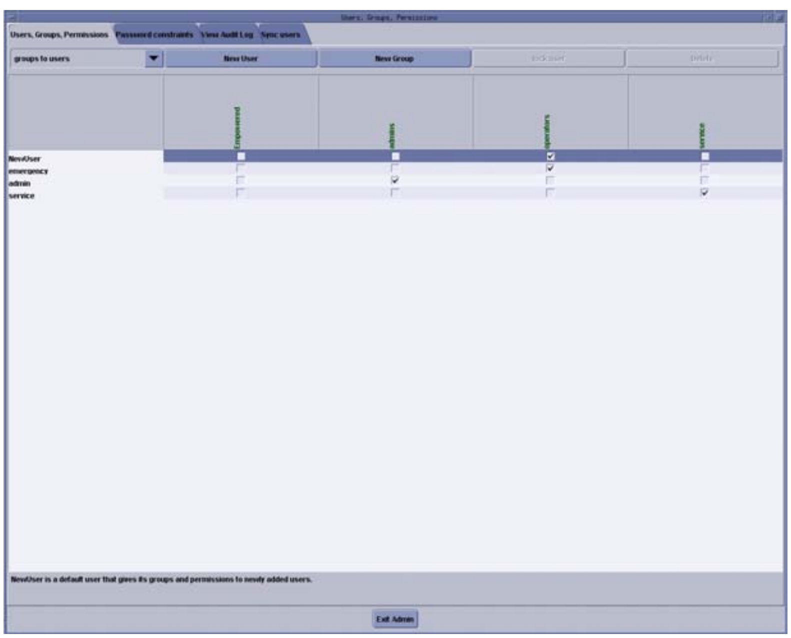


Figure 1-15 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select <u>NEW USER</u> button, which will pop up an entry to type a new user. Enter the user name and select <u>OK</u>.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a operator user has been added, select `groups to users` from the pull-down menu, and assign the operator user to group Operators by selecting the check-box under `operators` in the row of the new user name, e.g., testuser (see Figure 1-16).

Figure 1-16 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`** ENTER

`Password:` **`ctAdmin`** ENTER

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

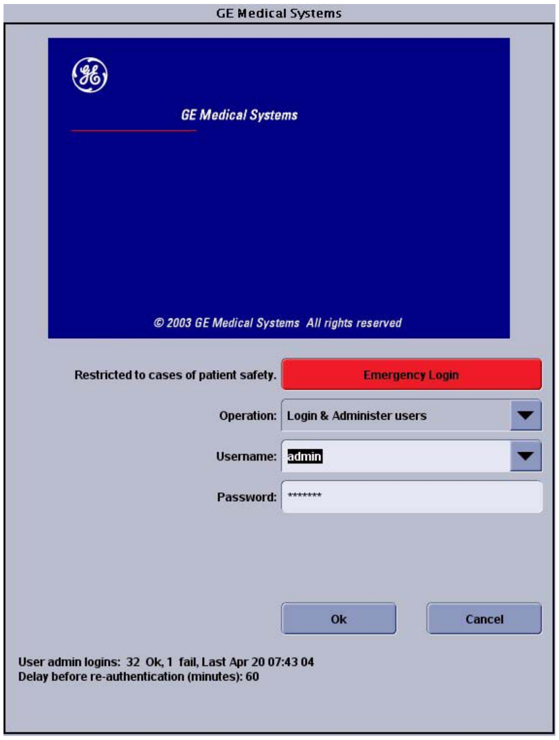


Figure 1-17 Login/Admin screen

3.8 Restoring MOD System State - Site Upgrading From IRIX To Linux Only

Note: This procedure is ONLY for upgrade sites with System State Information on a MOD (not a DVD).

1.) When applications are up, open a Unix Shell from the Service Desktop:

a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.

b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.

c.) Execute the following commands as *`ctuser`*.

`cd /usr/g/bin` ENTER

`sysstategui -MOD -IRX2LNX` ENTER

d.) When the system state GUI comes up, select ALL and then select RESTORE.

e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.

2.) In the same unix shell opened in step 1, execute the following command as ***ctuser***:
`gettubeusagedata` ENTER.

Note: If the above command is not found on the system due to older version software, then type the following set of commands as ***ctuser***:

a.) **`mountMOD -I2L`** ENTER
b.) **`mv /usr/g/service/.bb /usr/g/service/bb.lfc`** ENTER
c.) **`mkdir /usr/g/service/.bb`** ENTER
d.) **`rm -rf /tmp/*bb`** ENTER
e.) **`cp /MOD/service_mod_data/system_state/*bb /usr/g/service/.bb/`** ENTER
f.) **`/usr/g/bin/SwapTubeData`** ENTER
g.) **`/bin/cp -f /tmp/*bb /usr/g/service/.bb/`** ENTER

3.) Close the Unix Shell.
4.) Remove the MOD from the drive.
5.) Now you will re-save system state to DVD.
6.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
7.) Select: SERVICE DESKTOP.
8.) Select: UTILITIES.
9.) Select: SYSTEM STATE.
10.) Click ALL to select all the cals, characterizations, etc.
11.) Click SAVE.
12.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
13.) When completed select DISMISS.
14.) Close the window in the upper left corner of the screen.
15.) Continue with **Section 3.10**.

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.
2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: UTILITIES.
4.) Select: SYSTEM STATE.
5.) Click the ALL selection which highlights the cals, characterization, etc.
6.) Make certain to click RESTORE.
7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.
8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) On the Scan (left) Monitor, click RECON MANAGEMENT, RESTART QUEUE, then QUIT.
2.) On the Display (right) Monitor, verify idle status.
3.) On the Scan (left) Monitor:
   a.) Click RETRO RECON.
   b.) Click the **rat gold** series.
   c.) Click SELECT SERIES.
4.) On the Display (right) Monitor:
   a.) Click EXAM RX.
   b.) Select Autoview and 4-image layout.
5.) On the Scan (left) Monitor:
   a.) Click CONFIRM (25 times for 100 images).
   b.) Verify the image(s) reconstruct and are displayed.
   c.) Click on QUIT.

3.11 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

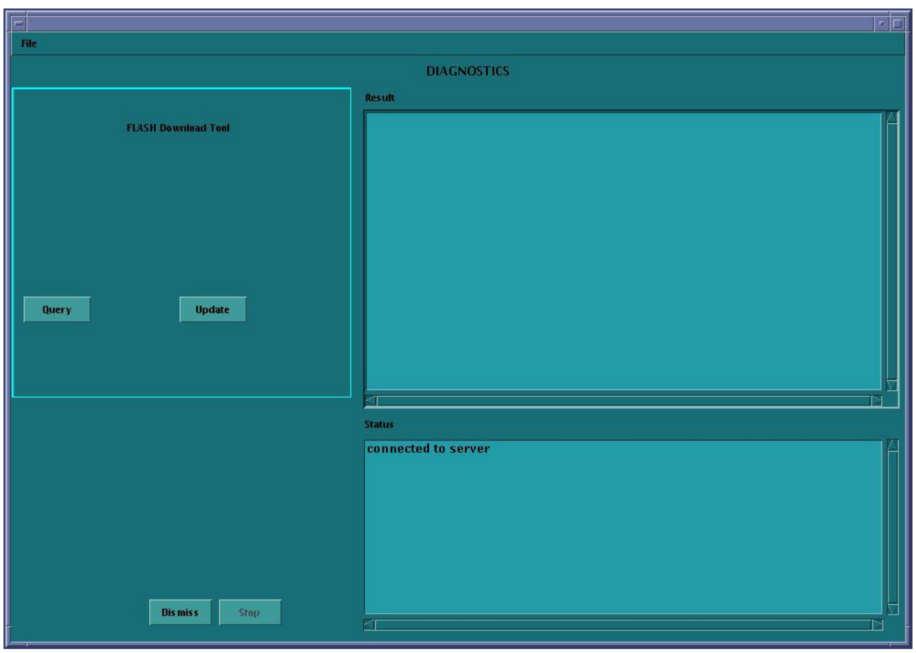


Figure 1-18 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

For LightSpeed 1.X, 2.X, 3.X LightSpeed QX/i, HiSpeed QX/i:

ping obcr ENTER

ping etc ENTER

ping stc ENTER

For LightSpeed 5.X Pro^{16} (100kW), Pro^{16} (80kW), LS^{16}, Ultra, Plus, and RT:

ping orp ENTER

ping tgp ENTER

3.12 Install Options

NOTICE for Perfusion and Denta Scan **For systems where CT Perfusion (e.g. CT Perfusion 2) or Denta Scan is included in Available Options, perform this Install Options procedure in the English language environment. (The LFC can be performed in other language environment than English.) Otherwise, the LFC procedure is possible to fail, or if you change the language setting after installing CT Perfusion or Deta Scan, this software option will not work; you will not even be able to start the program.**

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select CONFIGURATION.
4.) Select INSTALL.

Note: LS5.X and later products do not have an "Install" folder under the "Configuration" tab. Skip this step and go to the next step.

5.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 1-19) appears.

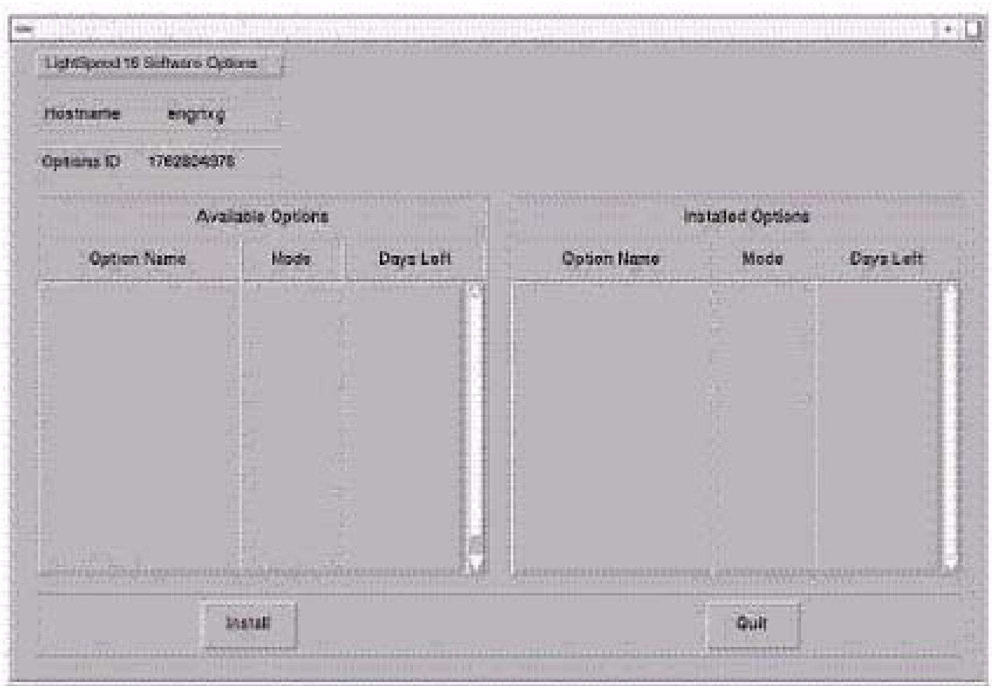


Figure 1-19 Install Options screen

6.) Click INSTALL. A Select Mechanism (Figure 1-20) window opens. Select the mechanism through which you want to install Option Keys.

Figure 1-20 Select Mechanism window

7.) Click PERMANENT. A Select Device (Figure 1-21) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 1-21 Select Device window

8.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 1-21 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

9.) Click OK. Available options appear on the Software Options screen (Figure 1-22).

Example:
Options screen
with Permanent
& FlexTrial
options.

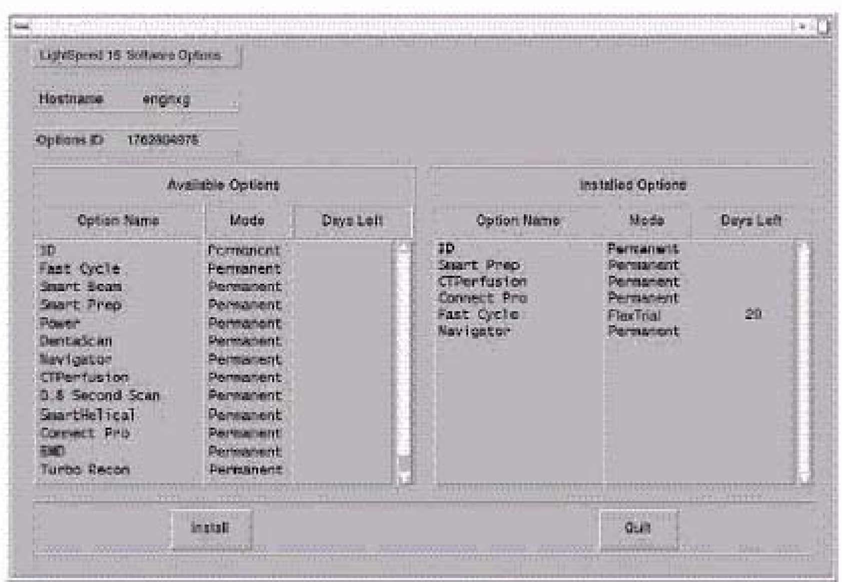


Figure 1-22 Software Options Screen (Example)

10.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

11.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **`su -`** ENTER.

Type the root password: **`#bigguy`** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press **ENTER**.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** ENTER.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Enable Extended CT Number Range (LightSpeed 5.X RT Only)

The extended CT number range (available from -31743 to +31743 Hounsfeld units) is configurable using a selection on the Image Works Desktop. It allows the user to distinguish between different types of metal more clearly and gives a better range when measuring the CT number on a patient. This feature is defaulted OFF for LightSpeed 5x systems because some of the cardiac and other applications do not do well with this extended CT number range. However, since the LightSpeed RT system does not use these cardiac applications, it should be turned ON for RT system type.

1.) For all LightSpeed RT Systems: Click on the TURN ON EXTEND HU button. See Figure 1-23.

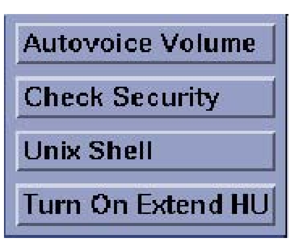


Figure 1-23 Turn On Extend HU

Note: If Extend HU is already turned-on, the Tool bar will display TURN OFF EXTEND HU, as shown in Figure 1-24.

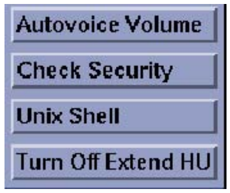


Figure 1-24 Turn OFF Extend HU

2.) A pop-up box appears, indicating that a system reboot is required to enable the change of modes from the current to mode selected in the Tool Bar. See Figure 1-25. Click OK to reboot.

Figure 1-25 System Reboot Notice Screen

3.) Click OK to reboot.

4.) After the system reboots, continue with Section 3.15.2.

3.15 Restart the System

3.15.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** ENTER

2.) Select OK as needed to answer messages. Note any Recon Failures.

3.) Select the red SHUTDOWN ICON. A set of options appears.

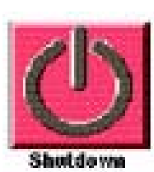


Figure 1-26 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.15.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.16 System Sanity Scanning

1.) Successfully perform a Scout scan.
2.) Successfully perform a Helical scan.
3.) Successfully perform an Axial scan.

Load Process is now complete!

Section 4.0
GRE Console Information Sheets

4.1 LightSpeed 1.X (QX/i) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: SDAS4
- ❑ Tube Type (15): Performix
- ❑ PDU Type: CT COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: H1
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.2 LightSpeed 2.X (Plus, QX/i) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: SDAS4
- ❑ Tube Type (15): Performix
- ❑ PDU Type: CT COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: H2
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.3 HiSpeed QX/i GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4
- ❑ Tube Type (15): Performix
- ❑ PDU Type: CT COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: QX
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.4 LightSpeed 3.X (MDAS4/8) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4 MDAS8
- ❑ Tube Type (15): Performix
- ❑ PDU Type: CT COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: H2
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.5 LightSpeed 4.X (LS 16) GRE M3.1 Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS16
- ❑ Tube Type (15): Performix
- ❑ PDU Type: CT COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: H2
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.6 LightSpeed 5.X (LS Pro^{16} (100kW)) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: GDAS16 or MDAS16
- ❑ Tube Type (16): Hercules
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: HP
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.7 LightSpeed 5.X (LS Pro^{16} (80kW)) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: GDAS16 or MDAS16
- ❑ Tube Type (16): OEM
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: HP
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

4.8 LightSpeed 5.X (LS^{16}, Ultra, and Plus) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type:
  - GDAS16 or MDAS16 for LightSpeed^{16}
  - GDAS8 for LightSpeed Ultra
  - GDAS4 for LightSpeed Plus
- ❑ Tube Type: 15
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: 2
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: hp
- ❑ Target Noise Index Table: Table1 Table 2 Table3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________

  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:

  CTTITLE: __________

  PORTNUM: __________

  AETITLE: __________

  IPADDR: ____ . ____ . ____ . ____

4.9 LightSpeed 5.X (RT) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4
- ❑ Tube Type (17): OEM
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ Gantry Type: WB
- ❑ Target Noise Index Table: Table 1 Table 2 Table 3
- ❑ Dose Info Display: On On Without Total DLP Off
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: On Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info:
  CTTITLE: __________
  PORTNUM: __________
  AETITLE: __________
  IPADDR: ____ . ____ . ____ . ____

Chapter 2
LFC Procedures – LightSpeed 3.X (LS 4/8 Slice Plus/Ultra M4) M4 & HS QX/i M4

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 3.X 4/8 Slice MDAS and HiSpeed QX/i GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed or HiSpeed QXi Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed /HiSpeed QXi DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 3.X (4/8 Slice Ultra/Plus) and HiSpeed QX/i**:

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **5111369 – LightSpeed 3.X Applications Software**
  Version 308L.8_H3.3M4
- **2401242 - HiSpeed QX/i M4 Applications Software**
  Version 103L.4.HSQXIM4
- **2394896 – LightSpeed / HiSpeed QXI DARC Applications M4**
  Version – H4.2/H3.3/HSQXI M4
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: **`cat /usr/g/config/INFO`** **ENTER**
   c.) Record screen information in Section 3.16. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 4/8 MDAS GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: **`cat /usr/g/config/INFO`** **ENTER**
- ❑ Install the Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – YES to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, darc will reboot then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: **`st`** **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **#bigguy**
- ctuser: **4$apps**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again`.
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Console Upgrade or a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.).

Console Upgrade: Follow the instructions below for the following console upgrade:

- GOC2/Linux Console (M3 software) to GOC3/GRE (M4 software)

Software Upgrade: Follow the instructions below for the following software upgrade:

- GOC3/GRE M3 or GOC3/GRE3.1 to GOC3/GRE M4

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing the above Console or software upgrades. This is a one time procedure necessary prior to loading M4 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M4 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

COMMON INSTRUCTIONS

Steps 1–4 below pertain to ALL of the above:

1.) Open up a Unix Shell.
2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.
3.) Type from the following to save CSA Protocols:
   a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER
   b.) **`mountDVD`** ENTER
   `Mounting DVD.`
   c.) **`\cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER
   d.) **`unmountDVD`** ENTER
   `Unmounting DVD.`
4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

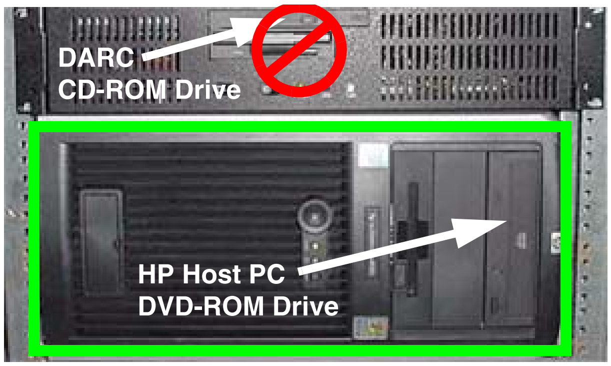


Figure 2-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 2-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed 3.X Applications Software CDROM** or the **HiSpeed QX/i M4 Applications Software CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the appropriate Applications Software CDROM (**LightSpeed 3.x Applications Software** CDROM or the **HiSpeed QX/i M4 Applications Software** CDROM) into the DVD-ROM drive on the HP Host PC.

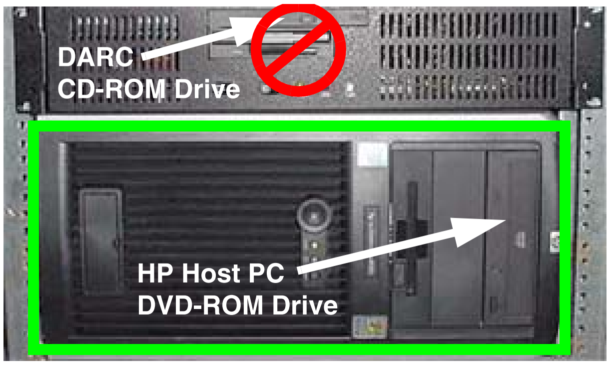


Figure 2-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

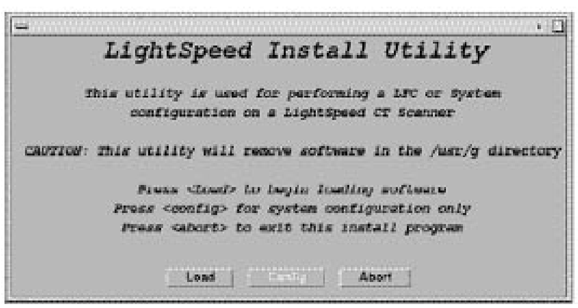


Figure 2-4 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

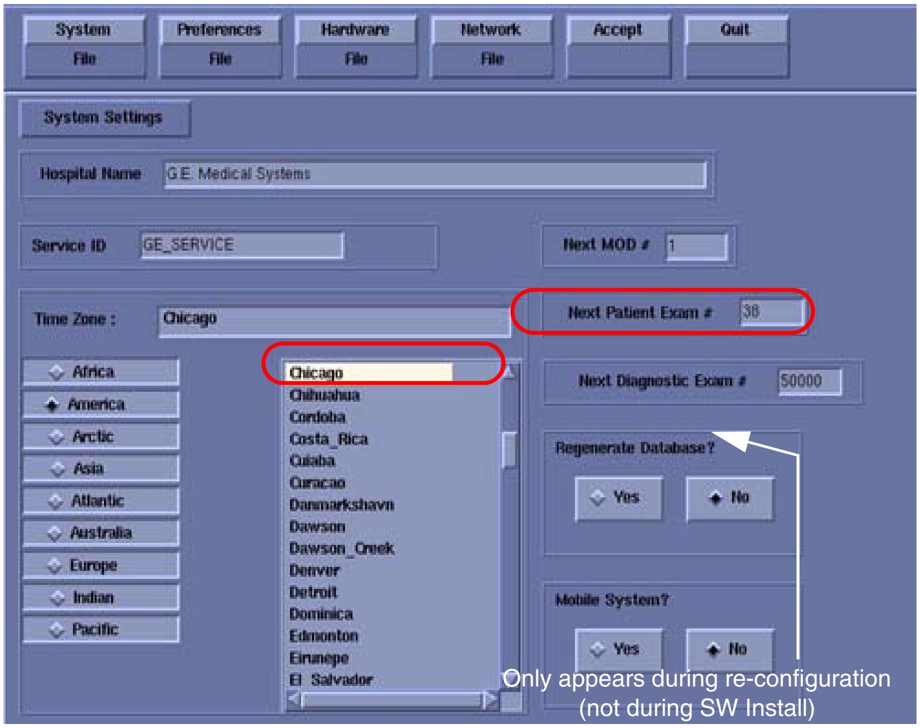


Figure 2-5 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

8.) Verify that the proper **Time Zone** is selected (example: Chicago).

9.) Click the PREFERENCES button.

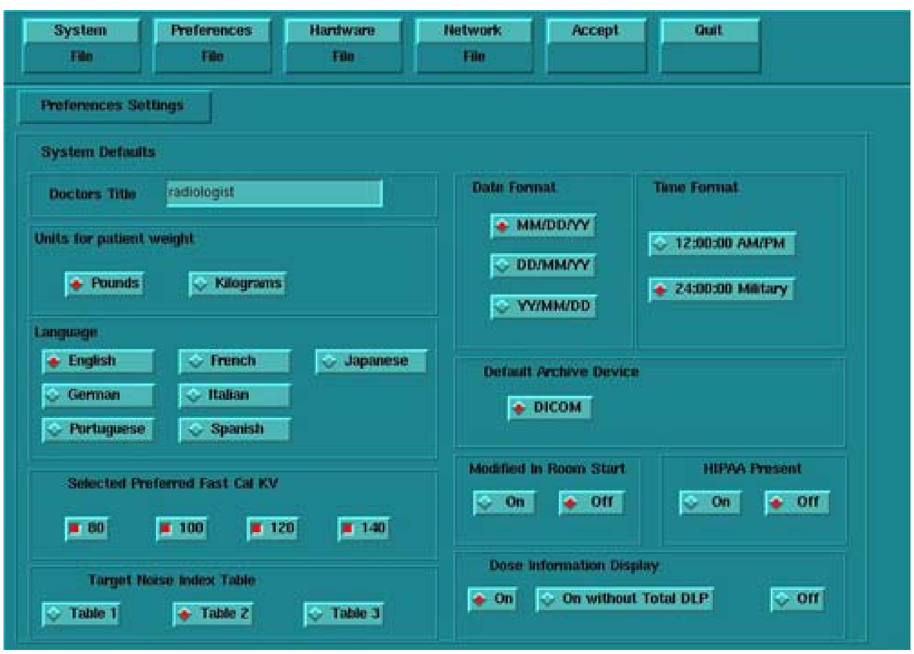


Figure 2-6 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

Figure 2-7 Hardware Settings screen

21.) Select the proper DAS Type: MDAS4 or MDAS8. (For HiSpeed QX/i, select MDAS4.)

22.) Select the proper Tube Type: MX_200MCT.

23.) Select the proper PDU Type: CT_COMPACT. (For HiSpeed QX/i, select NG PDU.)

24.) Select the proper Collimator/Filter Type: HELIOS G1.

25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Select the proper GANTRY TYPE: H2 GANTRY. (For HiSpeed QX/i, select HISPEED QX/I.)

Note: The DAS Type and Scan Recon Hardware Configuration selections are interactive. Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

28.) Click the NETWORK button (site will be different).

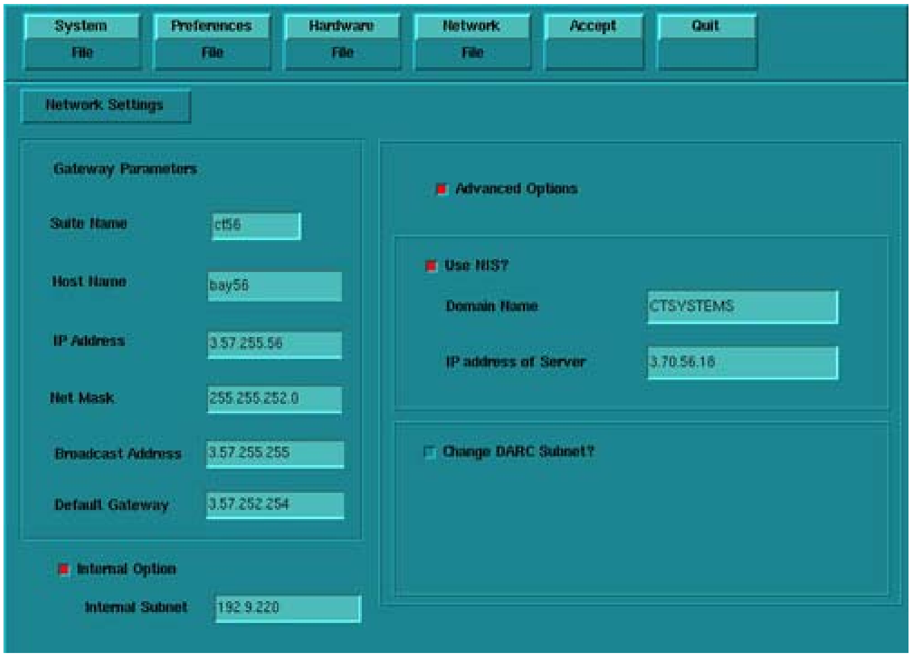


Figure 2-8 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

33.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

34.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

35.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

36.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

37.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

38.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

39.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

40.) Click the ACCEPT button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed 3.X Applications Software CDROM or the HiSpeed QX/i M4 Applications Software CDROM (as applicable) from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

44.) Press the button on the DVD ROM drive to remove the LightSpeed 3.x Applications Software CDROM or the HiSpeed QX/i M4 Applications Software CDROM (as applicable) from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat space /usr/g/config/INFO`** **ENTER**

Verify DAS type (MDAS4 or MDAS8), Tube type (15 for MX_200MCT), PDU type, Gantry type (h2), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darcOS`** **ENTER**

An Attention window appears. See Figure 2-9.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 2-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

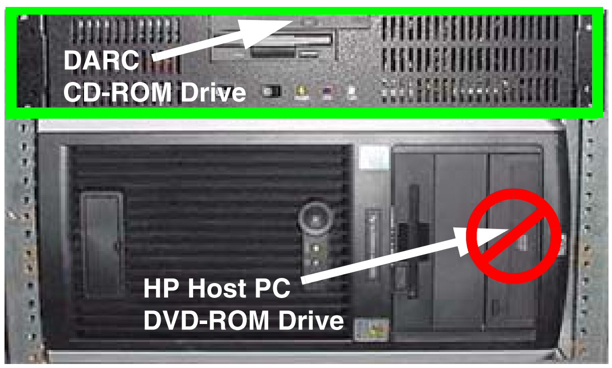


Figure 2-10 DARC CD-ROM Drive Location

5.) Click on **OK**..

A sh window displays initial information, repetitive dpccli> commands, then the message below:

```
The OS load on the DARC Node is started.  This takes about 9-11
minutes.
##.## % done.
```

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

```
The OS load is complete. Please remove the OS disk from the DARC disk
drive. DARC is rebooting. Wait for the DARC to boot up…
```

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch. Manually eject the CD after approximately 2 minutes. See Figure D-8 for location of DARC Reset Switch.

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 2-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

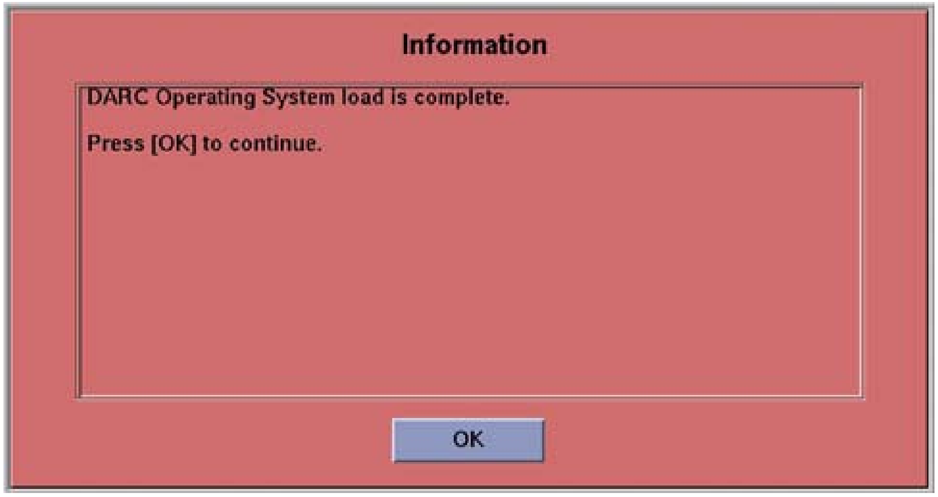


Figure 2-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully ioaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **LightSpeed / HiSpeed QXI DARC Applications M4** CDROM into the DVD-ROM drive on the HP Host Computer.

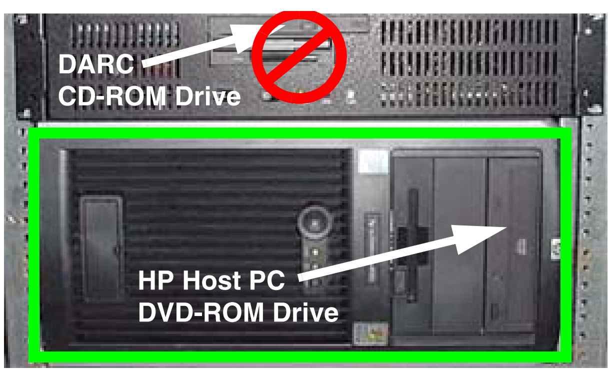


Figure 2-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** **-** **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

`root:` **rsh SPACE darc ENTER**

`[root@localhost root]` (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

`[root@localhost root]` **exit ENTER**

4.) If login was successful, continue with DARC Apps load:

`root:` **cd SPACE / ENTER**

`root:` **mount SPACE /mnt/cdrom ENTER**

`root:` **cd SPACE /mnt/cdrom ENTER**

`root:` **./copy_rpms ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

`root:` **cd SPACE / ENTER**

`root:` **umount SPACE /mnt/cdrom ENTER**

`root:` **eject ENTER**

6.) Remove the LightSpeed / HiSpeed QXI DARC Applications M4 CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

`root:` **cd SPACE /usr/g/scripts ENTER**

`root:` **./start_darc_load ENTER**

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **su SPACE - ENTER**

3.) Enter the root password: **#bigguy ENTER**

4.) Type: **sync ENTER**

5.) Type: **sync ENTER**

6.) Type: **reboot ENTER**
A message appears: `The system is going down for reboot NOW!.`

7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).

2.) Type: **st ENTER** or **startup ENTER**.

3.) A pink pop-up Attention Message appears: `OC initializing. Please wait`....

Read the Note(s) below.

4.) Click OK as needed to answer message prompts.

5.) If any Recon Selftest Failures are encountered, review the error log.

Note:

- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.

2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

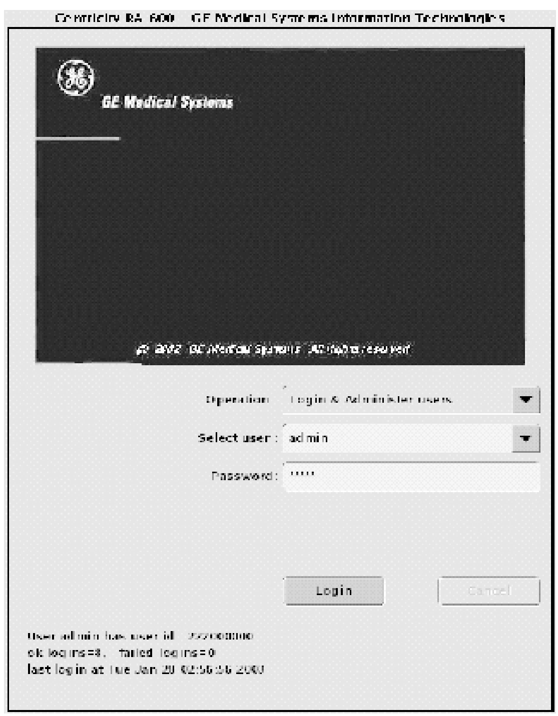


Figure 2-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

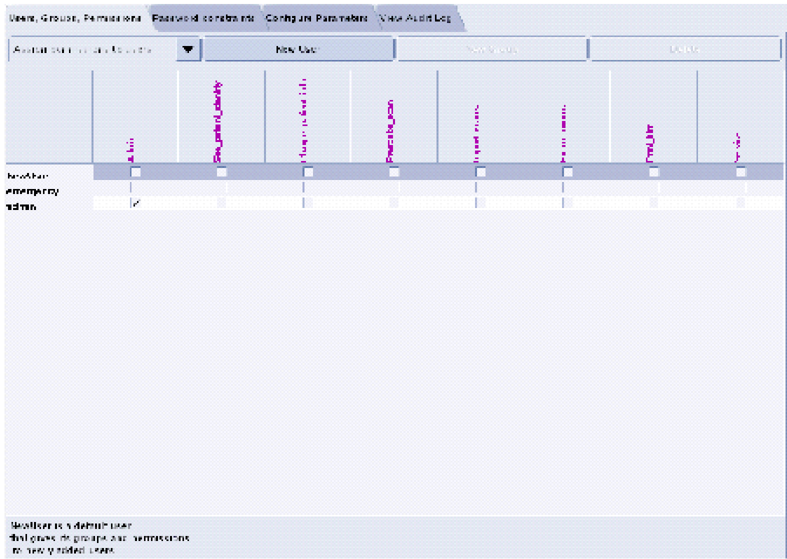


Figure 2-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 2-15).

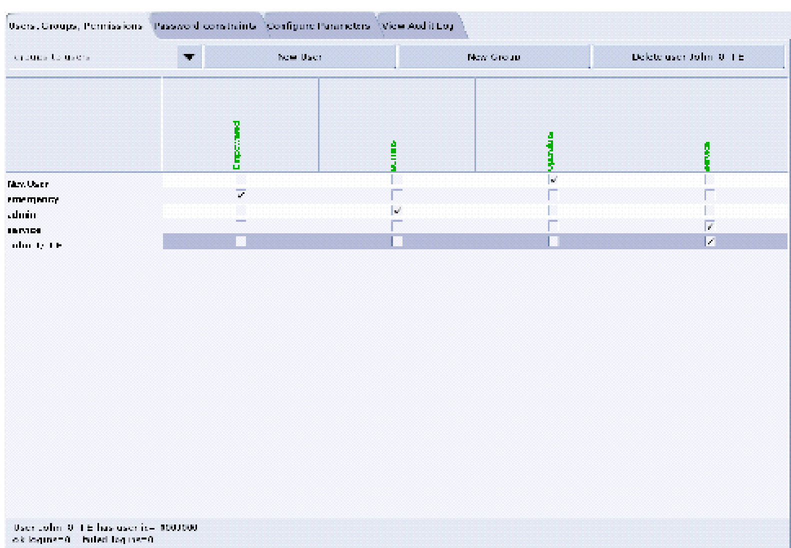


Figure 2-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 2-16 Login/Admin screen

3.8 Restoring MOD System State - Site Upgrading From IRIX To Linux Only

For sites that ONLY have System State Information on a MOD (not a DVD).

1.) When applications are up, open a Unix Shell from the Service Desktop:

a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.

b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.

c.) Execute the following commands as *`ctuser`*.

`cd /usr/g/bin` ENTER

`sysstategui -MOD -IRX2LNX` ENTER

d.) When the system state GUI comes up, select ALL and then select RESTORE.

e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.

2.) In the same unix shell opened in step 1, execute the following commands as ***ctuser***:

a.) **`mountMOD -I2L`** ENTER

b.) **`mv /usr/g/service/.bb  /usr/g/service/bb.lfc`** ENTER

c.) **`mkdir /usr/g/service/.bb`** ENTER

d.) **`rm -rf /tmp/*bb`** ENTER

e.) **`cp /MOD/service_mod_data/system_state/*bb  /usr/g/service/.bb/`** ENTER

f.) **`/usr/g/bin/SwapTubeData`** ENTER

g.) **`/bin/cp -f  /tmp/*bb /usr/g/service/.bb/`** ENTER

3.) Close the Unix Shell.

4.) Remove the MOD from the drive.

5.) Now you will re-save system state to DVD.

6.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.

7.) Select: SERVICE DESKTOP.

8.) Select: PM.

9.) Select: SYSTEM STATE.

10.) Click ALL to select all the cals, characterizations, etc.

11.) Click SAVE.

12.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`

13.) When completed select DISMISS.

14.) Close the window in the upper left corner of the screen.

15.) Continue with **Section 3.10**.

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.

8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.11 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

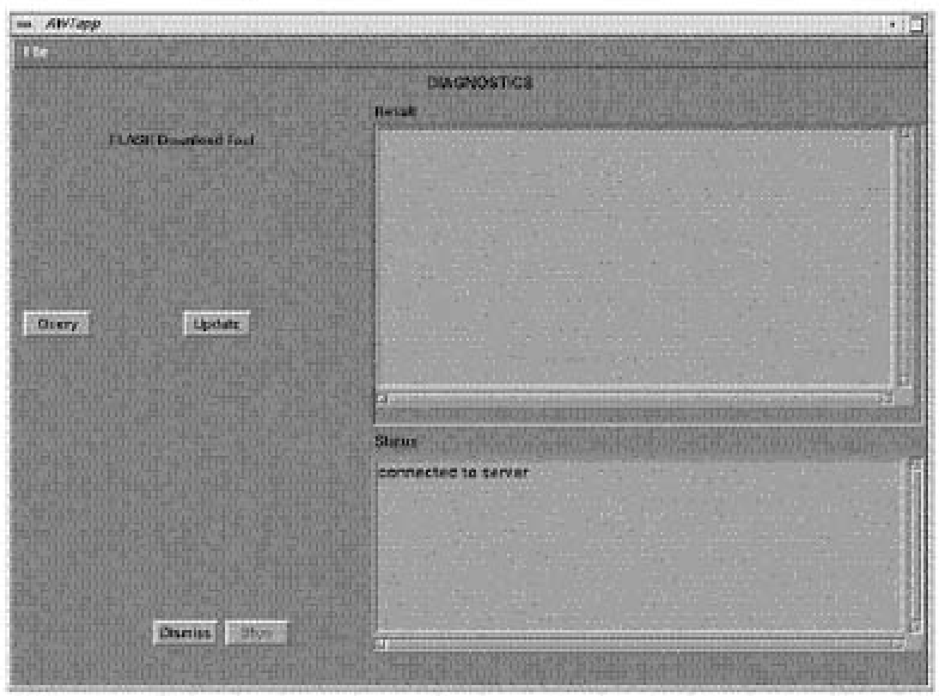


Figure 2-17 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping obcr` ENTER

`ping etc` ENTER

`ping stc` ENTER

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL.
5.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 2-18) appears.

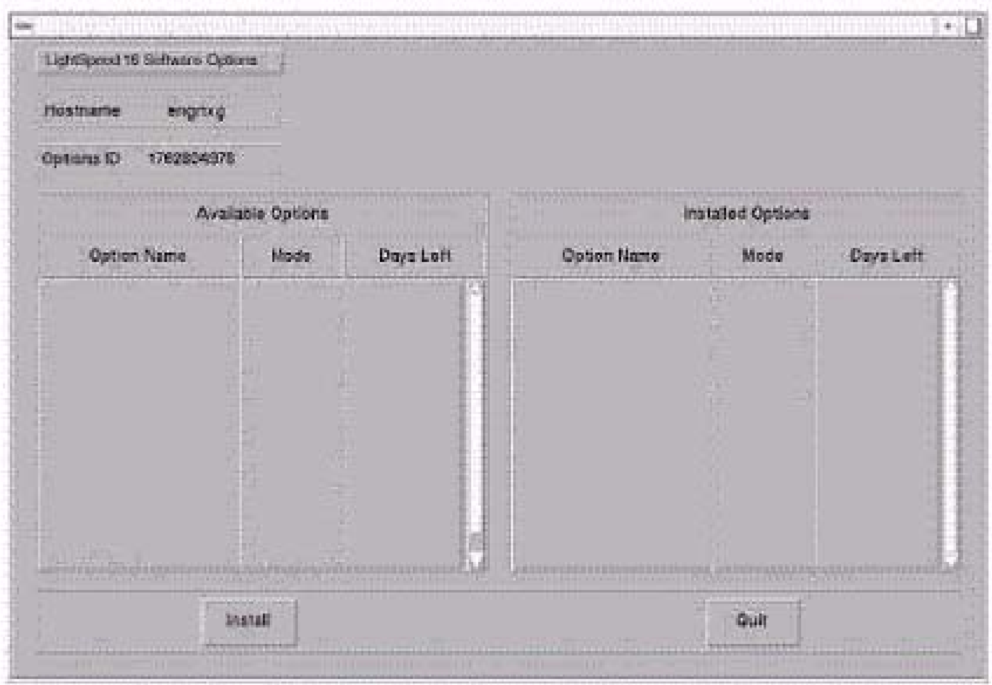


Figure 2-18 Install Options screen

6.) Click INSTALL. A Select Mechanism (Figure 2-19) window opens. Select the mechanism through which you want to install Option Keys.

Figure 2-19 Select Mechanism window

7.) Click PERMANENT. A Select Device (Figure 2-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 2-20 Select Device window

8.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 2-20 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

9.) Click OK. Available options appear on the Software Options screen (Figure 2-21).

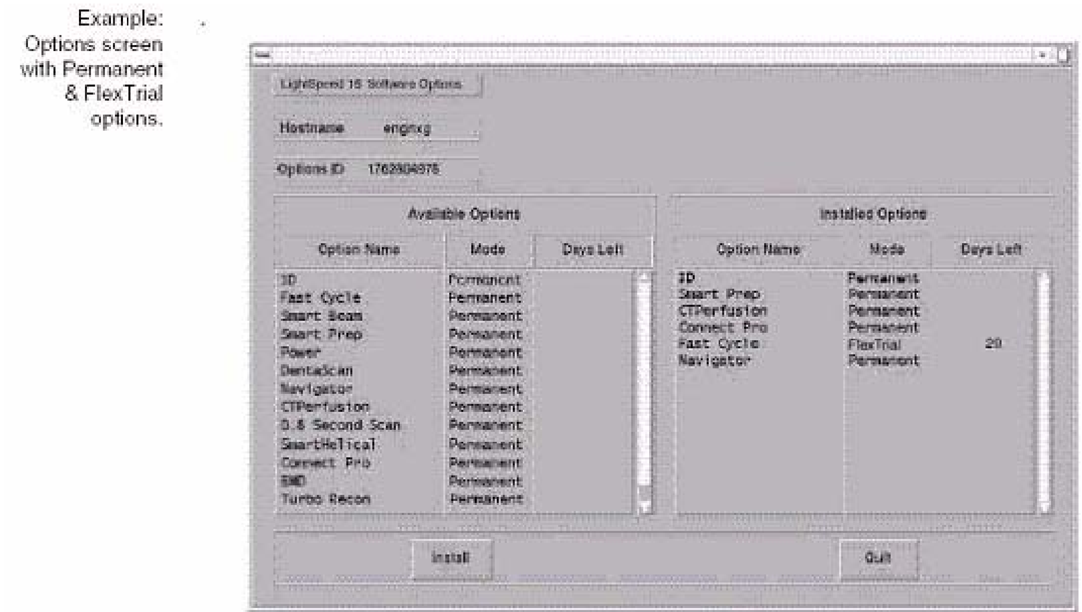


Figure 2-21 Software Options screen

10.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

11.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su** **SPACE** **-** **ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press **ENTER**.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** **ENTER**

2.) Select OK as needed to answer messages. Note any Recon Failures.

3.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 2-22 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.

3.) Successfully perform an Axial scan.

Load Process is now complete!

3.16 LightSpeed 3.X (4/8 MDAS) & HiSpeed QX/i GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4 MDAS8 (For HiSpeed QX/i, select MDAS4.)
- ❑ Tube Type (15): MX_200MCT
- ❑ PDU Type: CT_COMPACT (For HiSpeed QX/i, select NG PDU.)
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: H2 GANTRY (For HiSpeed QX/i, select HISPEED QX/I.)
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: CT## WLSNorm 4008 3.57.255.##

Chapter 3
LFC Procedures – LightSpeed 3.x (LS 4/8 Slice Plus/Ultra M3) & HS QX/i M3

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 3.X 4/8 Slice MDAS and HiSpeed QX/i GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 3.X (4/8 Slice Ultra/Plus M3) and HiSpeed QX/i M3**:

1.) **2393921 – LightSpeed / HiSpeed / Xtream Console Operating System**
Version – GEMS Linux 1.7.9
Supports: H2/H3/H4/H5 M3
2.) **2393929 – LightSpeed 4&8 MDAS Application Software**
Version 307L.2_H3.3_GREM3
3.) **2393920 – Lightspeed / HiSpeed QX/I DARC Applications**
Version – H5/H4/H3/H2 M3

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM does not have a CD inserted until instructed in the procedure.
4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: `cat /usr/g/config/INFO` **ENTER**
   c.) Record the information on the screen in Section 3.16. For an example screen output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 4/8 MDAS GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: `cat /usr/g/config/INFO` **ENTER**
- ❑ Install the OS on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – YES to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, darc will reboot then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: `st` **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.

- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **#bigguy**
- ctuser: **4$apps**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM OS Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.
2.) Remove the GRE Operator Console's front cover.
3.) Locate the HP PC's DVD-ROM drive.
4.) Insert the LightSpeed Console Operating System CD-ROM into the DVD ROM drive on the HP Host Computer.
5.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 3-1 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

6.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.
As the computer restarts, the booting process messages appear.
After the booting process completes, the `boot:` prompt appears.
7.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

8.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.
9.) The HP Host Computer reboots. **Do not insert the LightSpeed Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the LightSpeed Applications SW CDROM into the DVD-ROM drive on the HP Host PC.

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

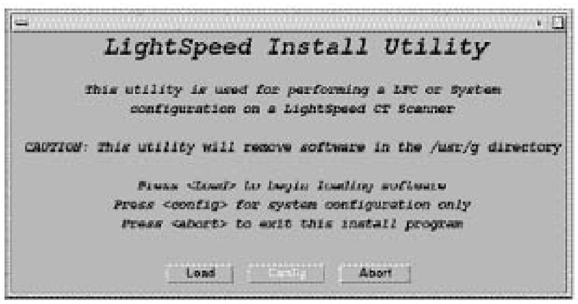


Figure 3-2 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

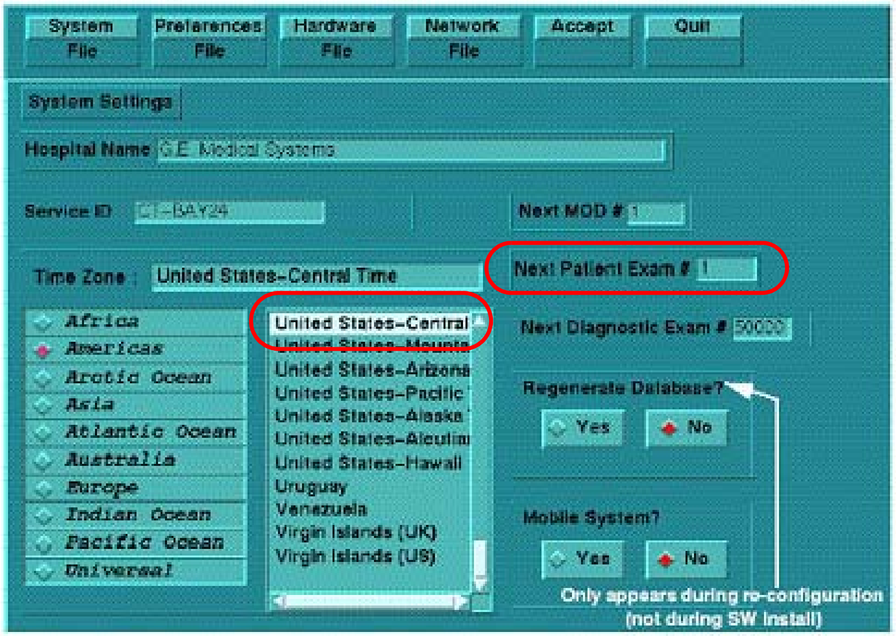


Figure 3-3 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is **1**.

8.) Verify that the proper **Time Zone** is selected (example: Americas and United States–Central and Chicago).

9.) Click the PREFERENCES button.

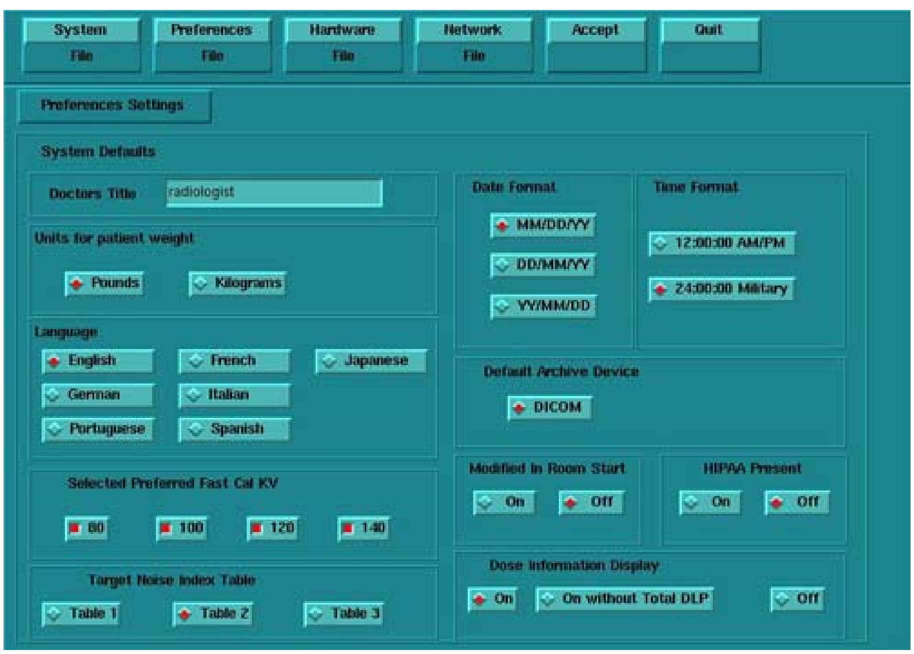


Figure 3-4 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.
14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).
15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).
16.) Default **Archive Device**: add your device here (Example: DICOM.)
17.) Verify **Modified in Room Start** is set to OFF.
18.) Set **HIPAA Present** to OFF.
19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:
   a.) Select ON (full CTDiw Display)
   b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
   c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

Figure 3-5 Hardware Settings screen

21.) Select the proper DAS Type: MDAS4 or MDAS8. (For HiSpeed QX/i, select MDAS4.)
22.) Select the proper Tube Type: MX_200MCT.
23.) Select the proper PDU Type: CT_COMPACT. (For HiSpeed QX/i, select NG PDU.)
24.) Select the proper Collimator/Filter Type: HELIOS G1.
25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.
26.) Select the proper GANTRY TYPE: H2 GANTRY. (For HiSpeed QX/i, select HISPEED QX/I.)

Note: The DAS Type and Scan Recon Hardware Configuration selections are interactive. Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.
28.) Click the NETWORK button (site will be different).

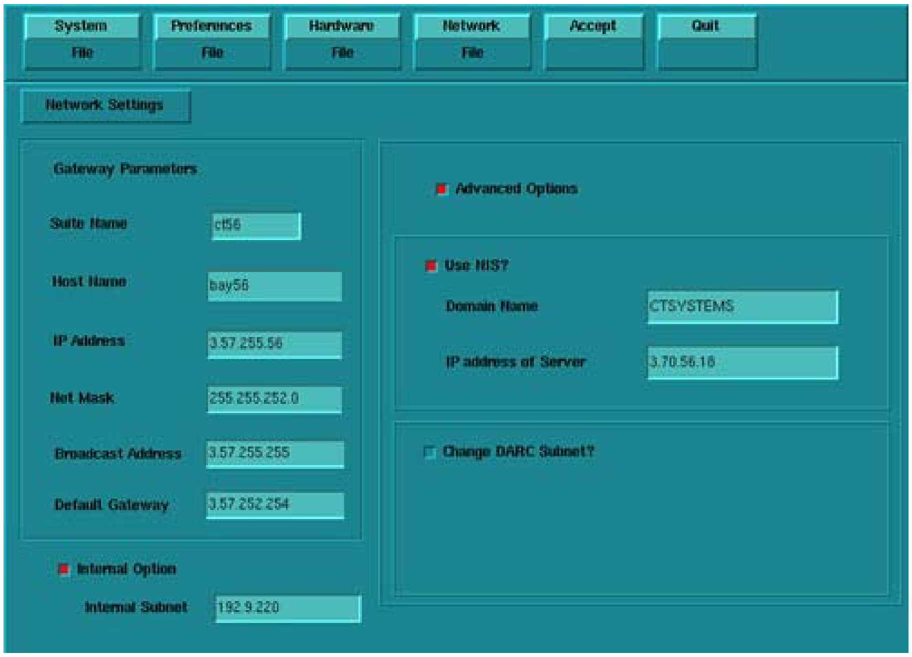


Figure 3-6 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

33.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

34.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

35.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

36.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

37.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

38.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

39.) Click the **ACCEPT** button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed Apps CDROM from the HP Host Computer until instructed to do so.

40.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load)`. A shell Installation screen appears and the application software starts loading.

41.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

42.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

43.) Press the button on the DVD ROM drive to remove the LightSpeed Apps SW CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node.

- If any problems arise, verify the correct CD is installed.
- If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **cat** **SPACE** **/usr/g/config/INFO** **ENTER**

Verify DAS type (MDAS4 or MDAS8), Tube type (15 for MX_200MCT), PDU type, Gantry type (h2), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **su** **SPACE** **-** **ENTER**

`password:` **#bigguy** **ENTER**

`root:` **cd** **SPACE** **/usr/g/scripts**

`root:` **./start_darcOS** **ENTER**

A sh window appears with the message below:

`Please insert the Operating System CD into the DARC Node disk drive.`

`Press <Enter> to start OS load.`

4.) Insert the LightSpeed Console Operating System CD-ROM into the DARC Node drive.

5.) Select the **ENTER** key.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

`The OS load on the DARC Node is started.  This takes about 9-11 minutes.`

`##.## % done.`

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. You may be instructed to reboot; open a Unix shell and type "halt", then power cycle the console.

6.) A `%done` status is displayed during the DARC OS load.

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

`The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…`

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully.

Note: If the OS disk CD does not eject or the DARC does not boot/reboot correctly, then press the DARC Node front panel Reset switch. Manually eject the CD after approximately 2 minutes.

9.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

10.) This completes the OS load on the DARC.

11.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

The DARC Applications Software CD-ROM will be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** `-` **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **`rsh`** **SPACE** **`darc`** **ENTER**

root@localhost login: **`exit`** **ENTER**

4.) If login was successful, continue with DARC Apps load:

`root:` **`cd`** **SPACE** `/` **ENTER**

`root:` **`mount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`cd`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`./copy_rpms`** **ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type:

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darc_load`** **ENTER**

A shell window pops up and the load begins. It takes 4–5 minutes.

Continue the procedure when the load window goes away.

6.) When the load displays a message that it is completed or window goes away, type the following:

`root:` **`cd`** **SPACE** `/` **ENTER**

`root:` **`umount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`eject`** **ENTER**

7.) Close the shell (do not use for the next process).

8.) Remove the DARC Applications SW CDROM from the HP Host Computer DVD-ROM drive.

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **`su`** **SPACE** `-` **ENTER**

3.) Enter the root password: **`#bigguy`** **ENTER**

4.) Type: **`sync`** **ENTER**

5.) Type: **`sync`** **ENTER**

6.) Type: `reboot` ENTER
A message appears: `The system is going down for reboot NOW!.`

7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: `st` ENTER or `startup` ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:

- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

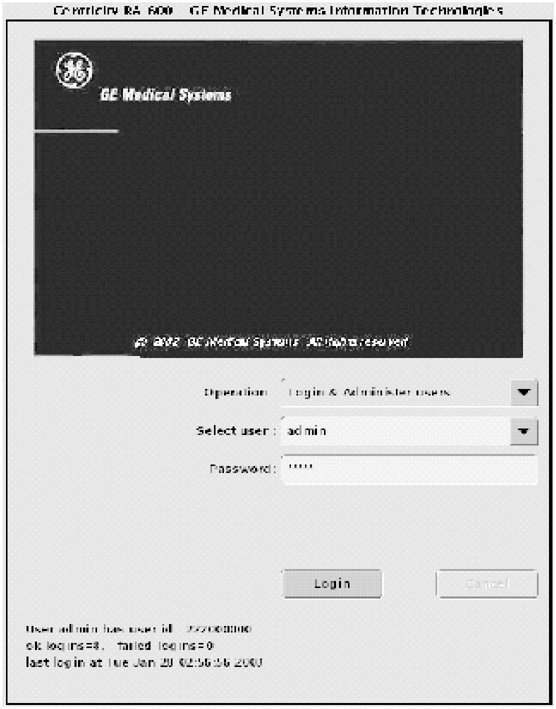


Figure 3-7 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

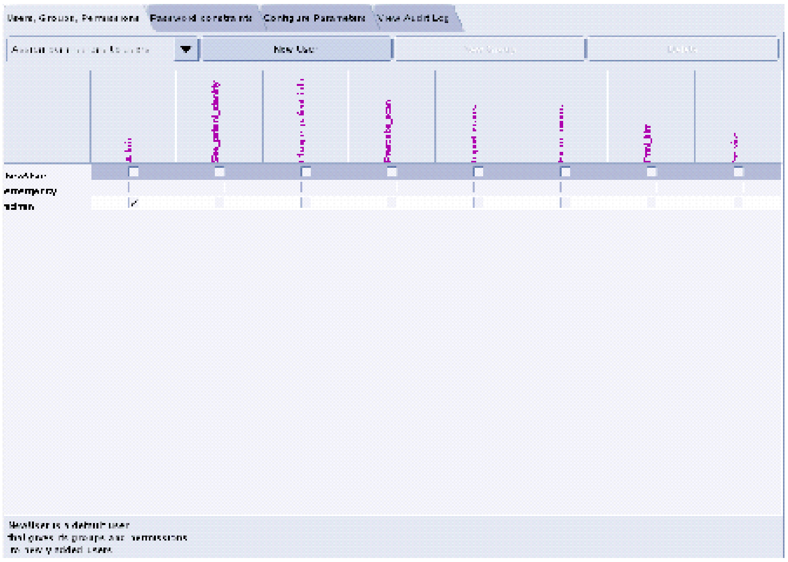


Figure 3-8 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 3-9).

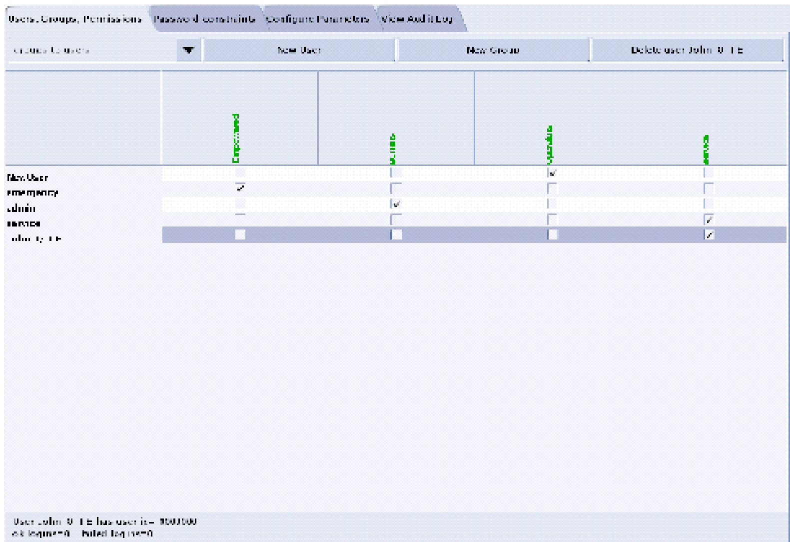


Figure 3-9 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 3-10 Login/Admin screen

3.8 Restoring MOD System State - Site Upgrading From IRIX To Linux Only

For sites that ONLY have System State Information on a MOD (not a DVD).

1.) When applications are up, open a Unix Shell:
   - a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.
   - b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.
   - c.) Execute the following commands as *`ctuser`*.

     `cd /usr/g/bin` ENTER

     `sysstategui -MOD -IRX2LNX` ENTER
   - d.) When the system state GUI comes up, select ALL and then select RESTORE.
   - e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.

2.) In the same unix shell opened in step 1, execute the following commands as ***`ctuser`***:
   - a.) **`mountMOD -I2L`** ENTER
   - **b.)** **`mv /usr/g/service/.bb  /usr/g/service/bb.lfc`** ENTER
   - **c.)** **`mkdir /usr/g/service/.bb`** ENTER
   - **d.)** **`rm -rf /tmp/*bb`** ENTER
   - e.) **`cp /MOD/service_mod_data/system_state/*bb  /usr/g/service/.bb/`** ENTER
   - f.) **`/usr/g/bin/SwapTubeData`** ENTER
   - g.) **`/bin/cp -f  /tmp/*bb /usr/g/service/.bb/`** ENTER

3.) Close the Unix Shell.

4.) Remove the MOD from the drive.

5.) Now you will re-save system state to DVD.

6.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.

7.) Select: SERVICE DESKTOP.
8.) Select: PM.
9.) Select: SYSTEM STATE.
10.) Click ALL to select all the cals, characterizations, etc.
11.) Click SAVE.
12.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
13.) When completed select DISMISS.
14.) Close the window in the upper left corner of the screen.
15.) Continue with **Section 3.10**.

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.
2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click the ALL selection which highlights the cals, characterization, etc.
6.) Make certain to click RESTORE.
7.) Verify the overall system state content has restored correctly.
8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.11 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

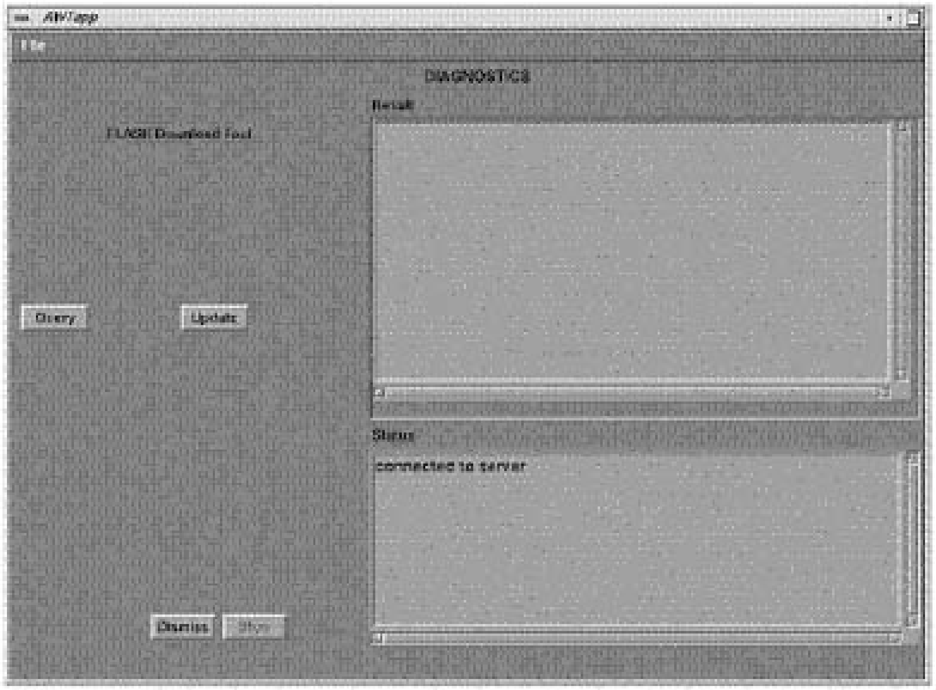


Figure 3-11 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping obcr` **ENTER**

`ping etc` **ENTER**

`ping stc` **ENTER**

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL.

5.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 3-12) appears.

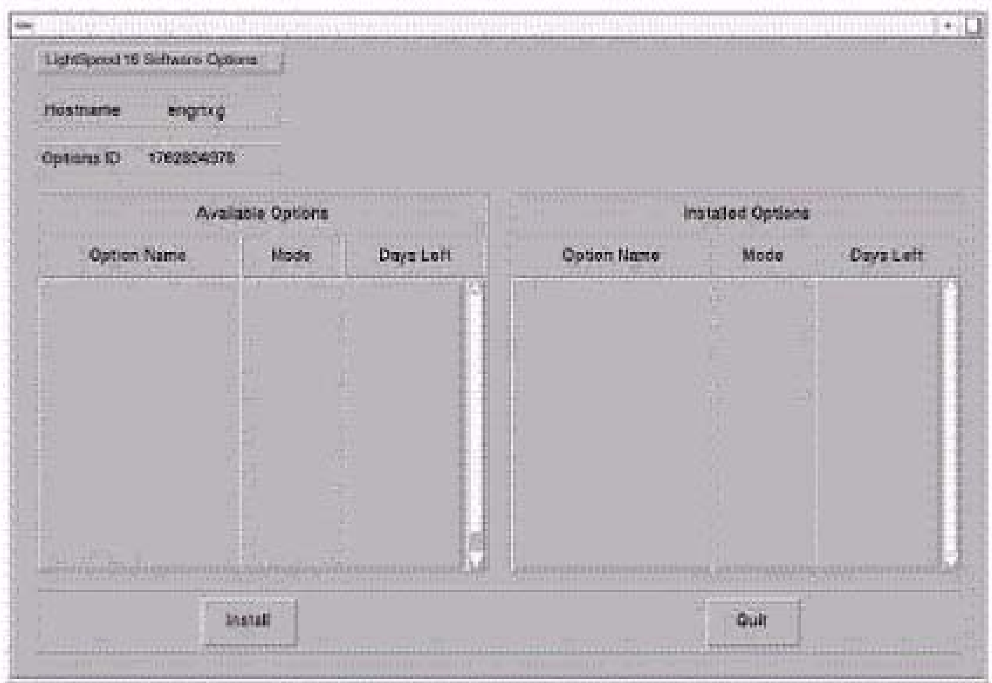


Figure 3-12 Install Options screen

6.) Click INSTALL. A Select Mechanism (Figure 3-13) window opens. Select the mechanism through which you want to install Option Keys.

Figure 3-13 Select Mechanism window

7.) Click PERMANENT. A Select Device (Figure 3-14) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 3-14 Select Device window

8.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 3-14 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

9.) Click OK. Available options appear on the Software Options screen (Figure 3-15).

Example: Options screen with Permanent & FlexTrial options.

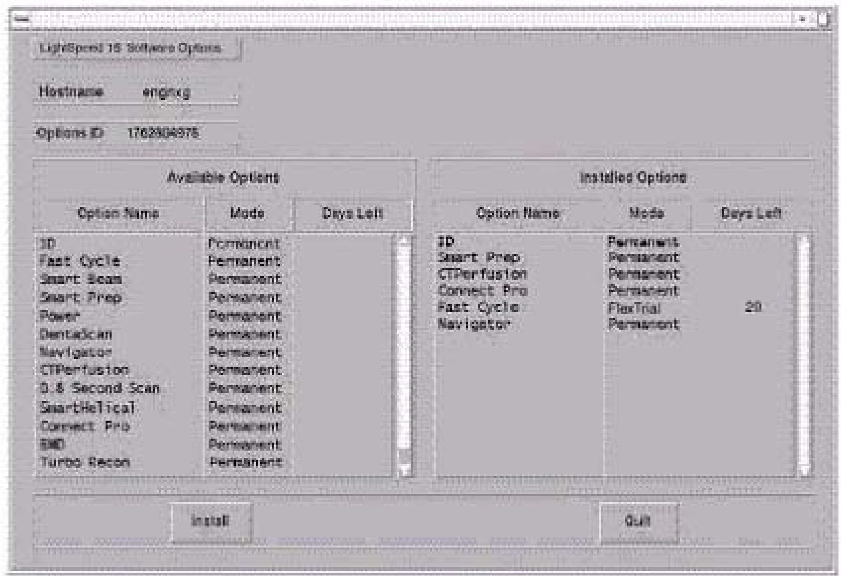


Figure 3-15 Software Options screen

10.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

11.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su** **SPACE** **-** **ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **setdate** ***mmddhhmmyyyy*** (e.g.: **setdate 050409002003**) and press **ENTER**.

Note: Alternate method exists

You may also type: setdate ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **setdate** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **exit** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

1.) To restart the system:

a.) Select OK as needed to answer messages. Note any Recon Failures.

b.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 3-16 Shutdown Icon

c.) Select: RESTART then OK to shutdown and Restart the system.

2.) If HIPAA Present SET TO ON --- Admin Screen

When the system comes up the following screen appears if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

a.) Select user: **`admin`**

b.) Password: **`ctAdmin`**

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.

3.) Successfully perform an Axial scan.

Load Process is now complete!

3.16 GRE Console Information Sheet

3.16.1 LightSpeed 3.X (4/8 MDAS) & HiSpeed QX/i GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4 MDAS8 (For HiSpeed QX/i, select MDAS4.)
- ❑ Tube Type (15): MX_200MCT
- ❑ PDU Type: CT_COMPACT (For HiSpeed QX/i, select NG PDU.)
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: H2 GANTRY (For HiSpeed QX/i, select HISPEED QX/I.)
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: CT## WLSNorm 4008 3.57.255.##

Chapter 4
LFC Procedures – LightSpeed 4.X (LS16 M4)

Section 1.0
Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 4.X 16 Slice MDAS GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0
Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 4.X (LS16)**:

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **5111371 – LightSpeed 4.x Applications Software**
  Version 405L.8_H4.2M4
- **2394896 – LightSpeed / HiSpeed QXI DARC Applications M4**
  Version – H4.2/H3.3/HSQXI M4
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.
    a.) Open a shell.
    b.) Type: **`cat /usr/g/config/INFO`** **ENTER**
    c.) Record screen information in Section 3.16. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 4.x (LS16) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: **`cat /usr/g/config/INFO`** **ENTER**
- ❑ Install the Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – YES to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, darc will reboot then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: **`st`** **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **#bigguy**
- ctuser: **4$apps**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Console Upgrade or a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.). CSA = Common Software Applications.

Console Upgrade: Follow the instruction below for the following console upgrade:

- GOC2/Linux Console (M3 or M3.1 software) to GOC3/GRE (M4 software)

Software Upgrade: Follow the instruction below for the following software upgrades:

- GOC3/GRE M3 or GOC3/GRE M3.1 to GOC3/GRE M4

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing the above Console or software upgrades. This is a one time procedure necessary prior to loading M4 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M4 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

COMMON INSTRUCTIONS

Steps 1–4 below pertain to ALL of the above:

1.) Open up a Unix Shell.

2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.

3.) Type from the following to save CSA Protocols:

a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER

b.) **`mountDVD`** ENTER

`Mounting DVD.`

c.) **`\cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER

d.) **`unmountDVD`** ENTER

`Unmounting DVD.`

4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

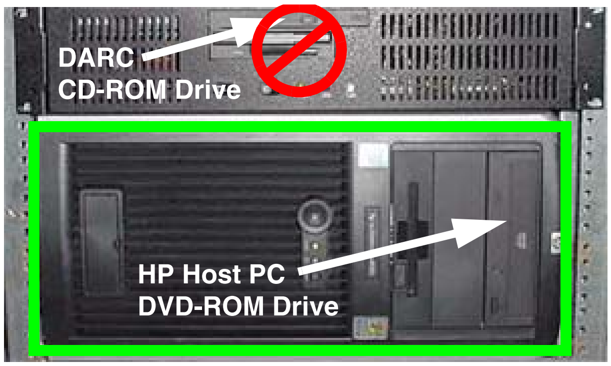


Figure 4-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 4-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed 4.x Applications Software CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the **LightSpeed 4.x Applications Software** CDROM into the DVD-ROM drive on the HP Host PC.

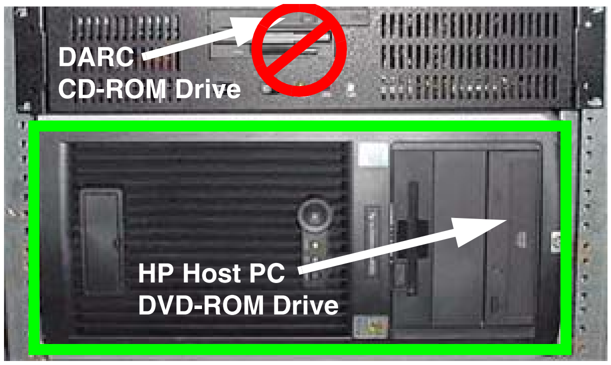


Figure 4-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

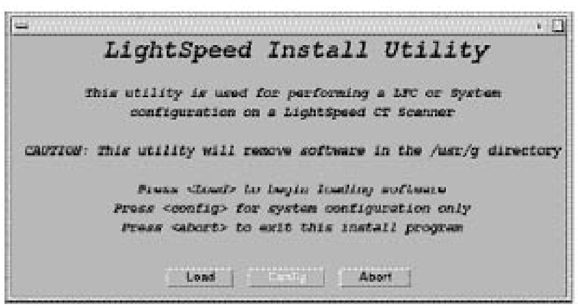


Figure 4-4 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

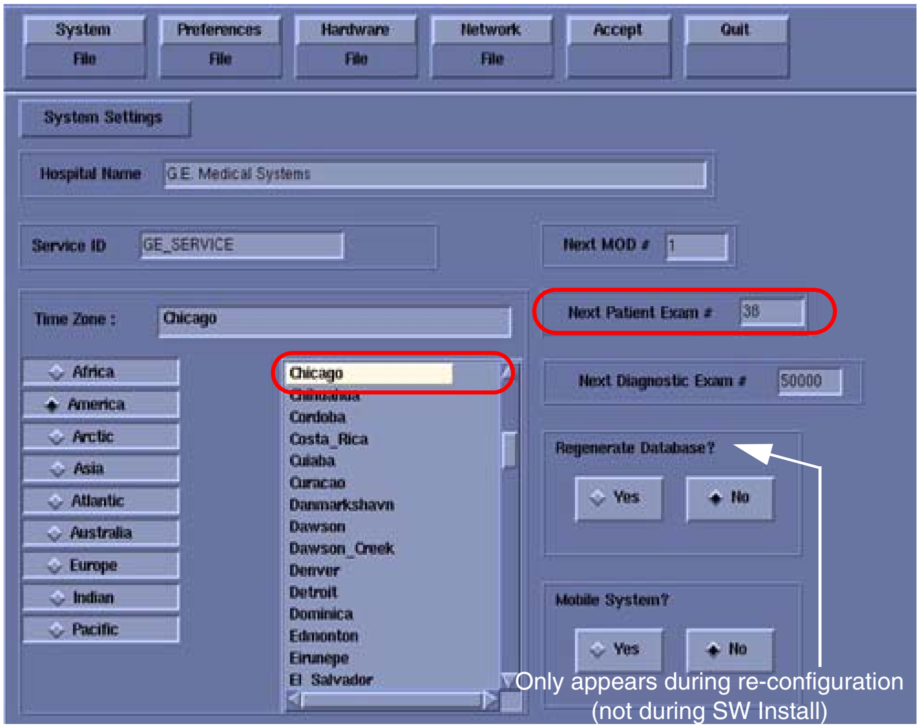


Figure 4-5 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

8.) Verify that the proper **Time Zone** is selected (example: Chicago).

9.) Click the PREFERENCES button.

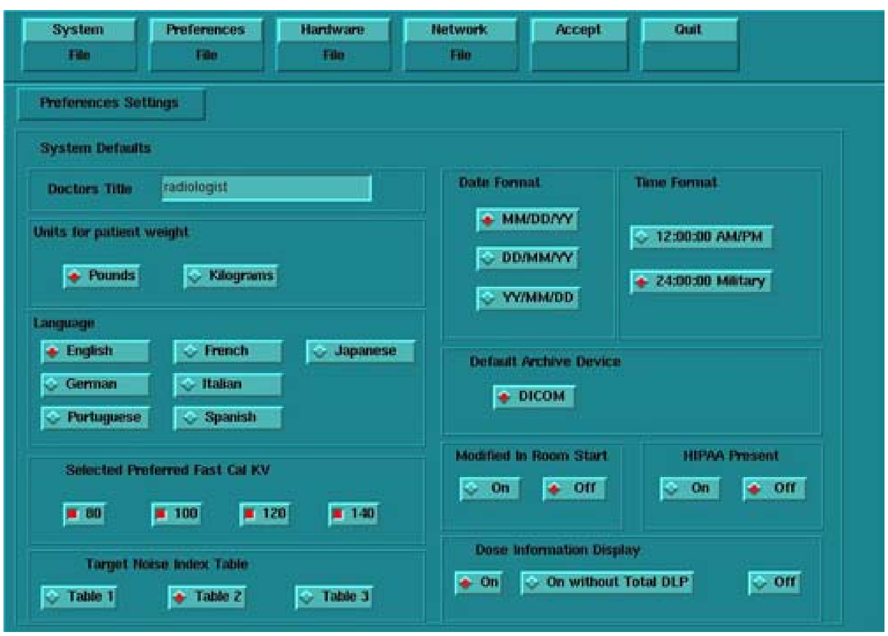


Figure 4-6 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

Figure 4-7 Hardware Settings screen

21.) Select the proper DAS Type: MDAS_H16_16.

22.) Select the proper Tube Type: MX_200MCT.

23.) Select the proper PDU Type: CT_COMPACT.

24.) Select the proper Collimator/Filter Type: HELIOS G1.

25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Select the proper GANTRY TYPE: H2 GANTRY.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active,

the software install will fail.

28.) Click the NETWORK button (site will be different).

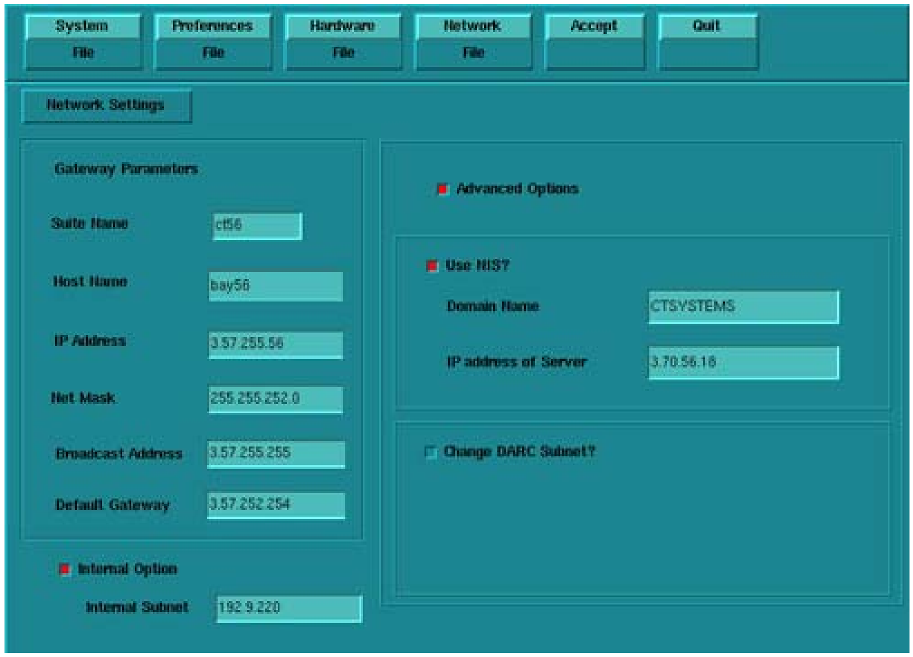


Figure 4-8 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

33.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

34.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

35.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

36.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

37.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

38.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

39.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

40.) Click the **ACCEPT** button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed 4.x Applications Software CDROM from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

44.) Press the button on the DVD ROM drive to remove the LightSpeed 4.x Applications Software CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat space /usr/g/config/INFO`** **ENTER**

Verify DAS type (MDAS_H16_16), Tube type (15 for MX_200MCT), PDU type, Gantry type (h2), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darcOS`** **ENTER**

An Attention window appears. See Figure 4-9.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 4-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

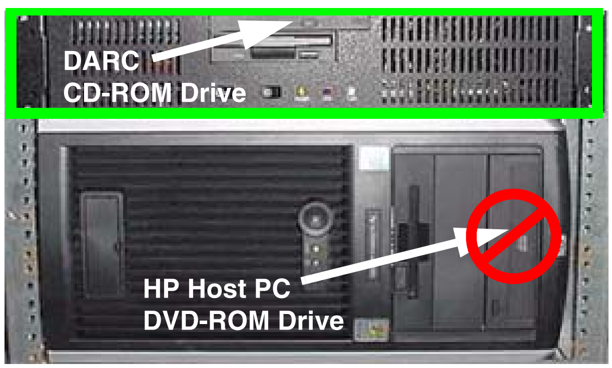


Figure 4-10 DARC CD-ROM Drive Location

5.) Click on **OK**.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

```
The OS load on the DARC Node is started.  This takes about 9-11 minutes.
##.## % done.
```

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

```
The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…
```

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch. Manually eject the CD after approximately 2 minutes. See Figure D-8 for location of DARC Reset Switch.

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 4-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

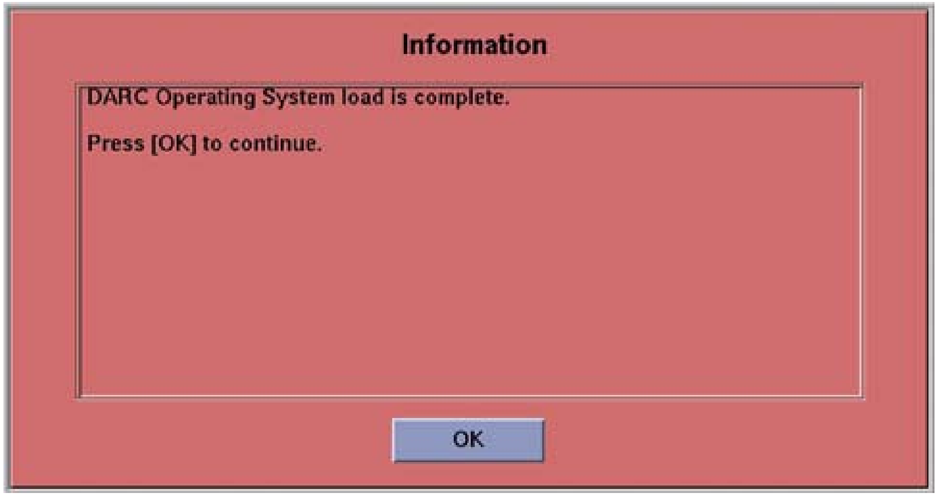


Figure 4-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully loaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **LightSpeed / HiSpeed QXI DARC Applications M4** CDROM into the DVD-ROM drive on the HP Host Computer.

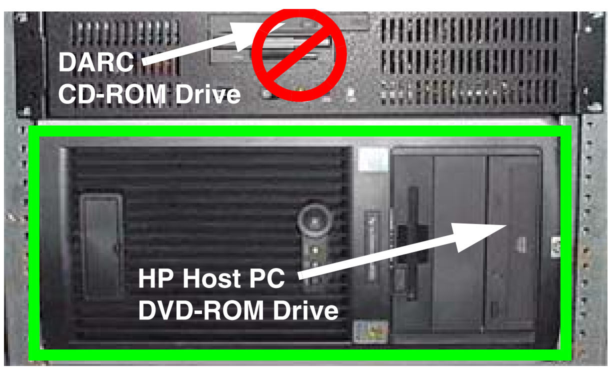


Figure 4-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** SPACE `-` ENTER

`password:` **`#bigguy`** ENTER

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **rsh** SPACE **darc** ENTER

[root@localhost root] (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

[root@localhost root] **exit** ENTER

4.) If login was successful, continue with DARC Apps load:

root: **cd** SPACE **/** ENTER

root: **mount** SPACE **/mnt/cdrom** ENTER

root: **cd** SPACE **/mnt/cdrom** ENTER

root: **./copy_rpms** ENTER

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

root: **cd** SPACE **/** ENTER

root: **umount** SPACE **/mnt/cdrom** ENTER

root: **eject** ENTER

6.) Remove the LightSpeed / HiSpeed QXI DARC Applications M4 CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

root: **cd** SPACE **/usr/g/scripts** ENTER

root: **./start_darc_load** ENTER

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **su** SPACE **-** ENTER

3.) Enter the root password: **#bigguy** ENTER

4.) Type: **sync** ENTER

5.) Type: **sync** ENTER

6.) Type: **reboot** ENTER
A message appears: `The system is going down for reboot NOW!`.

7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).

2.) Type: **st** ENTER or **startup** ENTER.

3.) A pink pop-up Attention Message appears: `OC initializing. Please wait`....

Read the Note(s) below.

4.) Click OK as needed to answer message prompts.

5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.

2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

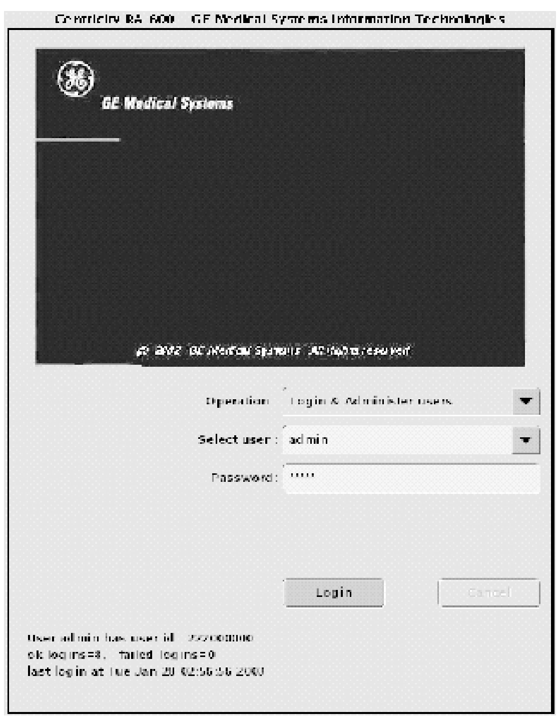


Figure 4-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

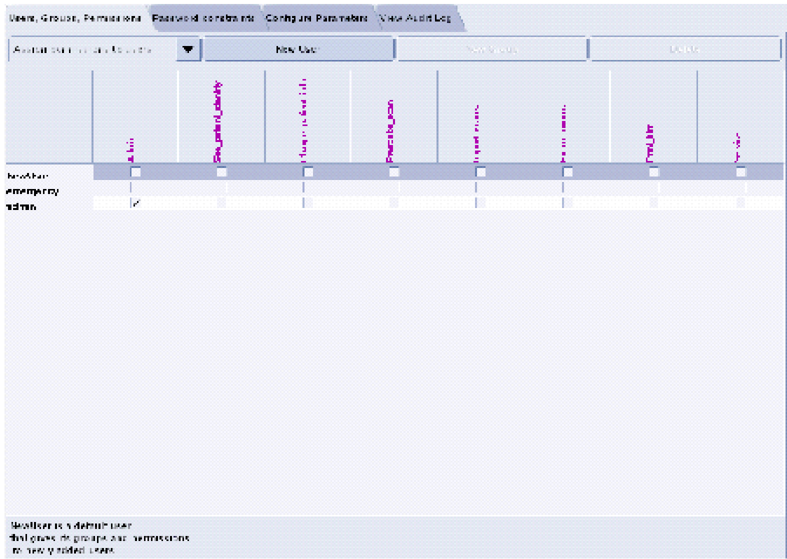


Figure 4-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 4-15).

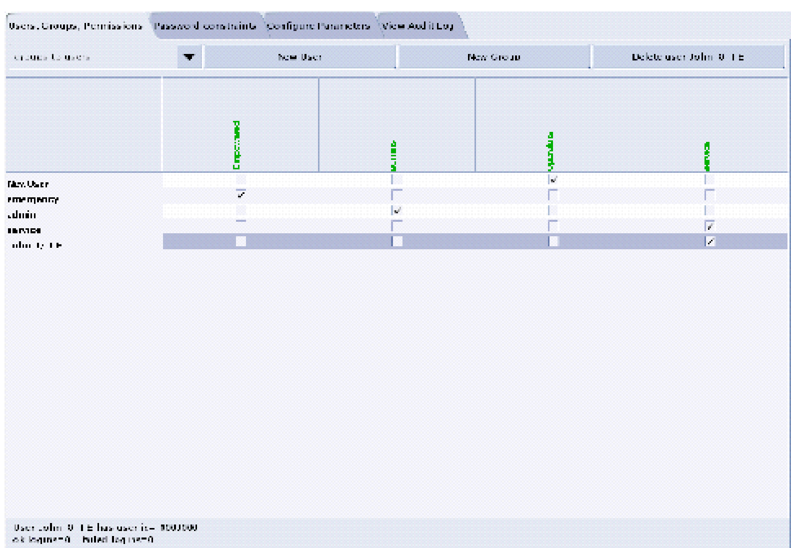


Figure 4-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 4-16 Login/Admin screen

3.8 Restoring MOD System State - Site Upgrading From IRIX To Linux Only

For sites that ONLY have System State Information on a MOD (not a DVD).

1.) When applications are up, open a Unix Shell from the Service Desktop:

a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.

b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.

c.) Execute the following commands as *`ctuser`*.

`cd /usr/g/bin` ENTER

`sysstategui -MOD -IRX2LNX` ENTER

d.) When the system state GUI comes up, select ALL and then select RESTORE.

e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.

2.) In the same unix shell opened in step 1, execute the following commands as ***ctuser***:

a.) **`mountMOD -I2L`** ENTER

b.) **`mv /usr/g/service/.bb  /usr/g/service/bb.lfc`** ENTER

c.) **`mkdir /usr/g/service/.bb`** ENTER

d.) **`rm -rf /tmp/*bb`** ENTER

e.) **`cp /MOD/service_mod_data/system_state/*bb  /usr/g/service/.bb/`** ENTER

f.) **`/usr/g/bin/SwapTubeData`** ENTER

g.) **`/bin/cp -f  /tmp/*bb /usr/g/service/.bb/`** ENTER

3.) Close the Unix Shell.

4.) Remove the MOD from the drive.

5.) Now you will re-save system state to DVD.

6.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.

7.) Select: SERVICE DESKTOP.

8.) Select: PM.

9.) Select: SYSTEM STATE.

10.) Click ALL to select all the cals, characterizations, etc.

11.) Click SAVE.

12.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`

13.) When completed select DISMISS.

14.) Close the window in the upper left corner of the screen.

15.) Continue with Section 3.10.

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.

8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.11 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

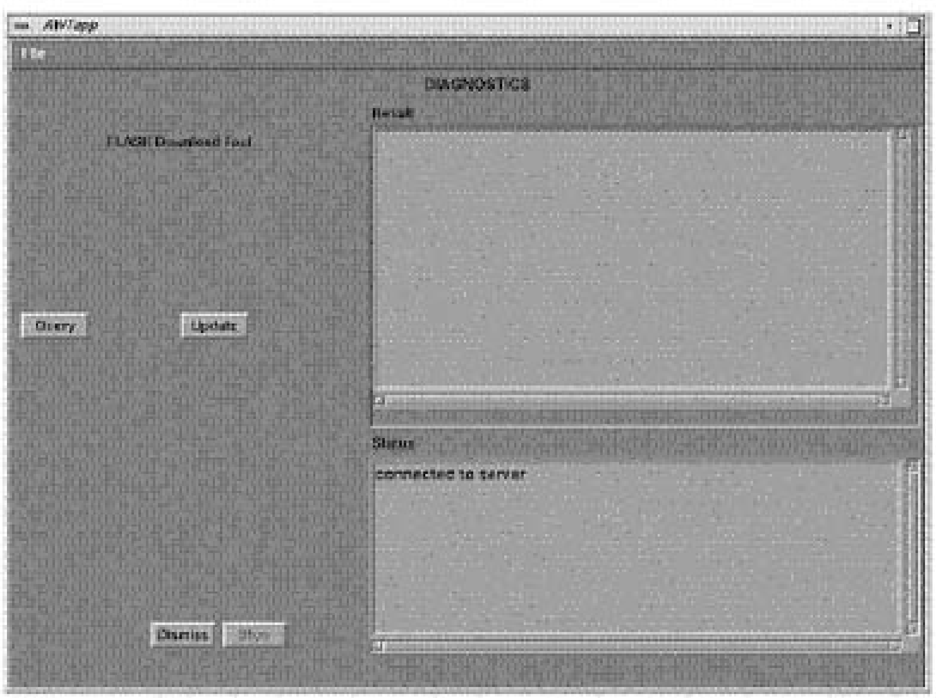


Figure 4-17 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping obcr` ENTER

`ping etc` ENTER

`ping stc` ENTER

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL.
5.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 4-18) appears.

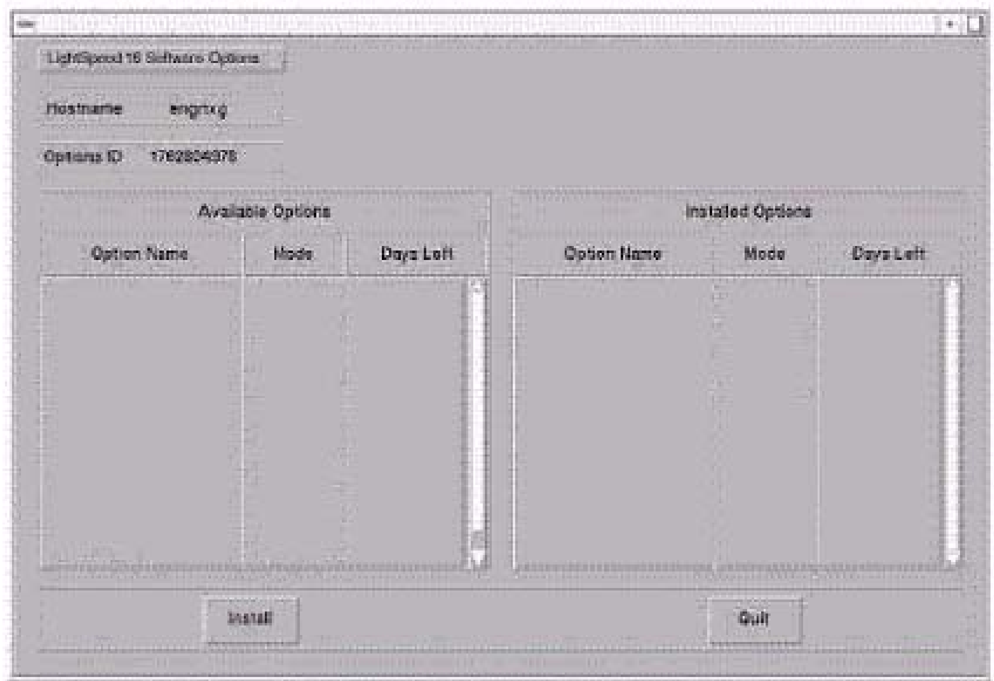


Figure 4-18 Install Options screen

6.) Click INSTALL. A Select Mechanism (Figure 4-19) window opens. Select the mechanism through which you want to install Option Keys.

Figure 4-19 Select Mechanism window

7.) Click PERMANENT. A Select Device (Figure 4-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 4-20 Select Device window

8.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 4-20 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

9.) Click OK. Available options appear on the Software Options screen (Figure 4-21).

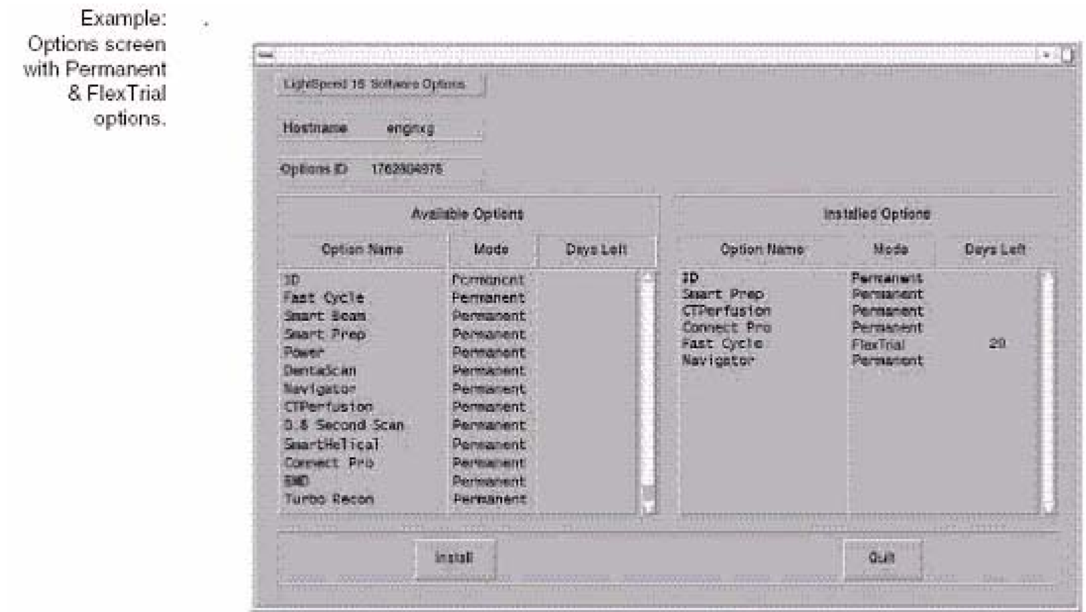


Figure 4-21 Software Options screen

10.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

11.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su** **SPACE** **-** **ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press ENTER.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** **ENTER**

2.) Select OK as needed to answer messages. Note any Recon Failures.

3.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 4-22 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.

3.) Successfully perform an Axial scan.

Load Process is now complete!

3.16 GRE Console Information Sheet

3.16.1 LightSpeed 4.X (LS 16) GRE M3.1 Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS_H16_16
- ❑ Tube Type: MX_200MCT
- ❑ PDU Type: CT_COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: H2 GANTRY
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: CT## WLSNorm 4008 3.57.255.##

Chapter 5
LFC Procedures – LightSpeed 4.X (LS16 M3.1)

Section 1.0
Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 4.X 16 Slice MDAS GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0
Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 4.X (LS16 M3.1)**:

1.) **2393921 – LightSpeed / HiSpeed / Xtream Console Operating System**
Version – GEMS Linux 1.7.9
Supports: H2/H3/H4/H5 M3
2.) **2393919 – LightSpeed 16 Applications Software**
Software – Version 404L.2_H4.2GREM3.1
3.) **2393920 – Lightspeed / HiSpeed QX/I DARC Applications**
Version – H5/H4/H3/H2 M3

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM does not have a CD inserted until instructed in the

procedure.

4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: `cat /usr/g/config/INFO` **ENTER**
   c.) Record the information on the screen in Section 3.16. For an example screen output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 4.x (LS16) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: `cat /usr/g/config/INFO` **ENTER**
- ❑ Install the OS on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – YES to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, darc will reboot then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: `st` **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: `#bigguy`
- ctuser: `4$apps`

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Console Upgrade or a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.). CSA = Common Software Applications.

Console Upgrade: Follow the instruction below for the following console upgrade:

- Linux/Liberty GOC2 Console (M3 software) to GRE (M3.1 software)

Software Upgrade: Follow the instruction below for the following software upgrades:

- GRE M3 to GRE M3.1

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing the above Console or software upgrades. This is a one time procedure necessary prior to loading M3.1 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M3.1 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

COMMON INSTRUCTIONS

Steps 1–4 below pertain to ALL of the above:

1.) Open up a Unix Shell.
2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.
3.) Type from the following to save CSA Protocols:
    a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER

b.) **`mountDVD`** ENTER

`Mounting DVD.`

c.) **`\cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER

d.) **`unmountDVD`** ENTER

`Unmounting DVD.`

4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM OS Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.
2.) Remove the GRE Operator Console's front cover.
3.) Locate the HP PC's DVD-ROM drive.
4.) Insert the LightSpeed Console Operating System CD-ROM into the DVD ROM drive on the HP Host Computer.
5.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 5-1 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

6.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

7.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

8.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.
9.) The HP Host Computer reboots. **Do not insert the LightSpeed Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the LightSpeed Applications SW CDROM into the DVD-ROM drive on the HP Host PC.

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

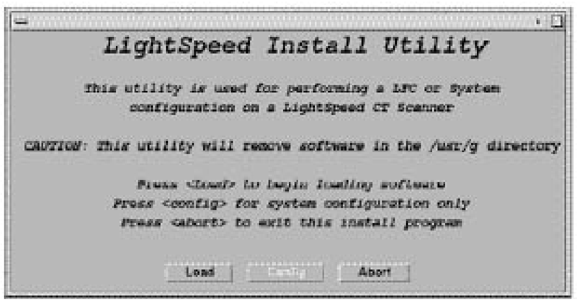


Figure 5-2 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

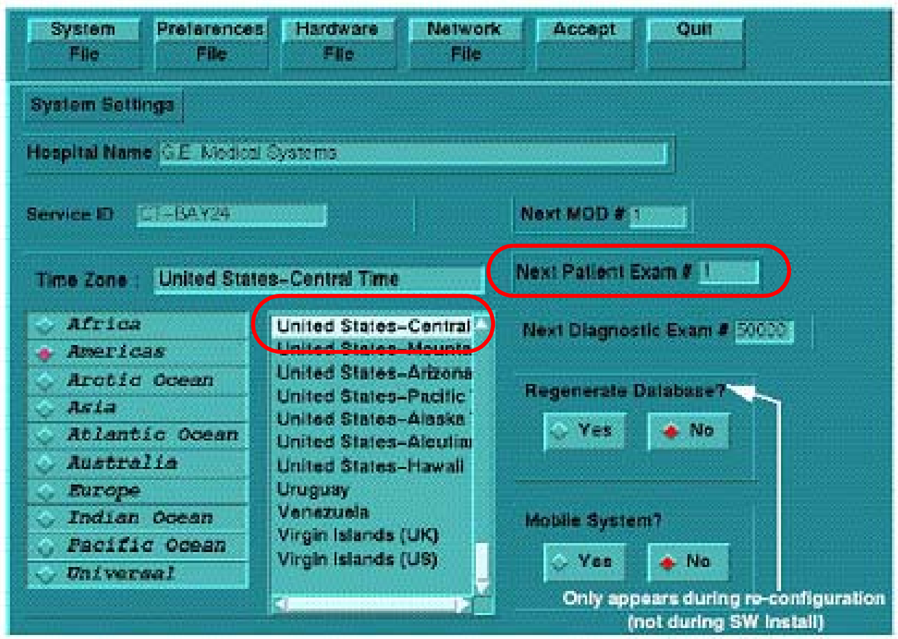


Figure 5-3 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is **1**.

8.) Verify that the proper **Time Zone** is selected (example: Americas and United States–Central and Chicago).

9.) Click the PREFERENCES button.

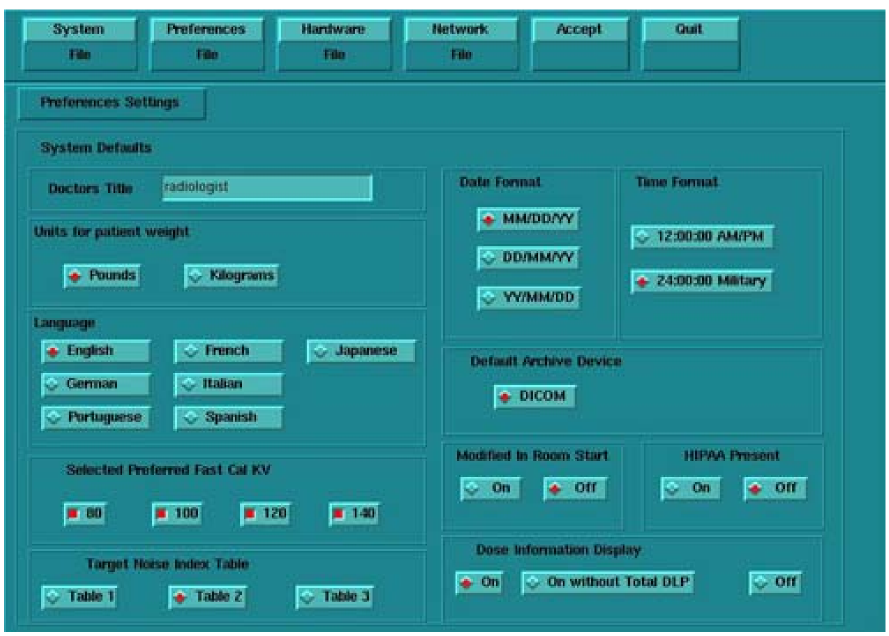


Figure 5-4 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

Figure 5-5 Hardware Settings screen

21.) Select the proper DAS Type: MDAS_H16_16.

22.) Select the proper Tube Type: MX_200MCT.

23.) Select the proper PDU Type: CT_COMPACT.

24.) Select the proper Collimator/Filter Type: HELIOS G1.

25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Select the proper GANTRY TYPE: H2 GANTRY.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

28.) Click the NETWORK button (site will be different).

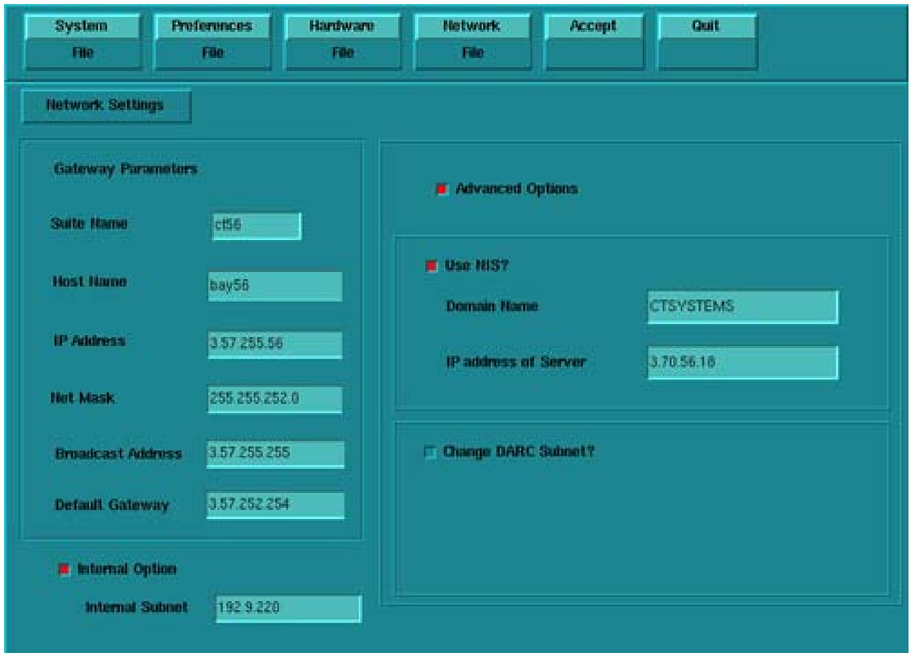


Figure 5-6 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

33.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

34.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

35.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

36.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

37.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

38.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

39.) Click the **ACCEPT** button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed Apps CDROM from the HP Host Computer until instructed to do so.

40.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

41.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

42.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

43.) Press the button on the DVD ROM drive to remove the LightSpeed Apps SW CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node.

- If any problems arise, verify the correct CD is installed.
- If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat`** **SPACE** **`/usr/g/config/INFO`** **ENTER**

Verify DAS type (MDAS_H16_16), Tube type (15 for MX_200MCT), PDU type, Gantry type (h2), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

`root:` **`cd`** **SPACE** **`/usr/g/scripts`**

`root:` **`./start_darcOS`** **ENTER**

A sh window appears with the message below:

`Please insert the Operating System CD into the DARC Node disk drive.`

`Press <Enter> to start OS load.`

4.) Insert the LightSpeed Console Operating System CD-ROM into the DARC Node drive.

5.) Select the **ENTER** key.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

`The OS load on the DARC Node is started.  This takes about 9-11 minutes.`

`##.## % done.`

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. You may be instructed to reboot; open a Unix shell and type "halt", then power cycle the console.

6.) A `%done` status is displayed during the DARC OS load.

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

`The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…`

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully.

Note: If the OS disk CD does not eject or the DARC does not boot/reboot correctly, then press the DARC Node front panel Reset switch. Manually eject the CD after approximately 2 minutes.

9.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

10.) This completes the OS load on the DARC.

11.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

The DARC Applications Software CD-ROM will be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **`rsh`** **SPACE** **`darc`** **ENTER**

root@localhost login: **`exit`** **ENTER**

4.) If login was successful, continue with DARC Apps load:

`root:` **`cd`** **SPACE** **`/`** **ENTER**

`root:` **`mount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`cd`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`./copy_rpms`** **ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type:

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darc_load`** **ENTER**

A shell window pops up and the load begins. It takes 4–5 minutes.

Continue the procedure when the load window goes away.

6.) When the load displays a message that it is completed or window goes away, type the following:

`root:` **`cd`** **SPACE** **`/`** **ENTER**

`root:` **`umount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`eject`** **ENTER**

7.) Close the shell (do not use for the next process).

8.) Remove the DARC Applications SW CDROM from the HP Host Computer DVD-ROM drive.

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **`su`** **SPACE** **`-`** **ENTER**

3.) Enter the root password: **`#bigguy`** **ENTER**

4.) Type: **`sync`** **ENTER**

5.) Type: **`sync`** **ENTER**

6.) Type: `reboot` ENTER
A message appears: `The system is going down for reboot NOW!.`

7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: `st` ENTER or `startup` ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

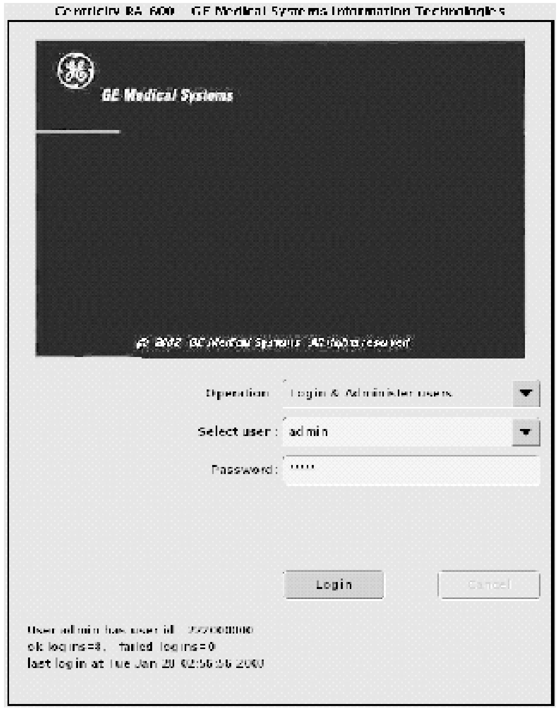


Figure 5-7 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

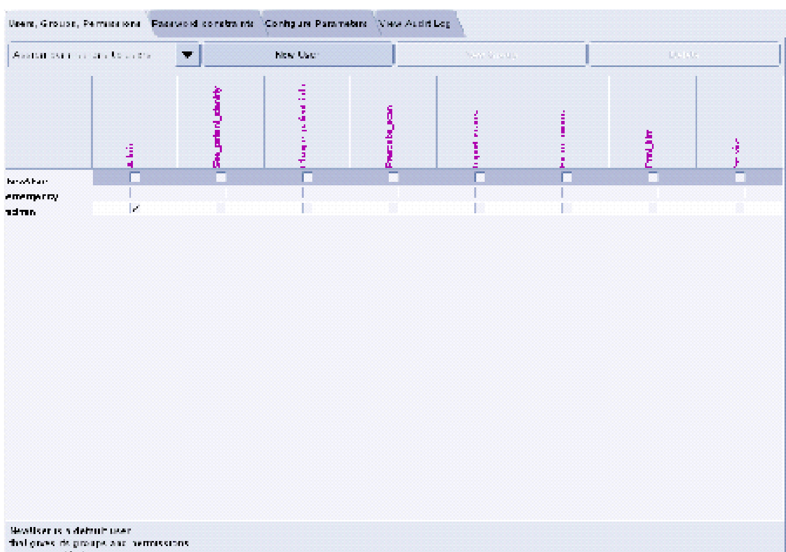


Figure 5-8 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 5-9).

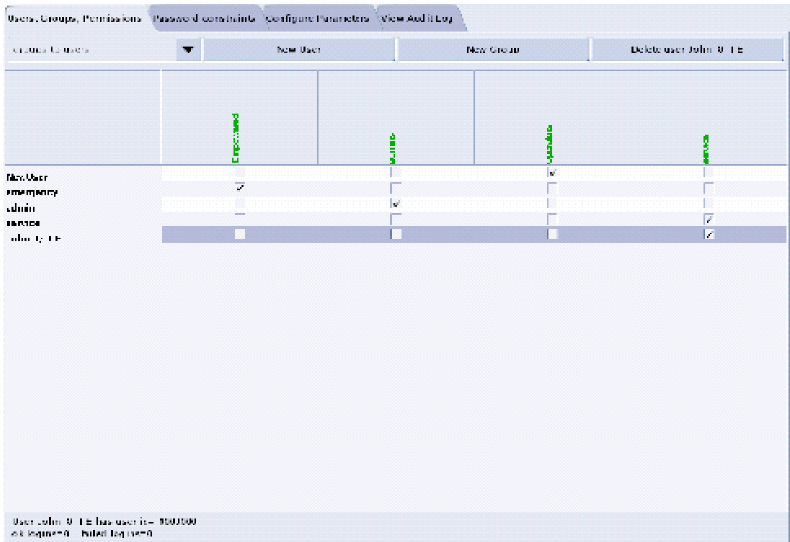


Figure 5-9 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

```
Select user: admin
Password: ctAdmin
```

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 5-10 Login/Admin screen

3.8 Restoring MOD System State - Site Upgrading From IRIX To Linux Only

For sites that ONLY have System State Information on a MOD (not a DVD).

1.) When applications are up, open a Unix Shell:
   a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.
   b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.
   c.) Execute the following commands as *`ctuser`*.
   `cd /usr/g/bin` ENTER
   `sysstategui -MOD -IRX2LNX` ENTER
   d.) When the system state GUI comes up, select ALL and then select RESTORE.
   e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.
2.) In the same unix shell opened in step 1, execute the following commands as ***`ctuser`***:
   a.) **`mountMOD -I2L`** ENTER
   b.) `mv /usr/g/service/.bb  /usr/g/service/bb.lfc` ENTER
   c.) `mkdir /usr/g/service/.bb` ENTER
   d.) `rm -rf /tmp/*bb` ENTER
   e.) **`cp /MOD/service_mod_data/system_state/*bb  /usr/g/service/.bb/`** ENTER
   f.) **`/usr/g/bin/SwapTubeData`** ENTER
   g.) **`/bin/cp -f  /tmp/*bb /usr/g/service/.bb/`** ENTER
3.) Close the Unix Shell.
4.) Remove the MOD from the drive.
5.) Now you will re-save system state to DVD.
6.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.

7.) Select: SERVICE DESKTOP.
8.) Select: PM.
9.) Select: SYSTEM STATE.
10.) Click ALL to select all the cals, characterizations, etc.
11.) Click SAVE.
12.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
13.) When completed select DISMISS.
14.) Close the window in the upper left corner of the screen.
15.) Continue with Section 3.10.

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.
2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click the ALL selection which highlights the cals, characterization, etc.
6.) Make certain to click RESTORE.
7.) Verify the overall system state content has restored correctly.
8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.11 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

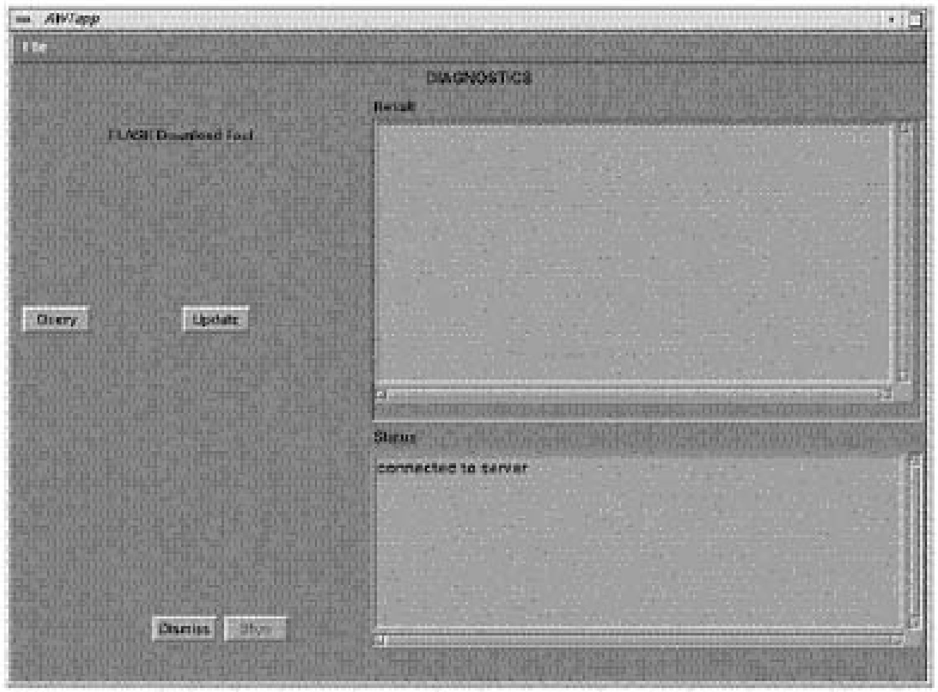


Figure 5-11 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping obcr` ENTER

`ping etc` ENTER

`ping stc` ENTER

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL.

5.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 5-12) appears.

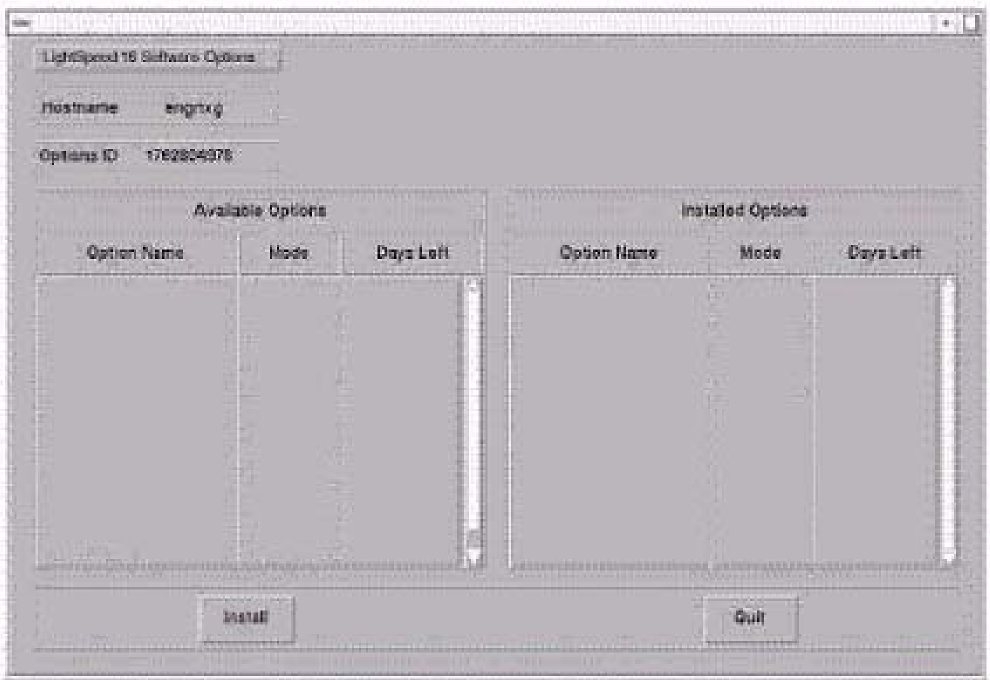


Figure 5-12 Install Options screen

6.) Click INSTALL. A Select Mechanism (Figure 5-13) window opens. Select the mechanism through which you want to install Option Keys.

Figure 5-13 Select Mechanism window

7.) Click PERMANENT. A Select Device (Figure 5-14) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 5-14 Select Device window

8.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 5-14 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

9.) Click OK. Available options appear on the Software Options screen (Figure 5-15).

Example: Options screen with Permanent & FlexTrial options.

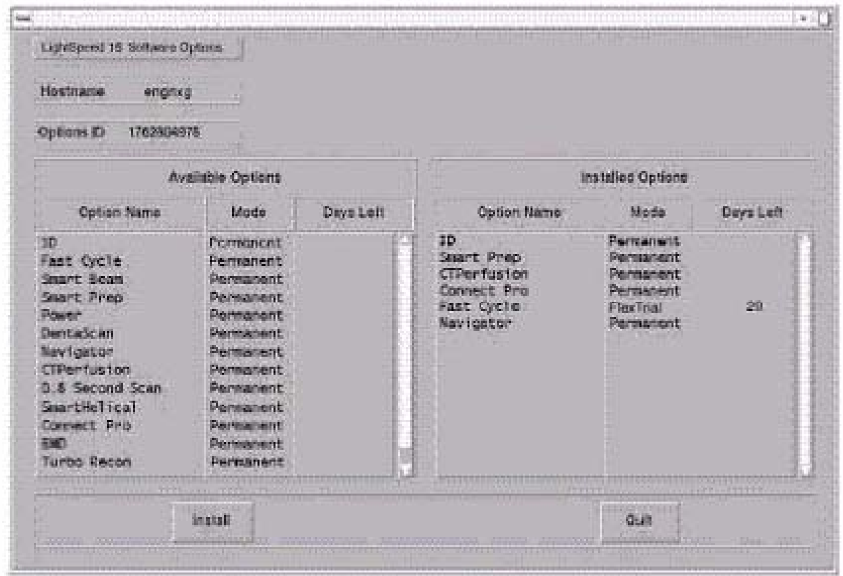


Figure 5-15 Software Options screen

10.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

11.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su SPACE - ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **setdate** ***mmddhhmmyyyy*** (e.g.: **setdate 050409002003**) and press **ENTER**.

Note: Alternate method exists

You may also type: setdate ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **setdate** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **exit ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

1.) To restart the system:

a.) Select OK as needed to answer messages. Note any Recon Failures.

b.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 5-16 Shutdown Icon

c.) Select: RESTART then OK to shutdown and Restart the system.

2.) If HIPAA Present SET TO ON --- Admin Screen

When the system comes up the following screen appears if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

a.) Select user: **`admin`**

b.) Password: **`ctAdmin`**

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.

3.) Successfully perform an Axial scan.

Load Process is now complete!

3.16 GRE Console Information Sheet

3.16.1 LightSpeed 4.X (LS 16) GRE M3.1 Console Information Sheet

- ❑ Site ID #: _____________ Console Serial Number _________________
- ❑ DAS Type: MDAS_H16_16
- ❑ Tube Type: MX_200MCT
- ❑ PDU Type: CT_COMPACT
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: H2 GANTRY
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ____________________
- ❑ Host Name: ____________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: ________________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: _____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: CT## WLSNorm 4008 3.57.255.##

Chapter 6
LFC Procedures – LS 5.X (LS Pro^{16} (100kW) M3.1)

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 5.X Pro^{16} (100kW) GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

If upgrading from an IRIX/Octane to GRE platform, a system state conversion is also described in Appendix A - "System State Conversion".

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 5.X (Pro^{16} (100kW)**:

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **2401265 – LightSpeed Pro 16 (100) Applications**
  Version 502HP100.2_H5.0M3.1
- **2400462 – LightSpeed Pro 16 DARC Applications**
  Version – H5.0 M3.1 (HP100)
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: **`cat /usr/g/config/INFO`** **ENTER**
   c.) Record screen information in Section 3.15. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 5.X (LS Pro[16] (100kW)) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: **`cat /usr/g/config/INFO`** **ENTER**
- ❑ Install the Console Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – **`Yes`** to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: **`st`** **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **`#bigguy`**
- ctuser: **`4$apps`**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again.`
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.). CSA = Common Software Applications.

Software Upgrade: Follow the instructions below for the following software upgrades:

- HPower ME4 or ME5 to M3.1 (ME7)

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing the above Console or software upgrades. This is a one time procedure necessary prior to loading M3.1 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M3.1 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

1.) Open up a Unix Shell.
2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.
3.) Type the following to save CSA Protocols:
   a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER
   b.) **`mountDVD`** ENTER
   `Mounting DVD.`

c.) **`\cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER

d.) **`unmountDVD`** ENTER

`Unmounting DVD.`

4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

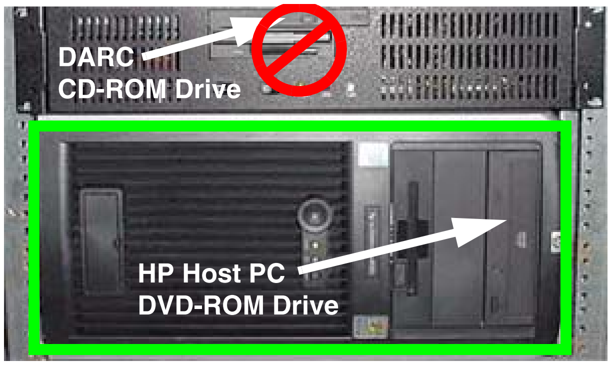


Figure 6-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 6-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed Pro 16 (100) Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the **LightSpeed Pro 16 (100) Applications** CDROM into the DVD-ROM drive on the HP Host PC.

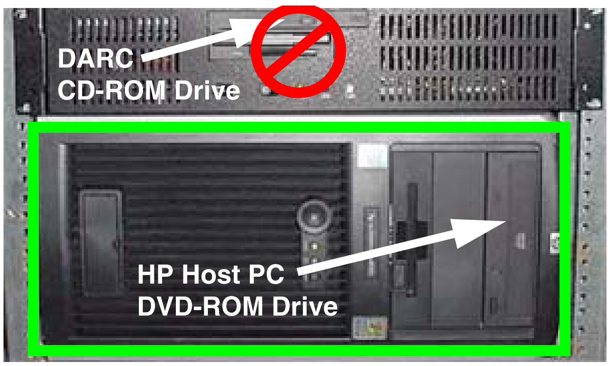


Figure 6-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

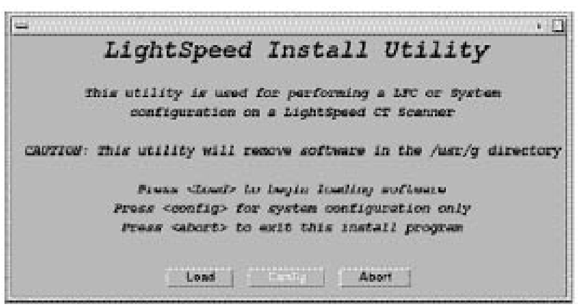


Figure 6-4 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

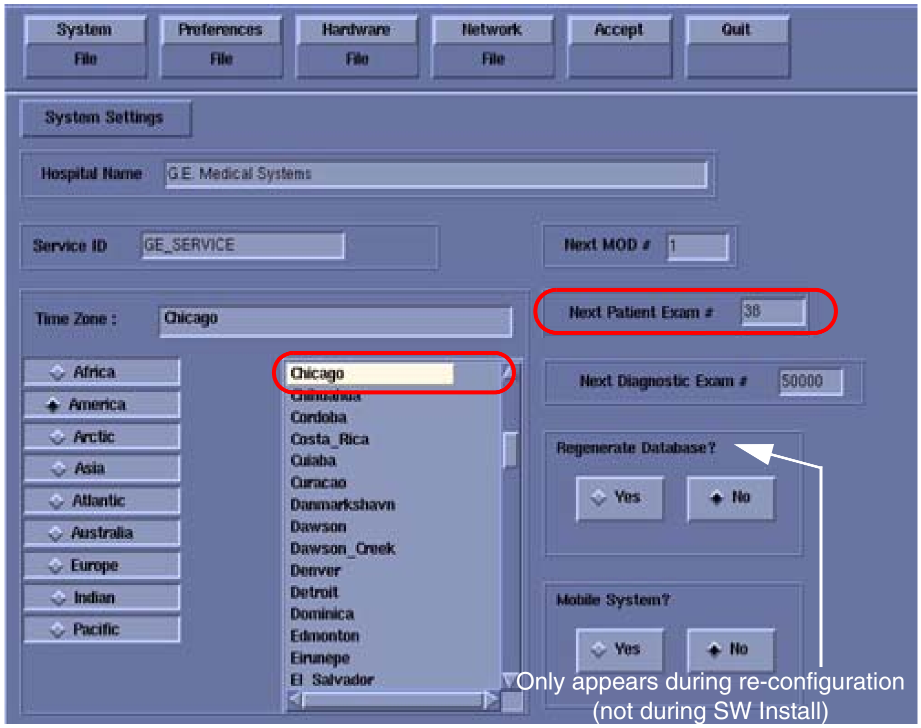


Figure 6-5 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

8.) Verify that the proper **Time Zone** is selected (example: Chicago).

9.) Click the PREFERENCES button.

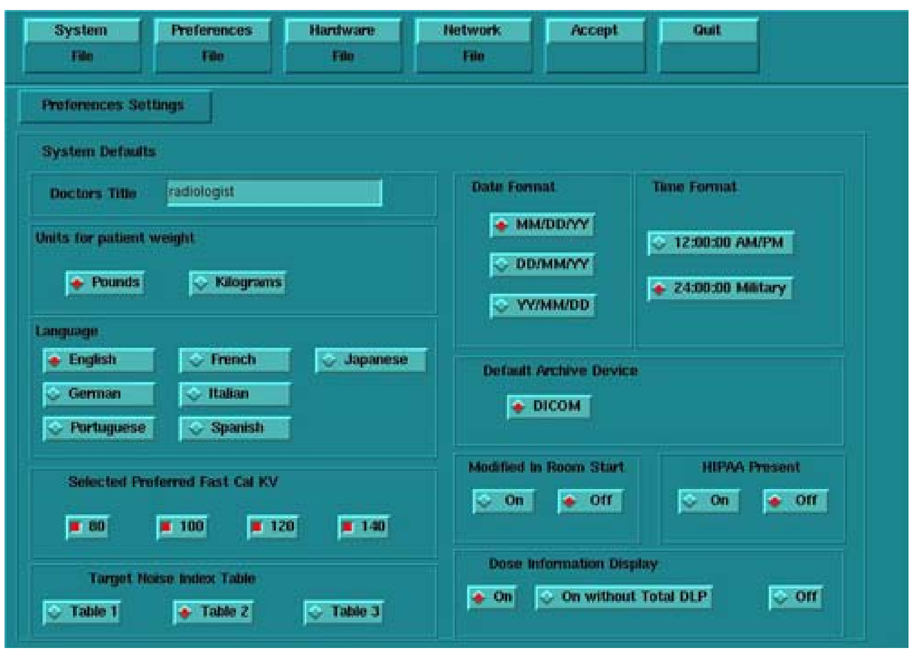


Figure 6-6 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

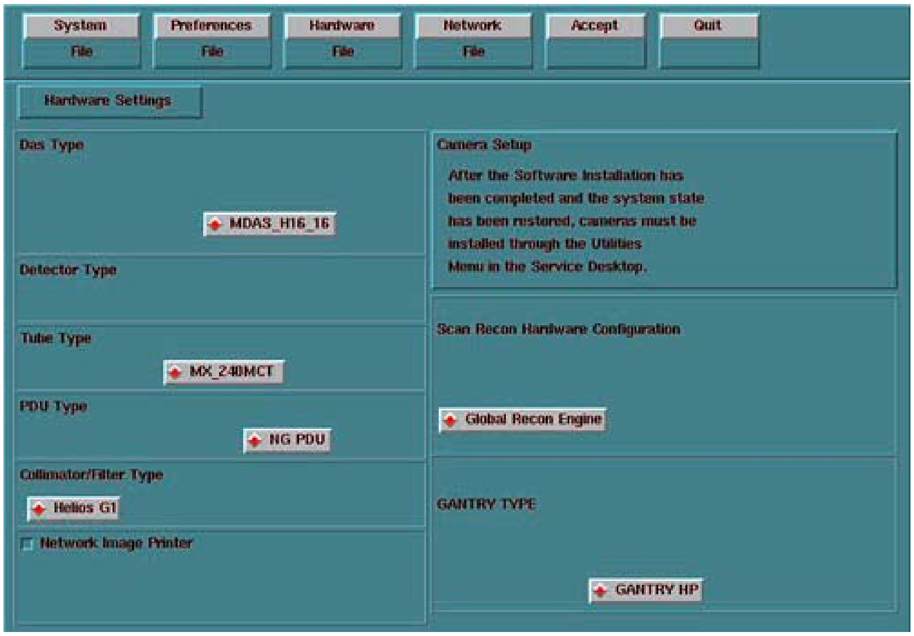


Figure 6-7 Hardware Settings screen

21.) Select the proper DAS Type: MDAS_H16_16.

22.) Select the proper Tube Type: MX_240MCT.

23.) Select the proper PDU Type: NG PDU.

24.) Select the proper Collimator/Filter Type: HELIOS G1.

25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Select the proper GANTRY TYPE: GANTRY HP.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active,

the software install will fail.

28.) Click the NETWORK button (site will be different).

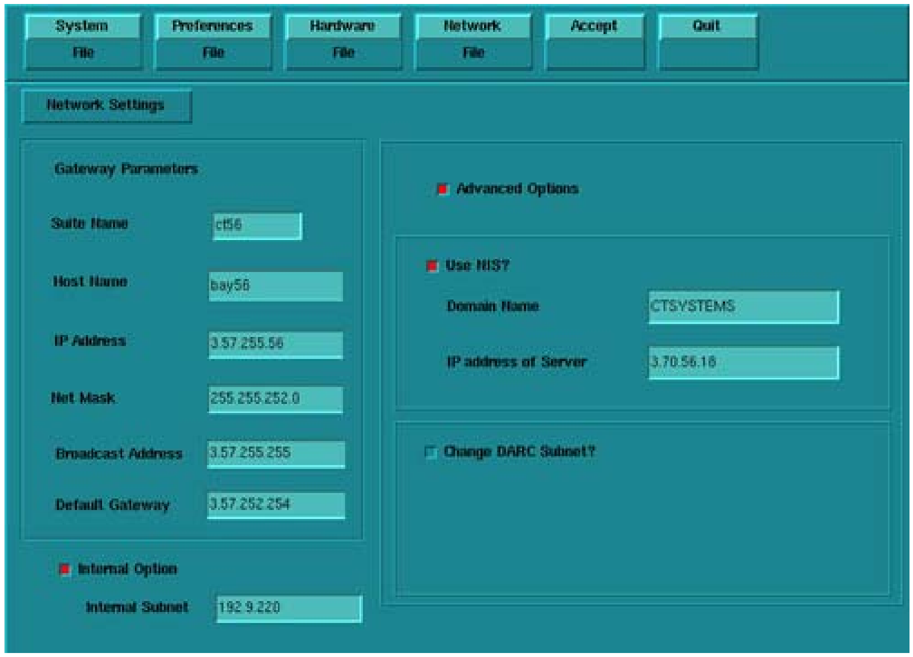


Figure 6-8 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

33.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

34.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

35.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

36.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

37.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

38.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

39.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

40.) Click the ACCEPT button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed Pro 16 (100) Applications CDROM from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

44.) Press the button on the DVD ROM drive to remove the LightSpeed Pro 16 (100) Applications CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat space /usr/g/config/INFO`** ENTER

Verify DAS type (MDAS_H16_16), Tube type (16 for MX_240MCT), PDU type (NG PDU), Gantry type (hp), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** SPACE **`-`** ENTER

`password:` **`#bigguy`** ENTER

`root:` **`cd`** SPACE **`/usr/g/scripts`** ENTER

`root:` **`./start_darcOS`** ENTER

An Attention window appears. See Figure 6-9.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 6-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

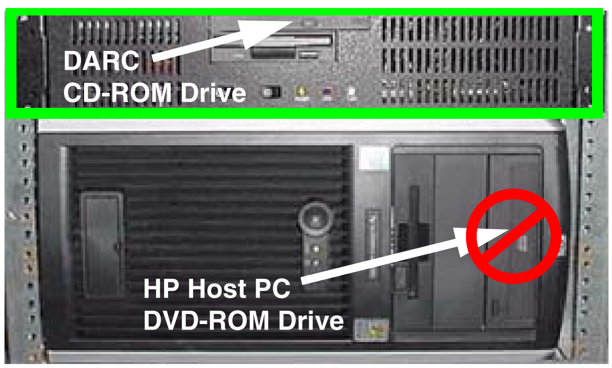


Figure 6-10 DARC CD-ROM Drive Location

5.) Click on **OK**.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

`The OS load on the DARC Node is started. This takes about 9-11 minutes.`

`##.## % done.`

Note: **The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.**

6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

`The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…`

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: **If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch . Manually eject the CD after approximately 2 minutes. See Figure D-8 for location of DARC Reset Switch.**

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 6-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

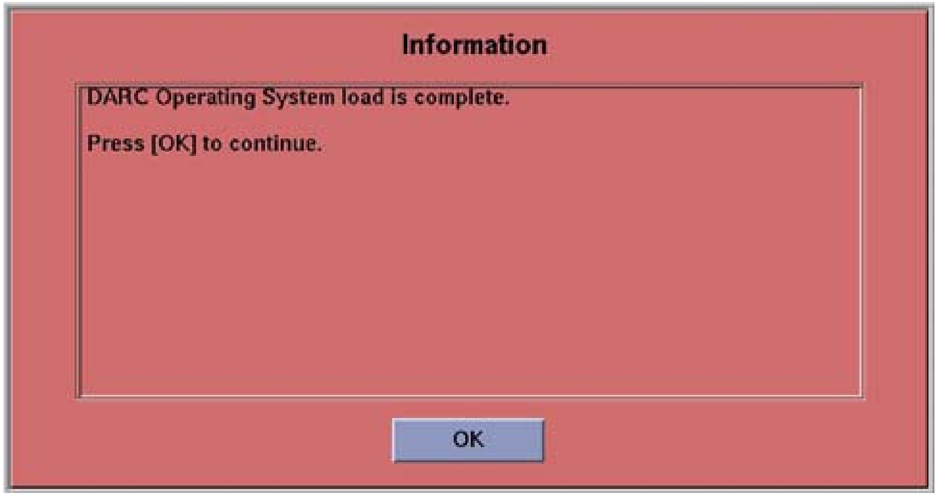


Figure 6-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully ioaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **LightSpeed Pro 16 DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

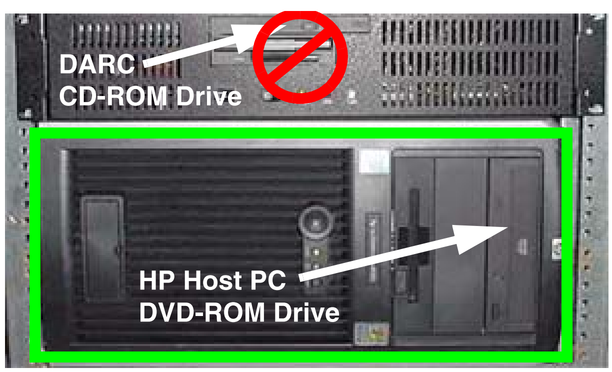


Figure 6-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** `-` **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **rsh SPACE darc ENTER**

[root@localhost root] (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

[root@localhost root] **exit ENTER**

4.) If login was successful, continue with DARC Apps load:

root: **cd SPACE / ENTER**

root: **mount SPACE /mnt/cdrom ENTER**

root: **cd SPACE /mnt/cdrom ENTER**

root: **./copy_rpms ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

root: **cd SPACE / ENTER**

root: **umount SPACE /mnt/cdrom ENTER**

root: **eject ENTER**

6.) Remove the LightSpeed Pro 16 DARC Applications CD-ROM from the HP Host Computer DVD-ROM drive.

7.) Type:

root: **cd SPACE /usr/g/scripts ENTER**

root: **./start_darc_load ENTER**

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.
2.) Type: **su SPACE - ENTER**
3.) Enter the root password: **#bigguy ENTER**
4.) Type: **sync ENTER**
5.) Type: **sync ENTER**
6.) Type: **reboot ENTER**
A message appears: The system is going down for reboot NOW!.
7.) After approximately 4–5 minutes a pink pop up window appears with the message:
CT Software Auto-Start Disabled.
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: **`st`** ENTER or **`startup`** ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

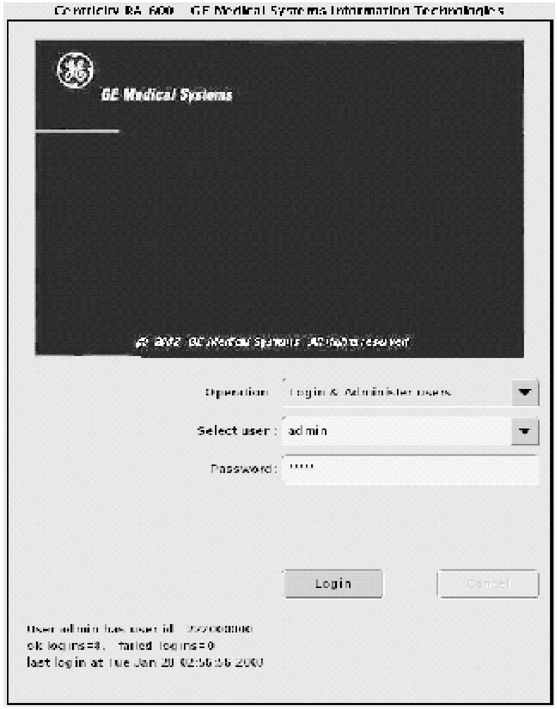


Figure 6-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

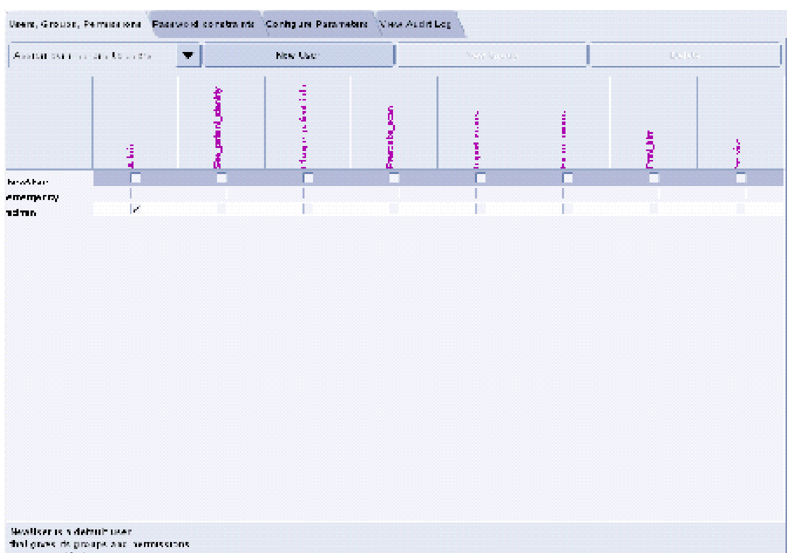


Figure 6-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 6-15).

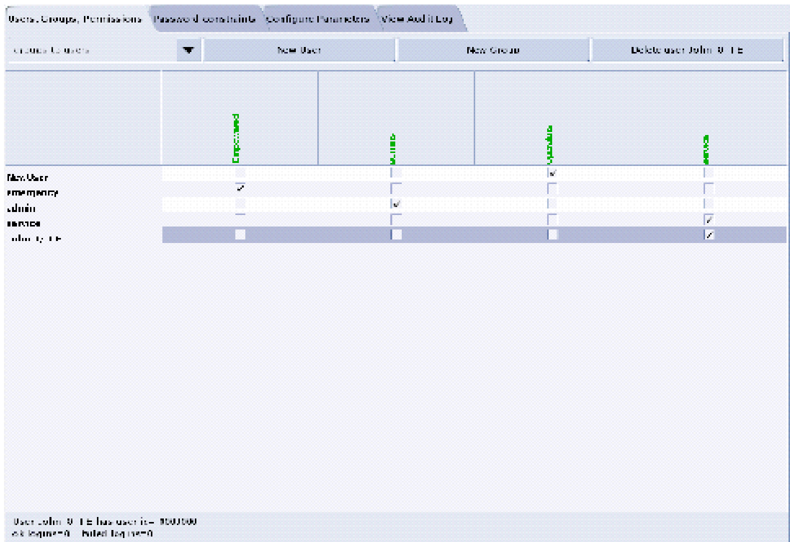


Figure 6-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 6-16 Login/Admin screen

3.8 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.

8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.9 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.

2.) Click the **rat gold** series.

3.) Click SELECT SERIES.

4.) Click on CONFIRM.

5.) Verify the image(s) reconstruct and are displayed.

6.) Click on QUIT.

3.10 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

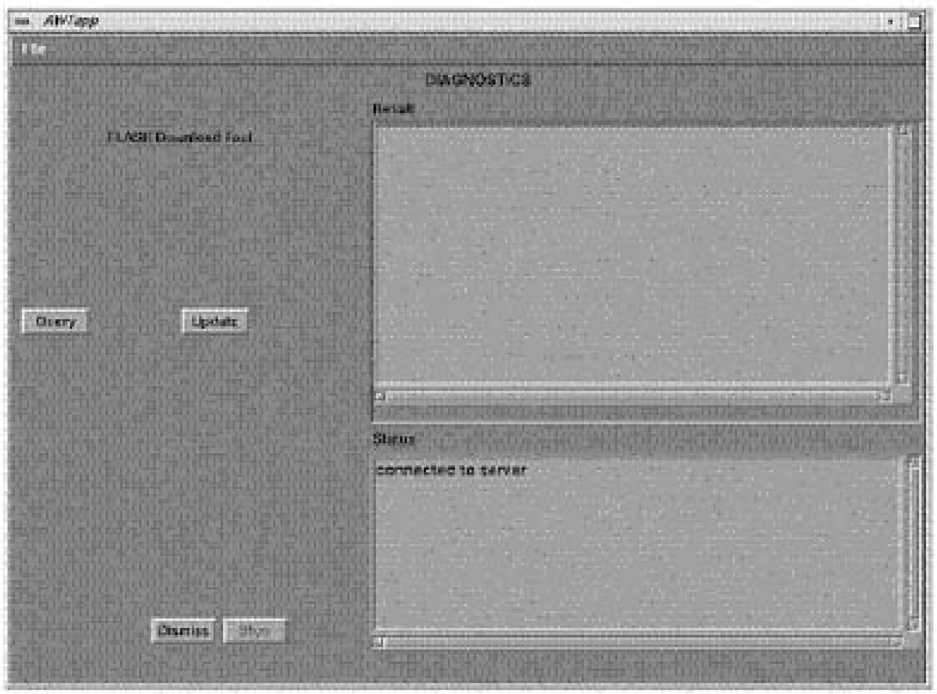


Figure 6-17 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping` SPACE `orp` ENTER

`ping` SPACE `stc` ENTER –OR– `ping` SPACE `tgp` ENTER

3.11 Tube and Generator Firmware Verification

Verify that the Tube and Generator firmware match before continuing with the options install.

1.) Visually inspect the Tube and heat exchanger: A LightSpeed Pro 16(100) with a Performix Pro 100 tube (mx240_mct.mx) will have **black** hoses running from the tube to the heat exchanger.
2.) Verify that the Jedi software installed on the system matches the tube that is mounted on the gantry by typing the following command in a Unix shell:
    cd SPACE /usr/g/ss_fw/jedi ENTER
    ls SPACE -l ENTER
3.) LightSpeed Pro 16(100) will have the following information displayed:

```
total 3732
-r--r--r--  1 ctuser  users  3291698 Nov 18 13:41 appl.mx
-r--r--r--  1 ctuser  users   494264 Nov 18 13:41 boot.mx
lrwxrwxrwx  1 root    sys         12 Nov 18 19:10 database.mx ->
mx240_mct.mx
-r--r--r--  1 ctuser  users    14360 Nov 18 13:41 mcs_7079.mx
-r--r--r--  1 ctuser  users    14360 Nov 18 13:41 mx240_mct.mx
```

Note: The database.mx is a symbolic link to mx240_mct.mx, which is the Performix Pro 100 database.

4.) If your Jedi directory does not match the information listed above, the wrong Host Applications Software may have been loaded. Verify that your software disk set is compatible with the system that you are working on. If the wrong Host Applications Software was installed, you must start the software install procedure over again.

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.

4.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 6-18) appears.

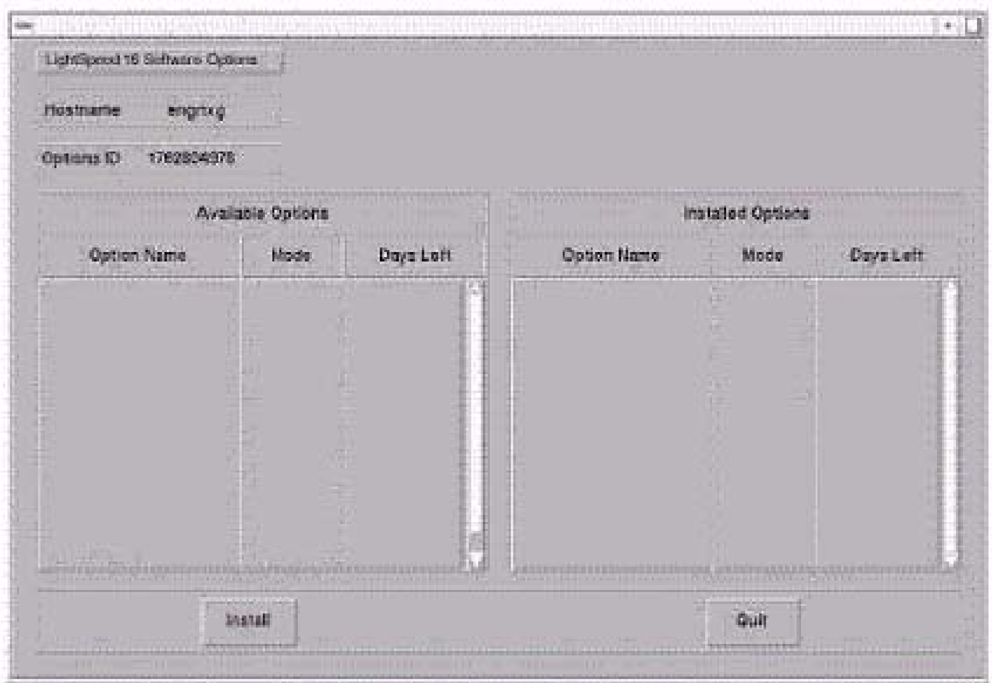


Figure 6-18 Install Options screen

5.) Click INSTALL. A Select Mechanism (Figure 6-19) window opens. Select the mechanism through which you want to install Option Keys.

Figure 6-19 Select Mechanism window

6.) Click PERMANENT. A Select Device (Figure 6-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 6-20 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 6-20 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

8.) Click OK. Available options appear on the Software Options screen (Figure 6-21).

Example:
Options screen
with Permanent
& FlexTrial
options.

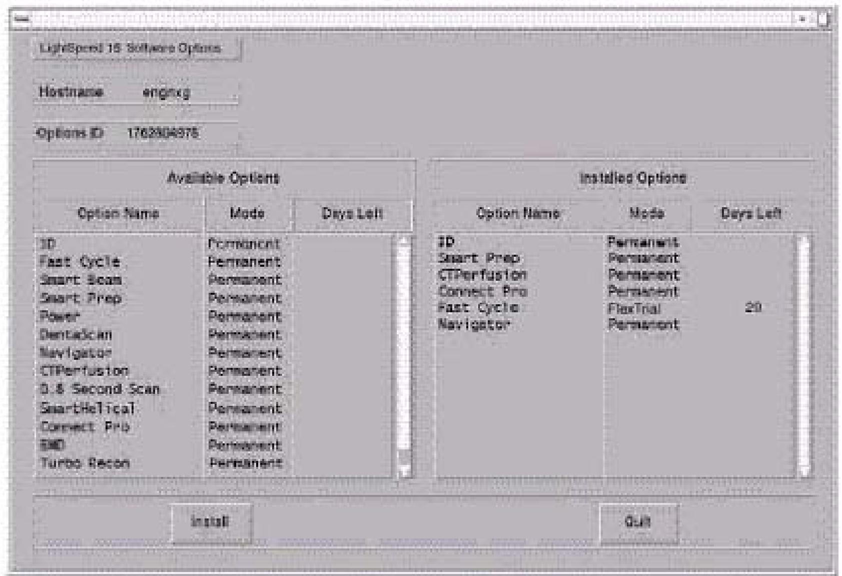


Figure 6-21 Software Options screen

9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

10.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su** **SPACE** **-** **ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **setdate** ***mmddhhmmyyyy*** (e.g.: **setdate 050409002003**) and press **ENTER**.

Note: Alternate method exists

You may also type: setdate ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **setdate** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **exit** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** **ENTER**

2.) Select OK as needed to answer messages. Note any Recon Failures.

3.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 6-22 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 LightSpeed 5.X (LS Pro^{16} (100kW)) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS_H16_16
- ❑ Tube Type: MX_240MCT
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: GANTRY HP
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: ***`CT##   WLSNorm   4008   3.57.255.##`***

Chapter 7
LFC Procedures – LS 5.X (LS Pro^{16} (80kW) M3)

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 5.X Pro^{16} (80kW) GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 5.X (Pro^{16} (80kW)**:

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **2400461 – LightSpeed Pro 16 (80) Applications**
  Version 502HP80.2_H5.0M3
- **2400462 – LightSpeed Pro 16 DARC Applications**
  Version – H5.0 M3 (HP80)
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.
    a.) Open a shell.
    b.) Type: `cat /usr/g/config/INFO` **ENTER**
    c.) Record screen information in Section 3.15. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 5.X (LS Pro16 (80kW)) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: `cat /usr/g/config/INFO` **ENTER**
- ❑ Install the Console Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – **Yes** to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: `st` **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **#bigguy**
- ctuser: **4$apps**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again`.
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

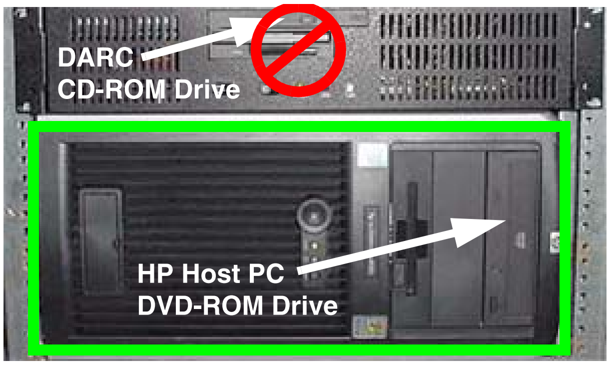


Figure 7-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 7-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** ENTER *(for English Keyboard only)*
or: **`iGEMS2`** ENTER *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed Pro 16 (80) Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the **LightSpeed Pro 16 (80) Applications** CDROM into the DVD-ROM drive on the HP Host PC.

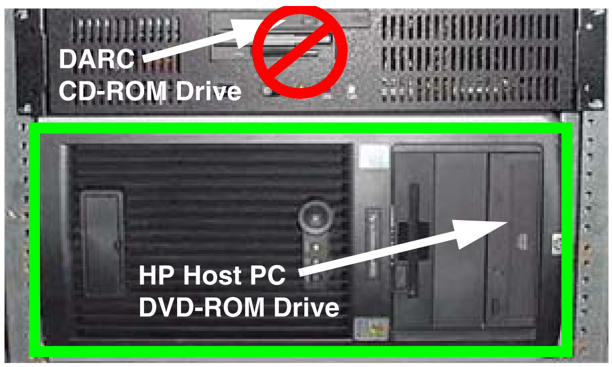


Figure 7-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

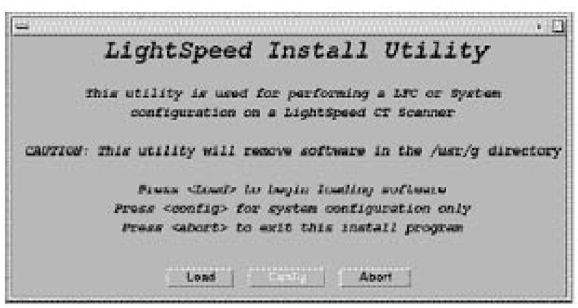


Figure 7-4 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

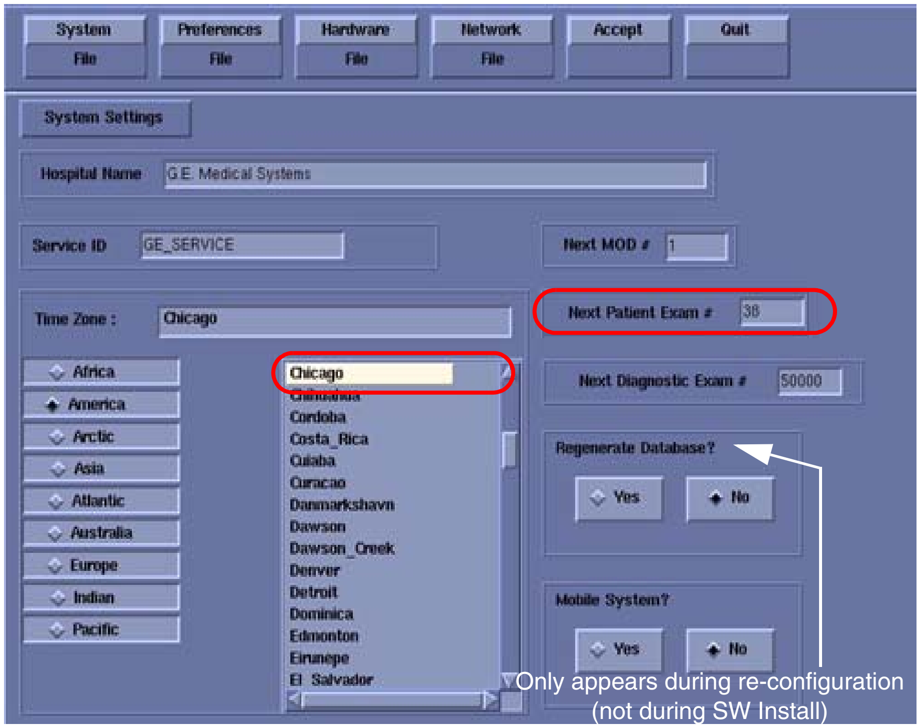


Figure 7-5 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

8.) Verify that the proper **Time Zone** is selected (example: Chicago).

9.) Click the PREFERENCES button.

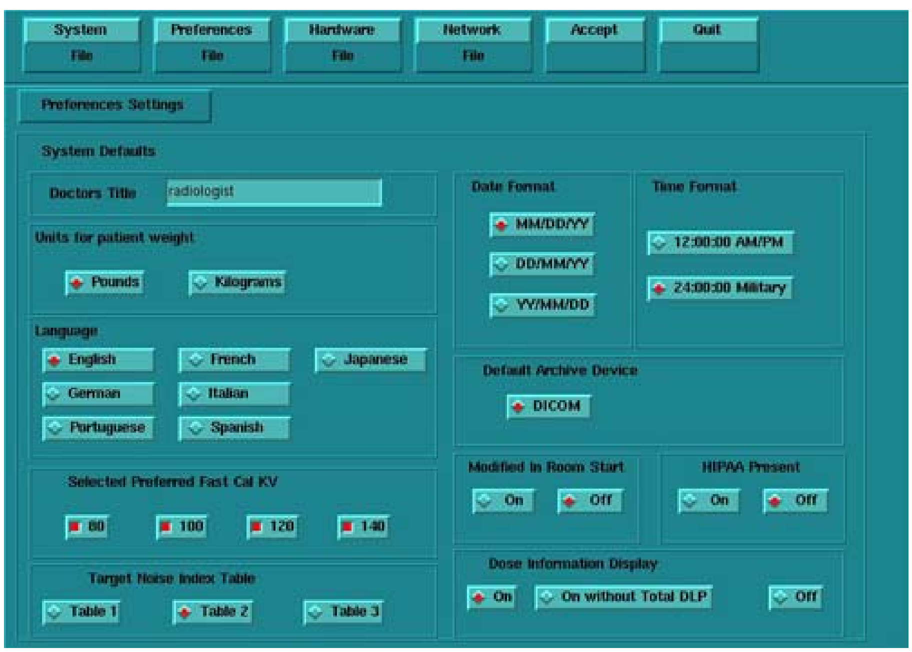


Figure 7-6 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

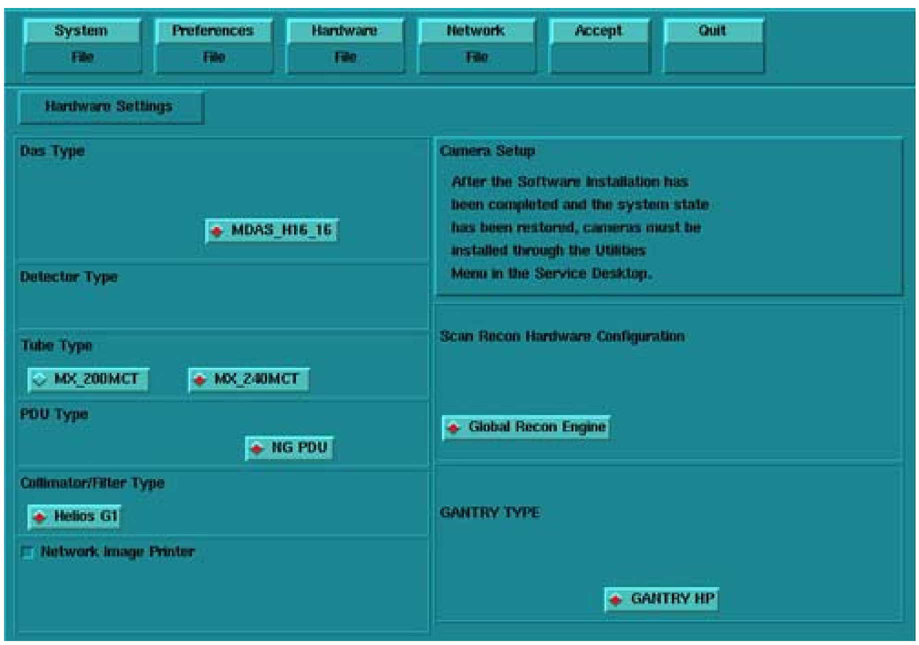


Figure 7-7 Hardware Settings screen

21.) Select the proper DAS Type: MDAS_H16_16.

22.) Select the proper Tube Type: MX_240MCT. (The LightSpeed Pro^{16}-80 Application software limits generator output to 80kW for this tube type.)

23.) Select the proper PDU Type: NG PDU.

24.) Select the proper Collimator/Filter Type: HELIOS G1.

25.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

26.) Select the proper GANTRY TYPE: GANTRY HP.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

28.) Click the NETWORK button (site will be different).

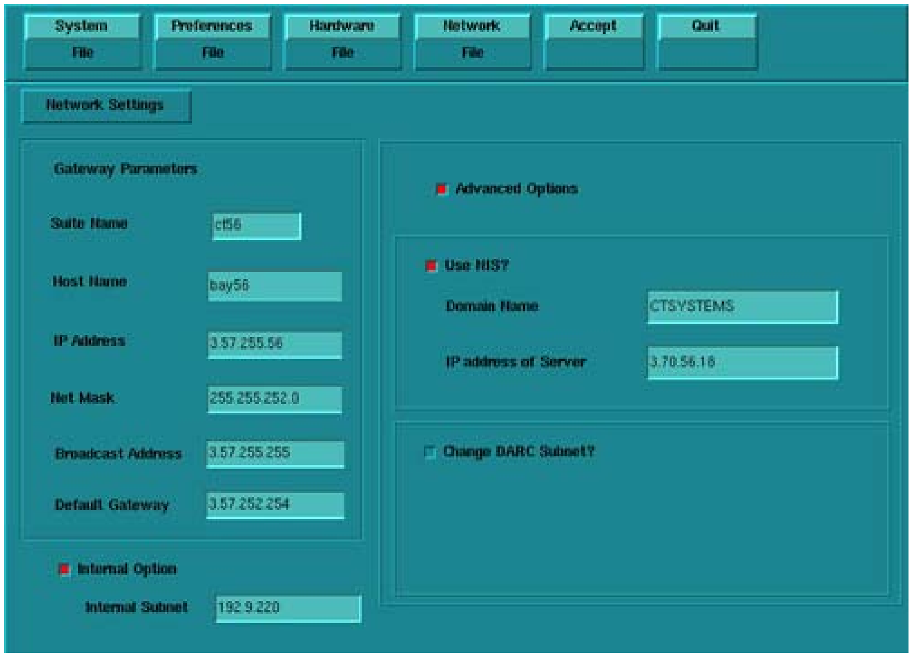


Figure 7-8 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

33.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

34.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

35.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

36.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

37.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

38.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

39.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

40.) Click the **ACCEPT** button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed Pro 16 (80) Applications CDROM from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

44.) Press the button on the DVD ROM drive to remove the LightSpeed Pro 16 (80) Applications CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat space /usr/g/config/INFO`** **ENTER**

Verify DAS type (MDAS_H16_16), Tube type (16 for MX_240MCT), PDU type (NG PDU), Gantry type (hp), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darcOS`** **ENTER**

An Attention window appears. See Figure 7-9.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 7-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

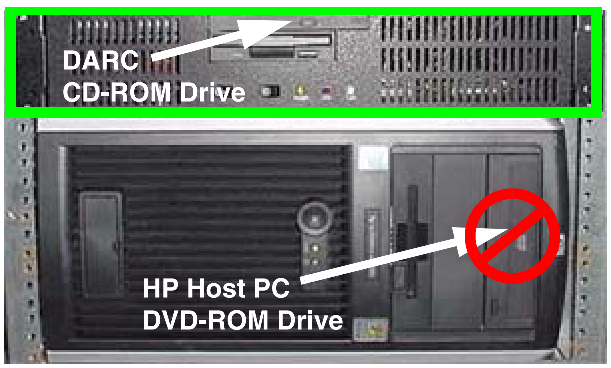


Figure 7-10 DARC CD-ROM Drive Location

5.) Click on **OK**.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

`The OS load on the DARC Node is started. This takes about 9-11 minutes.`

`##.## % done.`

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

`The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…`

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch . Manually eject the CD after approximately 2 minutes. See Figure D-8 for location of DARC Reset Switch.

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 7-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

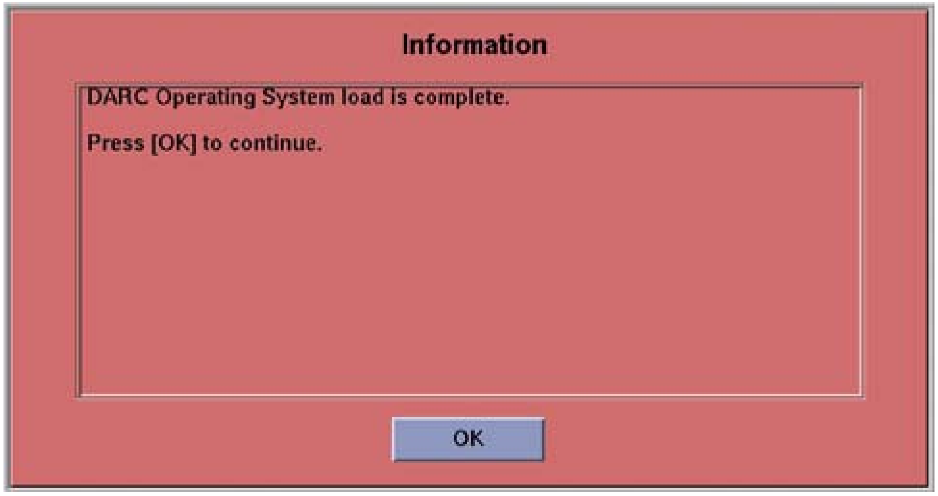


Figure 7-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully loaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **LightSpeed Pro 16 DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

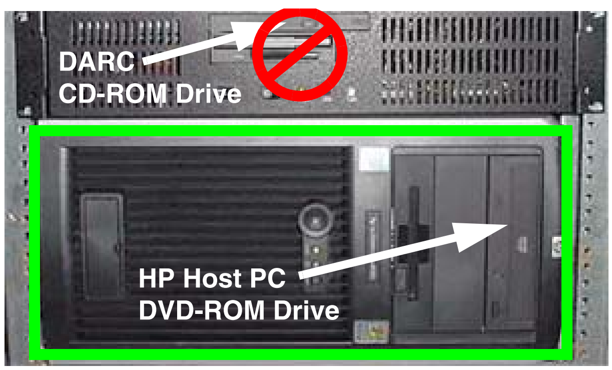


Figure 7-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **su** SPACE **-** ENTER

`password:` **#bigguy** ENTER

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **rsh** SPACE **darc** ENTER

[root@localhost root] (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

[root@localhost root] **exit** ENTER

4.) If login was successful, continue with DARC Apps load:

root: **cd** SPACE **/** ENTER

root: **mount** SPACE **/mnt/cdrom** ENTER

root: **cd** SPACE **/mnt/cdrom** ENTER

root: **./copy_rpms** ENTER

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

root: **cd** SPACE **/** ENTER

root: **umount** SPACE **/mnt/cdrom** ENTER

root: **eject** ENTER

6.) Remove the LightSpeed Pro 16 DARC Applications CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

root: **cd** SPACE **/usr/g/scripts** ENTER

root: **./start_darc_load** ENTER

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.

2.) Type: **su** SPACE **-** ENTER

3.) Enter the root password: **#bigguy** ENTER

4.) Type: **sync** ENTER

5.) Type: **sync** ENTER

6.) Type: **reboot** ENTER
A message appears: `The system is going down for reboot NOW!.`

7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: **`st`** ENTER or **`startup`** ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

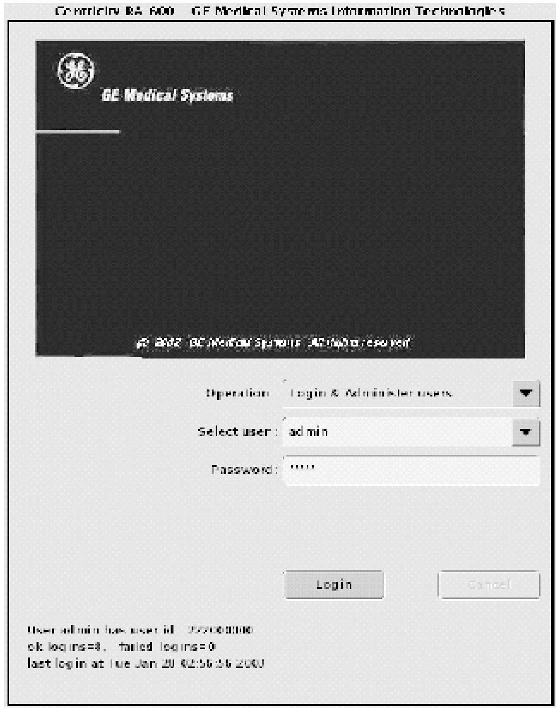


Figure 7-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

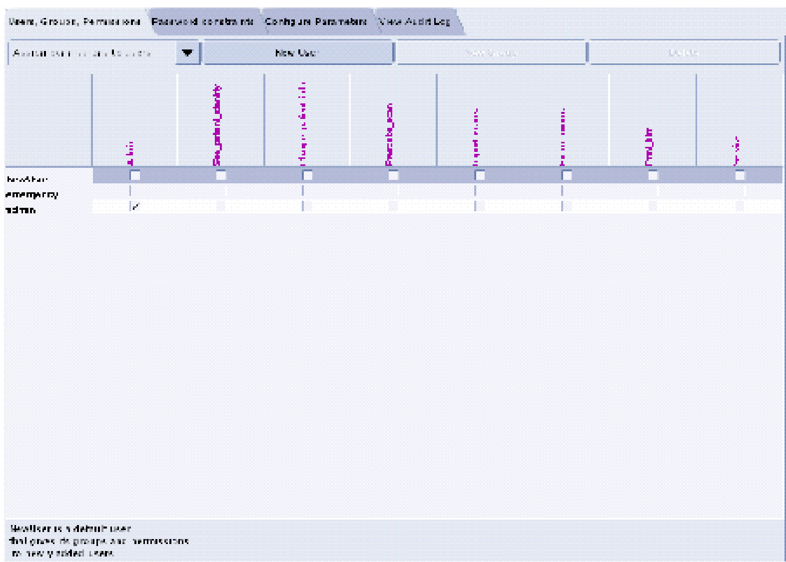


Figure 7-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 7-15).

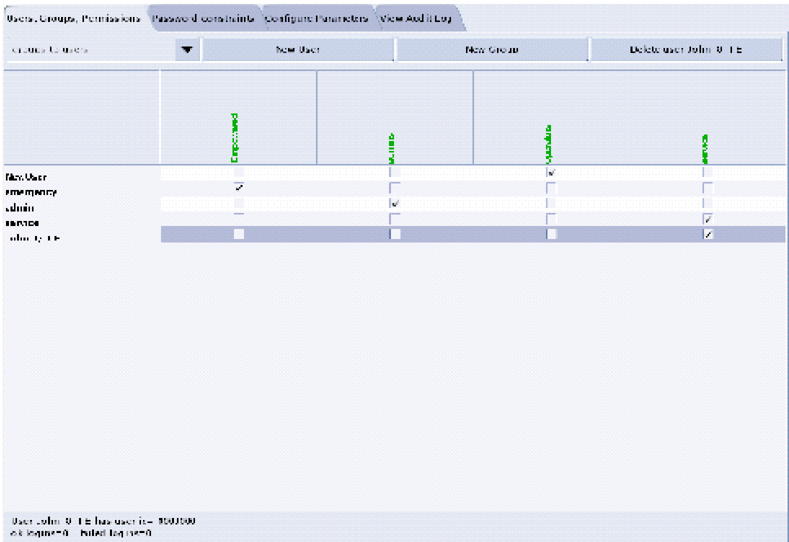


Figure 7-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 7-16 Login/Admin screen

3.8 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.

8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.9 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.

2.) Click the **rat gold** series.

3.) Click SELECT SERIES.

4.) Click on CONFIRM.

5.) Verify the image(s) reconstruct and are displayed.

6.) Click on QUIT.

3.10 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

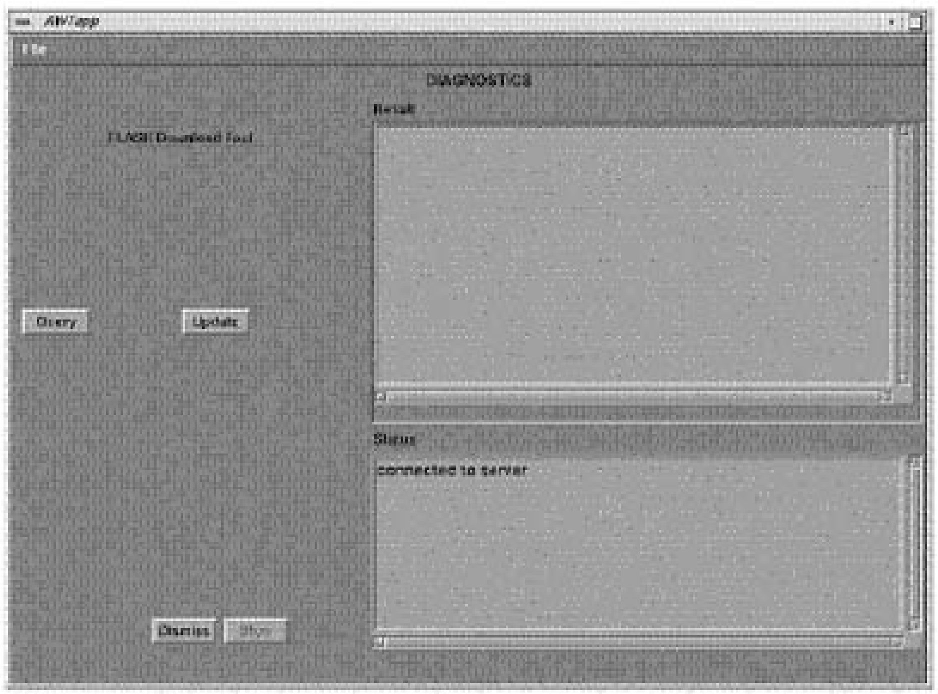


Figure 7-17 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping` SPACE `orp` ENTER

`ping` SPACE `stc` ENTER –OR– `ping` SPACE `tgp` ENTER

3.11 Tube and Generator Firmware Verification

Verify that the Tube and Generator firmware match before continuing with the options install.

1.) Visually inspect the Tube and heat exchanger: A LightSpeed Pro 16(80) with a Performix Pro tube (mcs_7079.mx) will have **blue** hoses running from the tube to the heat exchanger.
2.) Verify that the Jedi software installed on the system matches the tube that is mounted on the gantry by typing the following command in a Unix shell:

    cd SPACE /usr/g/ss_fw/jedi ENTER

    ls SPACE -l ENTER
3.) LightSpeed Pro 16(80) will have the following information displayed:

```
total 3748
-r--r--r--  1 ctuser  users    3291698 Nov 18 13:41 appl.mx
-r--r--r--  1 ctuser  users    494264 Nov 18 13:41 boot.mx
lrwxrwxrwx  1 ctuser  users       11 Nov 19 20:17 database.mx ->
mcs_7079.mx*
-rwxrwxrwx  1 root     root    14360 Nov 19 20:17 mcs_7079.mx*
-r--r--r--  1 ctuser  users    14360 Nov 18 13:41 mcs_7079.mx.lfc
-r--r--r--  1 ctuser  users    14360 Nov 18 13:41 mx240_mct.mx
```

Note: The database.mx is a symbolic link to mcs_7079.mx, which is the Performix Pro database.

4.) If your Jedi directory does not match the information listed above, the wrong Host Applications Software may have been loaded. Verify that your software disk set is compatible with the system that you are working on. If the wrong Host Applications Software was installed, you must start the software install procedure over again.

3.12 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.

4.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 7-18) appears.

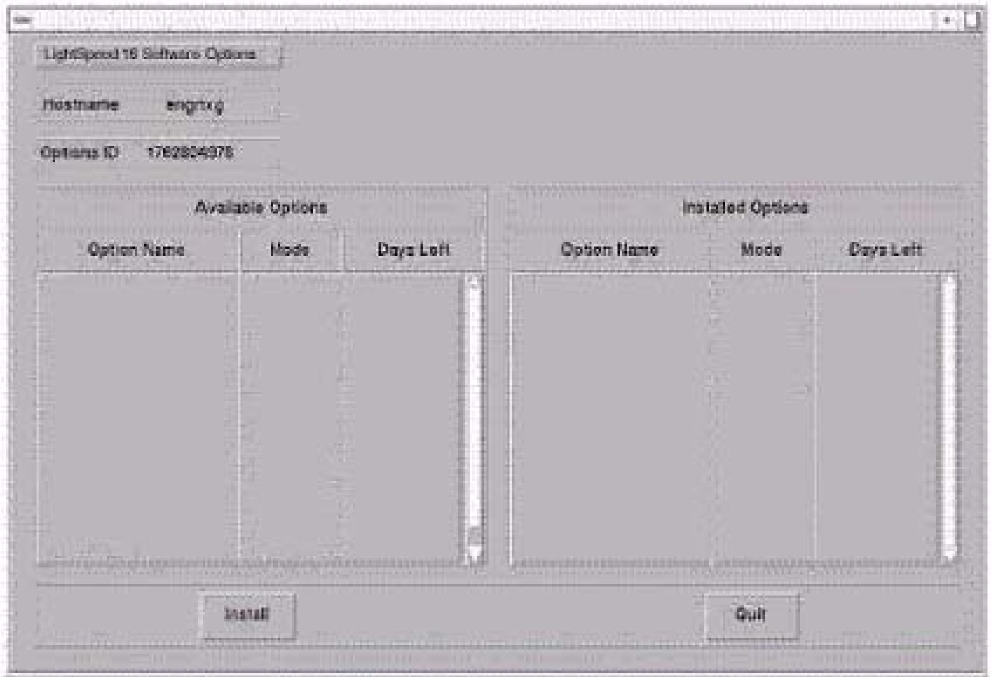


Figure 7-18 Install Options screen

5.) Click INSTALL. A Select Mechanism (Figure 7-19) window opens. Select the mechanism through which you want to install Option Keys.

Figure 7-19 Select Mechanism window

6.) Click PERMANENT. A Select Device (Figure 7-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 7-20 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 7-20 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

8.) Click OK. Available options appear on the Software Options screen (Figure 7-21).

Example:
Options screen
with Permanent
& FlexTrial
options.

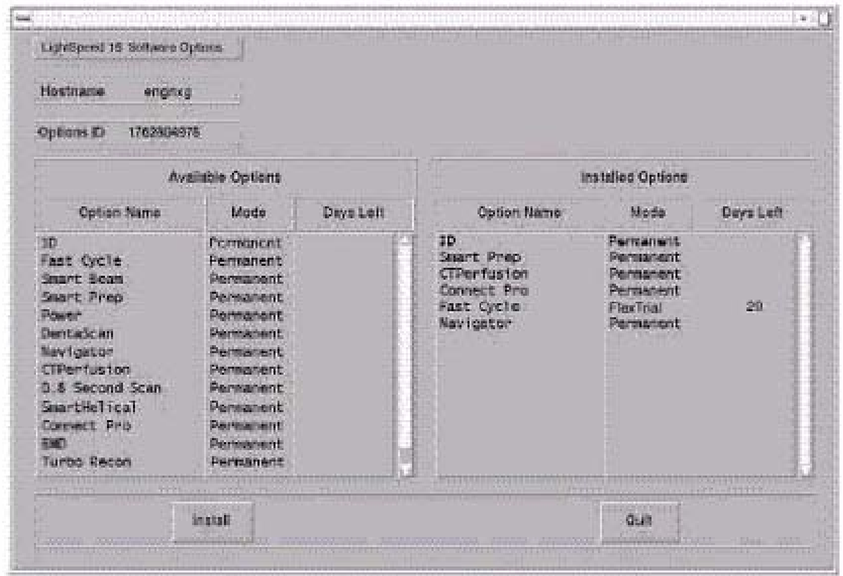


Figure 7-21 Software Options screen

9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

10.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **`su`** **SPACE** **`-`** **ENTER**.

Type the root password: **`#bigguy`** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press **ENTER**.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** ENTER
2.) Select OK as needed to answer messages. Note any Recon Failures.
3.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 7-22 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.15 LightSpeed 5.X (LS Pro^{16} (80kW)) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS_H16_16
- ❑ Tube Type: MX_240MCT
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: GANTRY HP
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: ***CT## WLSNorm 4008 3.57.255.##***

Chapter 8

LFC Procedures – LightSpeed 5.X (LS^{16}, Ultra & Plus 500.15M3)

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 5.x LS^{16}, Ultra & Plus GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 5.X (LS^{16}, Ultra & Plus)**:

1.) **2401507 – LightSpeed / HiSpeed / Linux Console Operating System**
Version - GEMS 1.7.10
2.) **2401706 – LightSpeed 16, Ultra & Plus Apps Software**
Version: 500.15_HP60_5.1M3
3.) **2402164 – LightSpeed 16, Ultra & Plus DARC Applications**
Version: 500.15_HP60_5.1M3

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD inserted until instructed in

the procedure.

4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: `cat /usr/g/config/INFO` **ENTER**
   c.) Record the information on the screen in Section 3.15. For an example screen output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Decide on Restore Protocols.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 5.X (LS16, Ultra & Plus) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: `cat /usr/g/config/INFO` **ENTER**
- ❑ Install the Console OS on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – **Yes** to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: `st` **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: `#bigguy`
- ctuser: `4$apps`

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again`.
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.). CSA = Common Software Applications.

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing a software upgrade. This is a one time procedure necessary prior to loading M3 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M3 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

1.) Open up a Unix Shell.
2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.
3.) Type the following to save CSA Protocols:
   a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER
   b.) **`mountDVD`** ENTER
   `Mounting DVD.`
   c.) **`cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER
   d.) **`unmountDVD`** ENTER
   `Unmounting DVD.`
4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM OS Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the LightSpeed Console Operating System CD-ROM into the DVD ROM drive on the HP Host Computer.

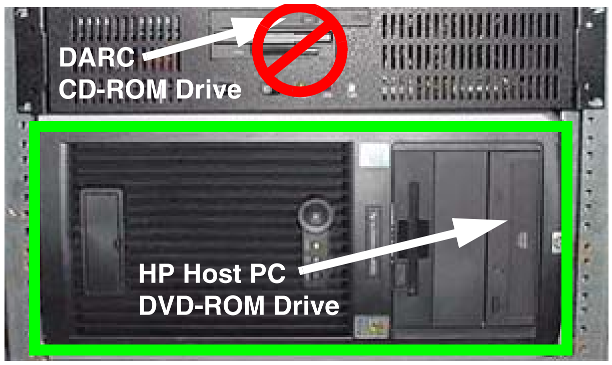


Figure 8-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 8-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** ENTER *(for English Keyboard only)*
or: **`iGEMS2`** ENTER *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the LightSpeed Applications SW CDROM into the DVD-ROM drive on the HP Host PC.

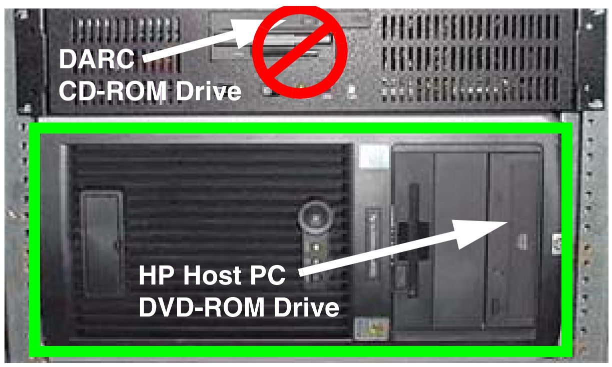


Figure 8-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

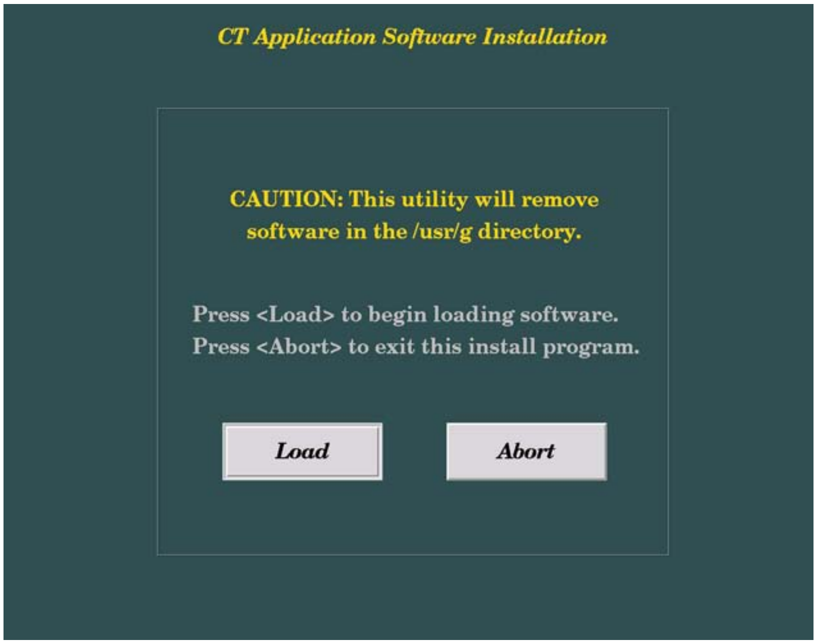


Figure 8-4 Lightspeed Install Utility window

4.) Click LOAD.

5.) Select the proper Product:

- LIGHTSPEED16 for LightSpeed[16]
- LIGHTSPEED8 for LightSpeed Ultra
- LIGHTSPEED4 for LightSpeed Plus

Then, cllick YES.

6.) A prompt message appears asking if you wish to load the INFO file from the System State DVD.
If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO, and then click OK.

7.) Click the SYSTEM button.

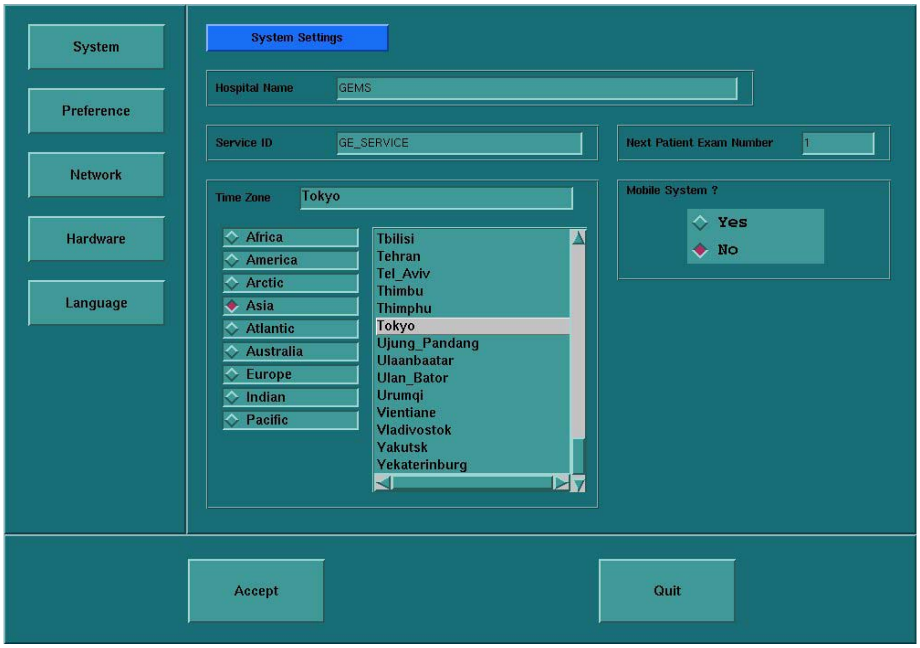


Figure 8-5 System Settings screen

8.) Verify or enter the Hospital Name.

9.) Verify or enter the Service ID.

10.) Verify that the **Next Patient Exam #** on the System Settings screen is correct (or, for the first installation, enter **1**.)

11.) Set Mobile System to NO.

12.) Verify that the proper **Time Zone** is selected.

13.) Click the PREFERENCE button.

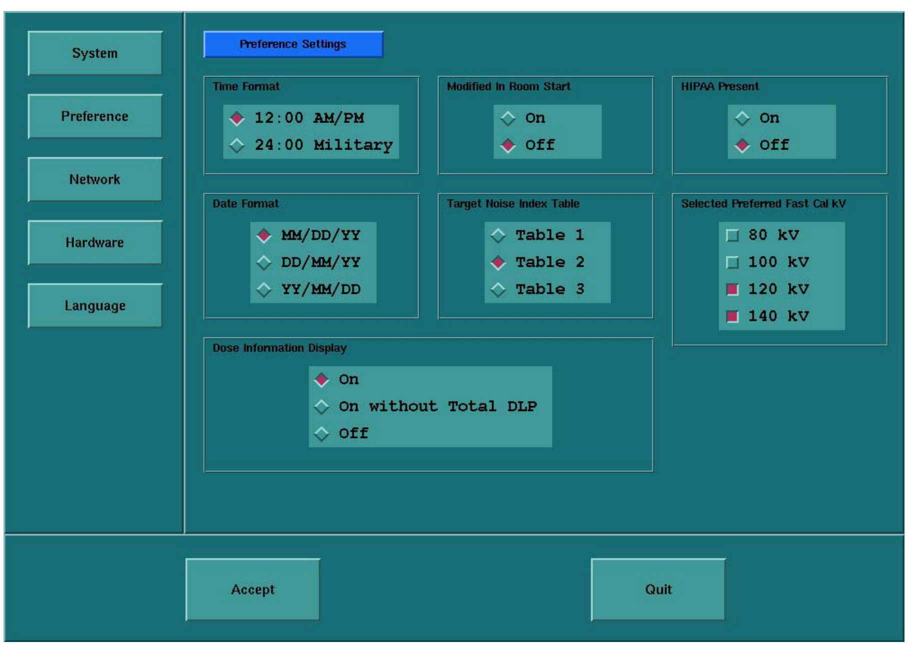


Figure 8-6 Preferences screen

14.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

15.) Select **Target Noise Index Table**: default is TABLE 2.

16.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

17.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

18.) Verify **Modified in Room Start** is set to OFF.

19.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

20.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

21.) Click the NETWORK button (site will be different).

Figure 8-7 Network Settings screen

22.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

23.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

24.) Set the **IP Address (OC Parameters)**: add your address here (Example: ***3.57.255.55***).

25.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

26.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

27.) Set the **IP Address (Gateway Parameters)**: add your address here (Example: ***3.57.252.254***).

28.) Click the HARDWARE button.

Figure 8-8 Hardware Settings screen example (your screen may not match exactly)

29.) Select the proper DAS Type:

- MDAS_H16_16 for for LightSpeed16
- MDAS8 for LightSpeed Ultra
- MDAS4 for LightSpeed Plus

30.) Select the proper Tube Type: MX_200MCT.

31.) Select the proper PDU Type: NG PDU.

32.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

33.) Select the proper GANTRY TYPE: HP GANTRY.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

34.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

35.) Click the LANGUAGE button.

36.) Select the proper language.

37.) Click the **ACCEPT** button at the bottom of the User Interface. The following message appears: `Please confirm a CT Application CD-ROM into a drive.`

38.) Click OK.

Note: Do not remove the LightSpeed Apps CDROM from the HP Host Computer until instructed to do so.

39.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load)`. A shell Installation screen appears and the application software starts loading.

40.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

41.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

42.) Press the button on the DVD ROM drive to remove the LightSpeed Apps SW CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

ctuser: **cat** SPACE **/usr/g/config/INFO** ENTER

Verify DAS type, Tube type, PDU type (NG PDU), Gantry type (hp), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

ctuser: **su** SPACE **-** ENTER

password: **#bigguy** ENTER

root: **cd** SPACE **/usr/g/scripts** ENTER

root: **./start_darcOS** ENTER

An Attention window appears. See Figure 8-9.

- If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

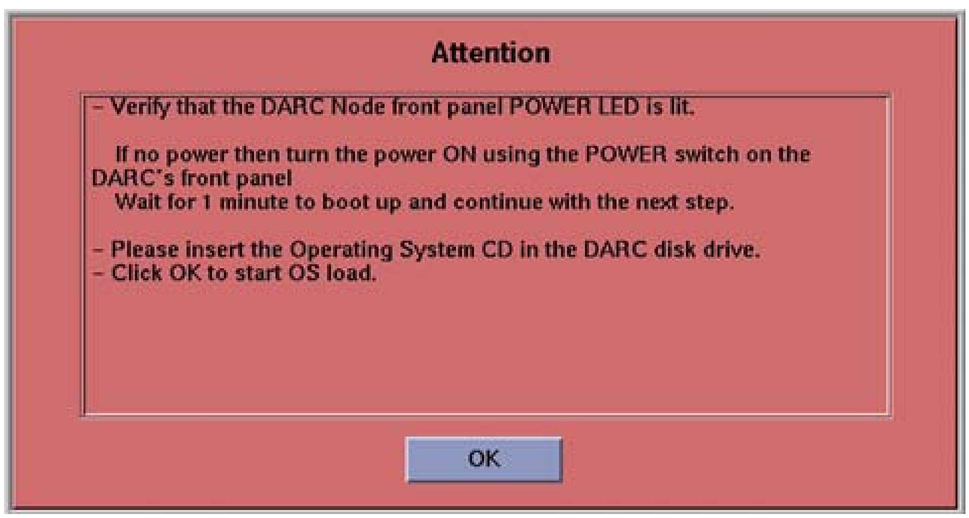


Figure 8-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the LightSpeed Console Operating System CD-ROM into the DARC Node drive.

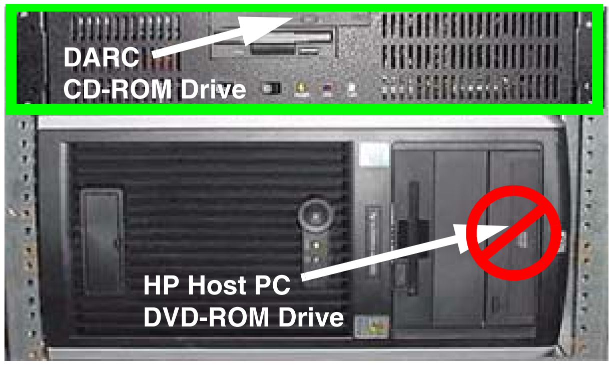


Figure 8-10 DARC CD-ROM Drive Location

5.) Click on **OK**.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

`The OS load on the DARC Node is started. This takes about 9-11 minutes.`

`##.## % done.`

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

6.) Refer to the `%done` status displayed during the DARC OS load. when the DARC OS load has completed, a the following message is displayed.

`The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up…`

7.) The OS disk ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel Reset switch. Manually eject the CD after approximately 2 minutes.

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 8-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

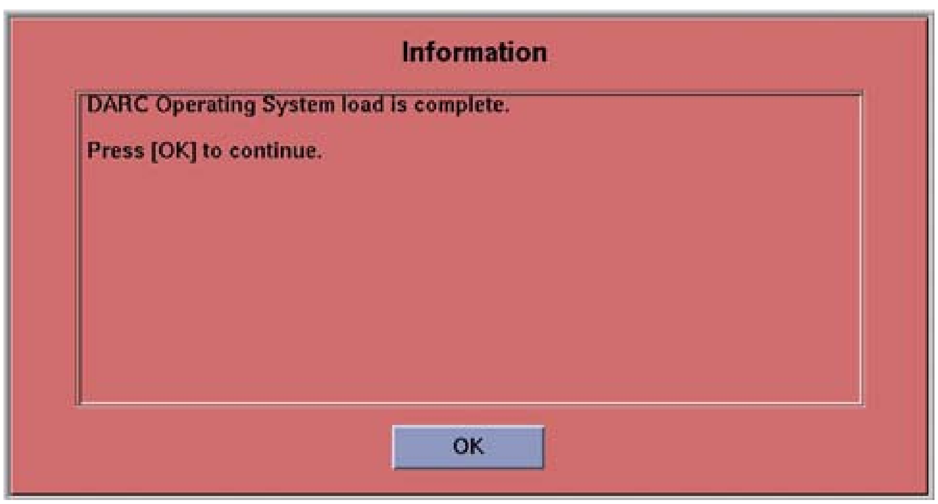


Figure 8-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully loaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

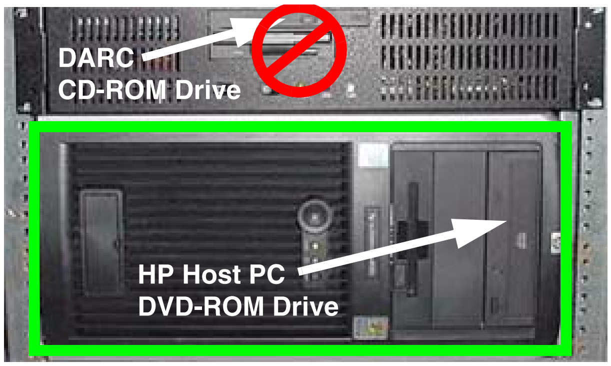


Figure 8-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Continue with DARC Apps load:

`root:` **`cd`** **SPACE** **`/`** **ENTER**

`root:` **`mount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`cd`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`./copy_rpms`** **ENTER**

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

4.) Type:

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darc_load`** **ENTER**

A shell window pops up and the load begins. A successful load takes 4–5 minutes.

Continue the procedure when the load window goes away.

5.) When the load displays a message that it is completed or window goes away, type the following:

`root:` **`cd`** **SPACE** **`/`** **ENTER**

`root:` **`umount`** **SPACE** **`/mnt/cdrom`** **ENTER**

`root:` **`eject`** **ENTER**

6.) Close the shell (do not use for the next process).

7.) Remove the DARC Applications SW CDROM from the HP Host Computer DVD-ROM drive.

3.4 Reboot the System

1.) Open a Unix Shell.
2.) Type: `su` SPACE - ENTER
3.) Enter the root password: `#bigguy` ENTER
4.) Type: `sync` ENTER
5.) Type: `sync` ENTER
6.) Type: `reboot` ENTER
A message appears: `The system is going down for reboot NOW!.`
7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: `st` ENTER or `startup` ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`
Read the Note(s) below.
4.) Click OK as needed to answer message prompts.
5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.
2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

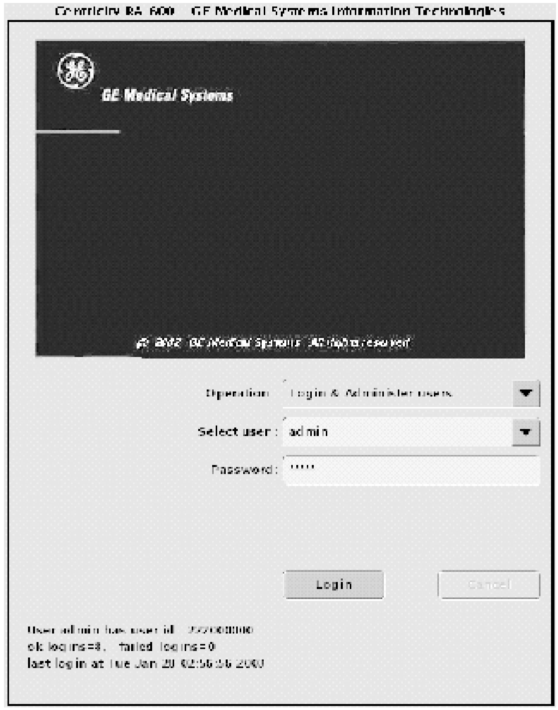


Figure 8-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

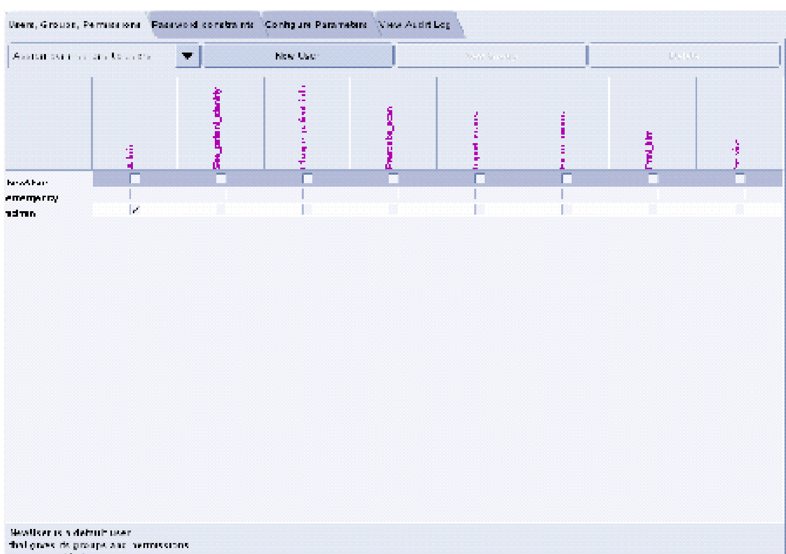


Figure 8-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 8-15).

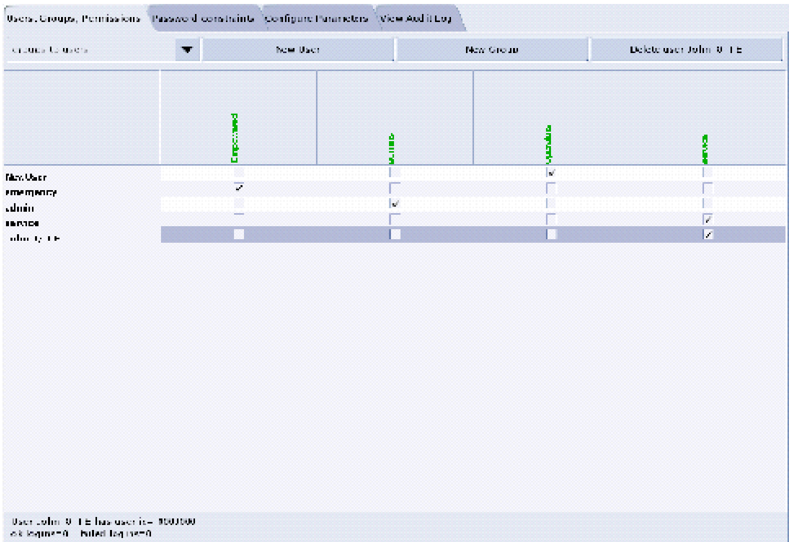


Figure 8-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 8-16 Login/Admin screen

3.8 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.
2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click the ALL selection which highlights the cals, characterization, etc.
6.) Make certain to click RESTORE.
7.) Verify the overall system state content has restored correctly, then select CANCEL.
8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.9 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.10 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

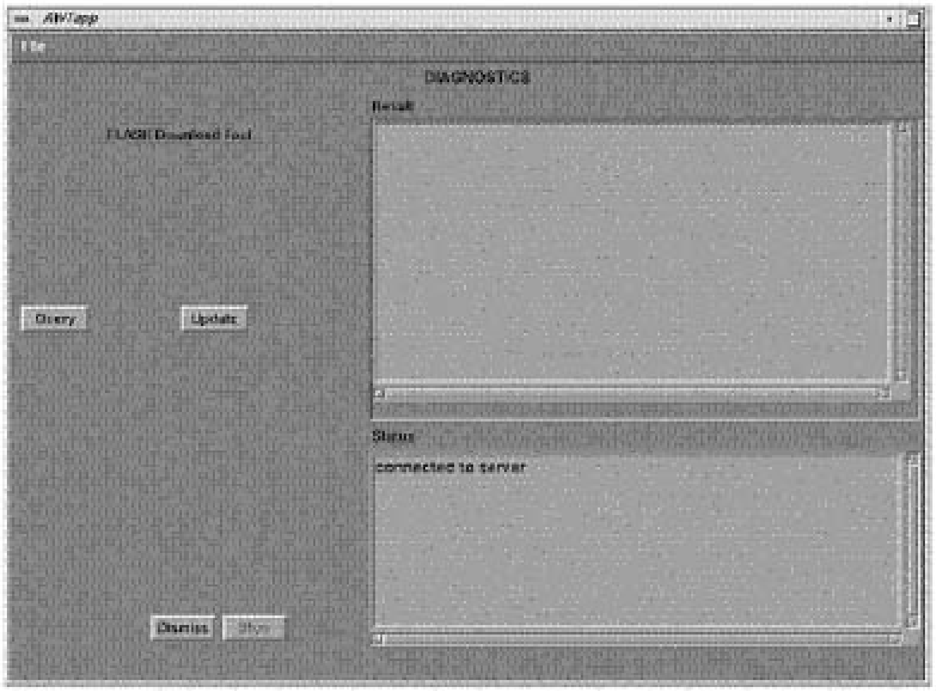


Figure 8-17 Diagnostics screen

5.) If there are any issues with the FLASH Download Tool update, verify the two Console Transceiver DIP switch settings (SW 1 up and SW2 down), and verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping` SPACE `orp` ENTER

`ping` SPACE `stc` ENTER –OR– `ping` SPACE `tgp` ENTER

3.11 Install Options

NOTICE **For systems where CT Perfusion 2 is included in Available Options, perform this Install Options procedure in the English language environment. (The LFC, however, can be performed in other language environment than English.) Otherwise, if you change the language setting after installing CT Perfusion 2, this option will not work; you will not even be able to start the program.**

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.

4.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 8-18) appears.

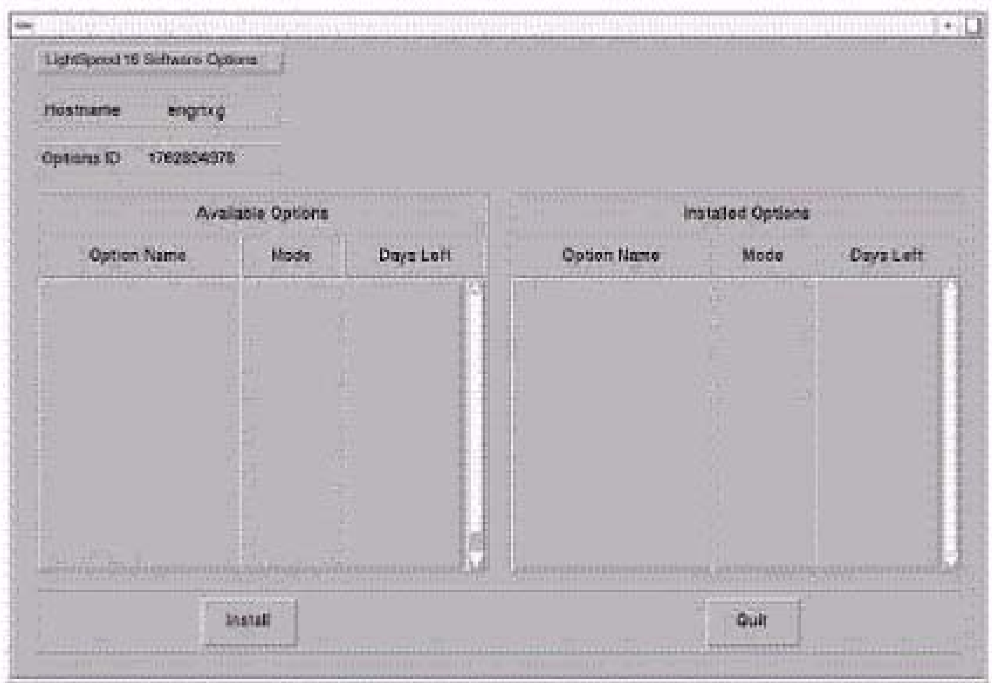


Figure 8-18 Install Options screen

5.) Click INSTALL. A Select Mechanism (Figure 8-19) window opens. Select the mechanism through which you want to install Option Keys.

Figure 8-19 Select Mechanism window

6.) Click PERMANENT. A Select Device (Figure 8-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 8-20 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

8.) Click OK. Available options appear on the Software Options screen (Figure 8-21).

Example:
Options screen
with Permanent
& FlexTrial
options.

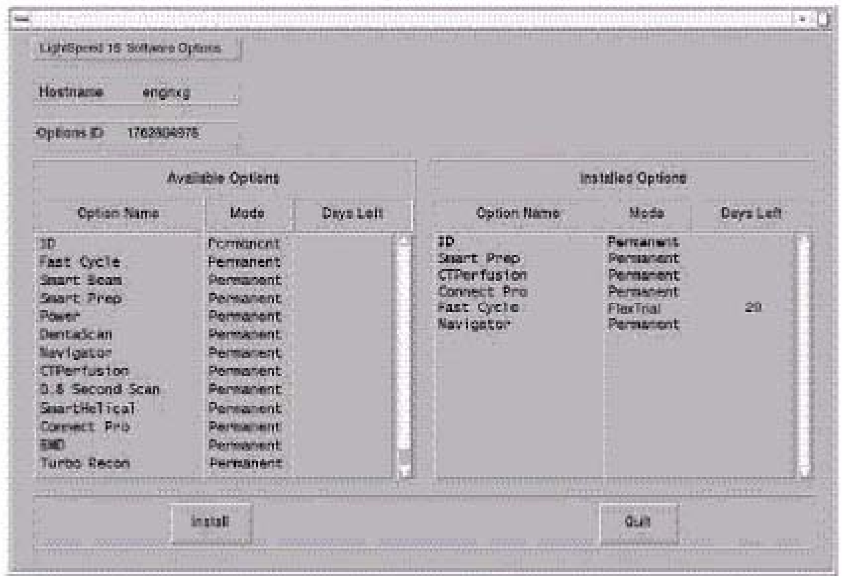


Figure 8-21 Software Options screen

9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

10.) When the process is complete, click QUIT then QUIT to close the window.

3.12 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su SPACE - ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **setdate *mmddhhmmyyyy*** (e.g.: **setdate 050409002003**) and press **ENTER**.

Note: Alternate method exists

You may also type: setdate ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **setdate** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **exit ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.13 Restart the System

3.13.1 Restarting the System

1.) To restart the system:

a.) Select OK as needed to answer messages. Note any Recon Failures.

b.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 8-22 Shutdown Icon

c.) Select: RESTART then OK to shutdown and Restart the system.

2.) If HIPAA Present SET TO ON --- Admin Screen

When the system comes up the following screen appears if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

a.) Select user: **`admin`**

b.) Password: **`ctAdmin`**

3.13.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.14 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.

3.) Successfully perform an Axial scan.

3.15 LightSpeed 5.X (LS^{16}, Ultra, Plus) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type:
  - MDAS_H16_16 for LightSpeed^{16}
  - MDAS8 for LightSpeed Ultra
  - MDAS4 for LightSpeed Plus
- ❑ Tube Type: MX_200MCT
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: GANTRY HP
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: ***CT## WLSNorm 4008 3.57.255.##***

Chapter 9
LFC Procedures – LightSpeed 5.X (LS^{16}, Ultra & Plus M3)

Section 1.0 Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 5.x LS^{16}, Ultra & Plus GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

Section 2.0 Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 5.X (LS^{16}, Ultra & Plus ME3)**:

1.) **2393757 – LightSpeed Series Xtream Console Operating System**
Supports: HP60_5.1
2.) **2393756 – LightSpeed Series Applications Software**
Version: 500.13_HP60_5.1M3
3.) **2393761 – Lightspeed Series DARC Applications Software**
Version: 500.13_HP60_5.1M3

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM does not have a CD inserted until instructed in the

procedure.

4.) Verify and record specific system hardware configuration.
   a.) Open a shell.
   b.) Type: `cat /usr/g/config/INFO` **ENTER**
   c.) Record the information on the screen in Section 3.15. For an example screen output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.
2.) Select PM, SAVE SYSTEM STATE with Applications up. Decide on Restore Protocols.
3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 5.X (LS16, Ultra & Plus) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: `cat /usr/g/config/INFO` **ENTER**
- ❑ Install the Console OS on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – **Yes** to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: `st` **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: `#bigguy`
- ctuser: `4$apps`

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State and CSA Protocol Retention

3.2.1 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again`.
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.2.2 CSA Protocol Retention (Upgrades Only)

Sites performing a Software Upgrade should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.). CSA = Common Software Applications.

Note: The manual CSA Protocol storage onto the "Save System State" DVD is only required when performing a software upgrade. This is a one time procedure necessary prior to loading M3 software. You do not need to manually restore the CSA protocols after performing this upgrade. Once loaded, the M3 Save/Restore System State software will automatically Save and Restore the CSA Protocols for the user.

1.) Open up a Unix Shell.
2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.
3.) Type the following to save CSA Protocols:
   a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER
   b.) **`mountDVD`** ENTER
   `Mounting DVD.`
   c.) **`cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER
   If an "overwrite" inquiry appears, respond "Yes".
   d.) **`unmountDVD`** ENTER
   `Unmounting DVD.`
4.) Close the Unix Shell and continue with the Console Operating System LFC Procedure.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM OS Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.
2.) Remove the GRE Operator Console's front cover.
3.) Locate the HP PC's DVD-ROM drive.
4.) Insert the LightSpeed Console Operating System CD-ROM into the DVD ROM drive on the HP Host Computer.
5.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 9-1 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

6.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

7.) At the boot prompt, type the following depending on system type:
Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed. Typing **`GEMS`** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

8.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.
9.) The HP Host Computer reboots. **Do not insert the LightSpeed Applications CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the LightSpeed Applications SW CDROM into the DVD-ROM drive on the HP Host PC.

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

Figure 9-2 Lightspeed Install Utility window

4.) Click LOAD.

5.) Select the proper Product:

- LIGHTSPEED16 for LightSpeed16
- LIGHTSPEED8 for LightSpeed Ultra
- LIGHTSPEED4 for LightSpeed Plus

Then, cllick YES.

6.) A prompt message appears asking if you wish to load the INFO file from the System State DVD.
If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO, and then click OK.

7.) Click the SYSTEM button.

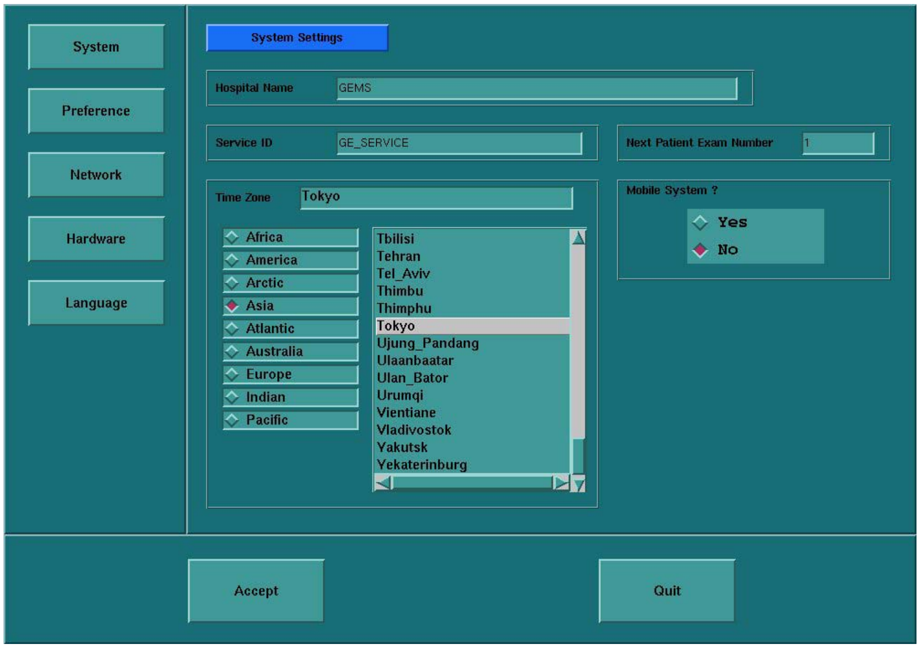


Figure 9-3 System Settings screen

8.) Verify or enter the Hospital Name.

9.) Verify or enter the Service ID.

10.) Verify that the **Next Patient Exam #** on the System Settings screen is correct (or, for the first installation, enter **1**.)

11.) Set Mobile System to NO.

12.) Verify that the proper **Time Zone** is selected.

13.) Click the PREFERENCE button.

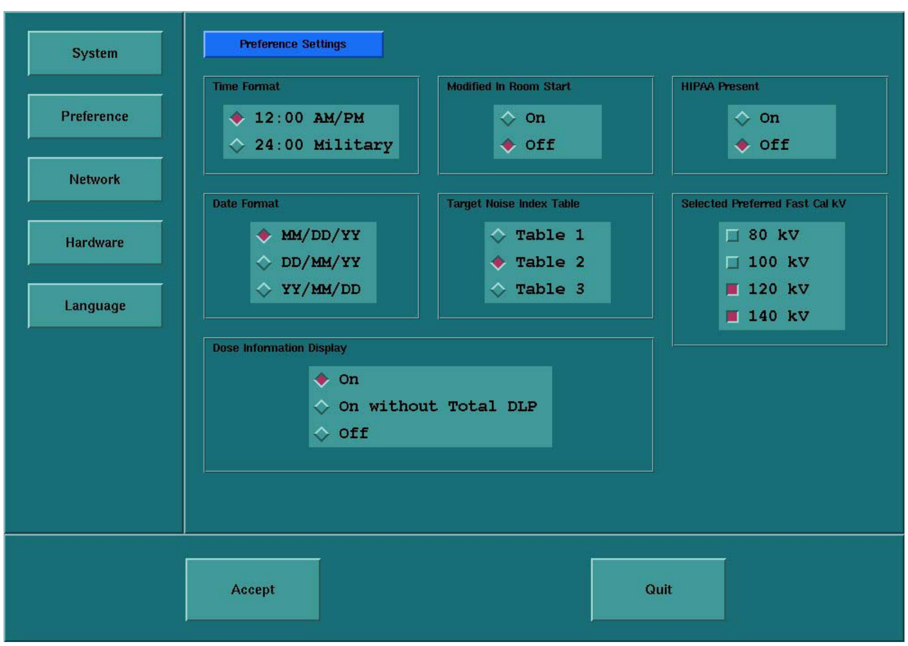


Figure 9-4 Preferences screen

14.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

15.) Select **Target Noise Index Table**: default is TABLE 2.

16.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

17.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

18.) Verify **Modified in Room Start** is set to OFF.

19.) Set **HIPAA Present** to OFF.

20.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

21.) Click the NETWORK button (site will be different).

Figure 9-5 Network Settings screen

22.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

23.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

24.) Set the **IP Address (OC Parameters)**: add your address here (Example: ***3.57.255.55***).

25.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

26.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

27.) Set the **IP Address (Gateway Parameters)**: add your address here (Example: ***`3.57.252.254`***).

28.) Click the HARDWARE button.

Figure 9-6 Hardware Settings screen example (your screen may not match exactly)

29.) Select the proper DAS Type:

- MDAS_H16_16 for for LightSpeed[16]
- MDAS8 for LightSpeed Ultra
- MDAS4 for LightSpeed Plus

30.) Select the proper Tube Type: MX_200MCT.

31.) Select the proper PDU Type: NG PDU.

32.) Select the proper Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

33.) Select the proper GANTRY TYPE: HP GANTRY.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

34.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

35.) Click the LANGUAGE button.

36.) Select the proper language.

37.) Click the **ACCEPT** button at the bottom of the User Interface. The following message appears: `Please confirm a CT Application CD-ROM into a drive.`

38.) Click OK.

Note: Do not remove the LightSpeed Apps CDROM from the HP Host Computer until instructed to do so.

39.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

40.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

41.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

42.) Press the button on the DVD ROM drive to remove the LightSpeed Apps SW CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node.

- If any problems arise, verify the correct CD is installed.
- If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat`** SPACE **`/usr/g/config/INFO`** ENTER

Verify DAS type, Tube type, PDU type (NG PDU), Gantry type (hp), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** SPACE **`-`** ENTER

`password:` **`#bigguy`** ENTER

`root:` **`cd`** SPACE **`/usr/g/scripts`**

`root:` **`./start_darcOS`** ENTER

A sh window appears with the message below:

```
Please insert the Operating System CD into the DARC Node disk drive.
Press <Enter> to start OS load.
```

4.) Insert the LightSpeed Console Operating System CD-ROM into the DARC Node drive.

5.) Select the ENTER key.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

```
The OS load on the DARC Node is started.  This takes about 9-11 minutes.
##.## % done.
```

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. You may be instructed to reboot; open a Unix shell and type "halt", then power cycle the console.

6.) A `%done` status is displayed during the DARC OS load.

7.) The OS disk ejects automatically. Remove the OS disk from the drive and close the drive.

```
The OS load is complete. Please remove the OS disk from the DARC disk
drive. DARC is rebooting. Wait for the DARC to boot up…
```

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully.

Note: **If the OS disk CD does not eject or the DARC does not boot/reboot correctly, then press the DARC Node front panel Reset switch. Manually eject the CD after approximately 2 minutes.**

9.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

10.) This completes the OS load on the DARC.

11.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

The DARC Applications Software CD-ROM will be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.
2.) Open a Unix Shell and type the following:
    `ctuser:` **`su`** **SPACE** `-` **ENTER**
    `password:` **`#bigguy`** **ENTER**
3.) Continue with DARC Apps load:
    `root:` **`cd`** **SPACE** `/` **ENTER**
    `root:` **`mount`** **SPACE** **`/mnt/cdrom`** **ENTER**
    `root:` **`cd`** **SPACE** **`/mnt/cdrom`** **ENTER**
    `root:` **`./copy_rpms`** **ENTER**
    The following messages appear.
    `Copying darc rpms...`
    `Please wait...`
    `Done...`
4.) Type:
    `root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**
    `root:` **`./start_darc_load`** **ENTER**
    A shell window pops up and the load begins. It takes 4–5 minutes.
    Continue the procedure when the load window goes away.
5.) When the load displays a message that it is completed or window goes away, type the following:
    `root:` **`cd`** **SPACE** `/` **ENTER**
    `root:` **`umount`** **SPACE** **`/mnt/cdrom`** **ENTER**
    `root:` **`eject`** **ENTER**
6.) Close the shell (do not use for the next process).
7.) Remove the DARC Applications SW CDROM from the HP Host Computer DVD-ROM drive.

3.4 Reboot the System

1.) Open a Unix Shell.
2.) Type: **`su`** **SPACE** `-` **ENTER**
3.) Enter the root password: **`#bigguy`** **ENTER**
4.) Type: **`sync`** **ENTER**
5.) Type: **`sync`** **ENTER**
6.) Type: **`reboot`** **ENTER**
    A message appears: `The system is going down for reboot NOW!.`
7.) After approximately 4–5 minutes a pink pop up window appears with the message:
    `CT Software Auto-Start Disabled.`
    Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).

2.) Type: `st` ENTER or `startup` ENTER.

3.) A pink pop-up Attention Message appears: `OC initializing. Please wait….`

Read the Note(s) below.

4.) Click OK as needed to answer message prompts.

5.) If any Recon Selftest Failures are encountered, review the error log.

Note:

- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.

2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

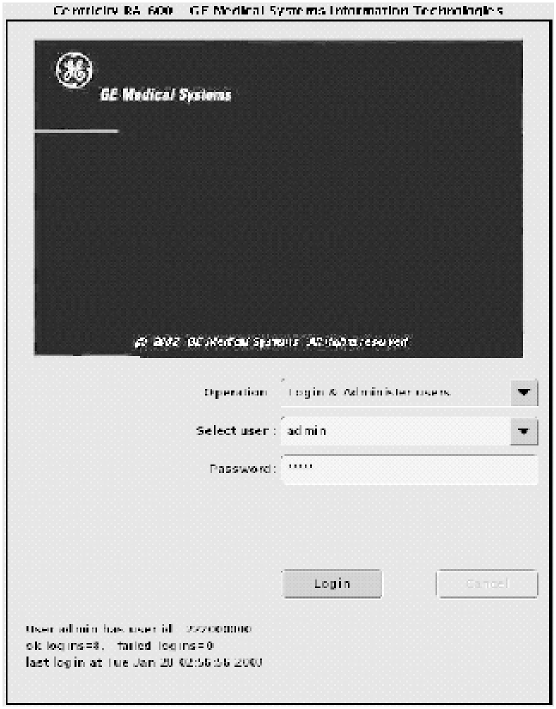


Figure 9-7 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

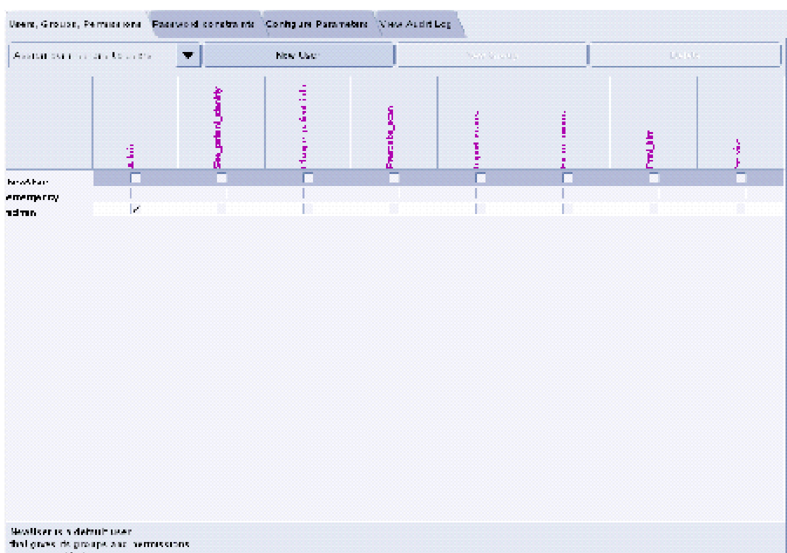


Figure 9-8 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 9-9).

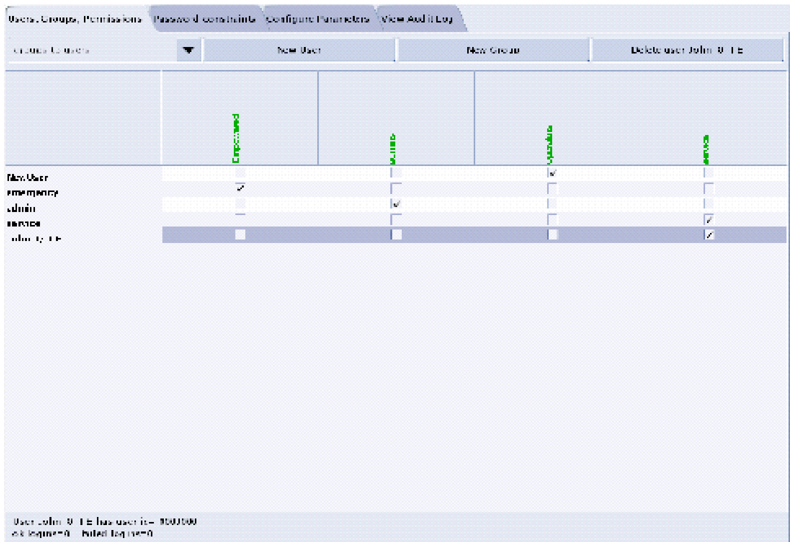


Figure 9-9 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 9-10 Login/Admin screen

3.8 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) **Software Reload**: Follow the instruction below for the following software reloads:
   - HPower ME4 to HPower ME4

8.) Click YES to Reset Hardware.
   If "FLASH Download" appears, ignore this, since you will execute it in Section 3.10.

9.) Do not remove the system state media (DVD) from the drive after the Restore operation.

10.) Open a shell and type the following:
   a.) **`mountDVD`** ENTER
   b.) **`cp`** SPACE **`-f`** SPACE **`/DVD/service_mod_data/system_state/vxtlBackup.tar`** SPACE **`/data/vxtlBackup.tar`** ENTER
   c.) **`tar`** SPACE **`Pxvf`** SPACE **`/data/vxtlBackup.tar`** ENTER
   d.) **`unmountDVD`** ENTER

11.) Close the shell and continue with the software load.

12.) Verify that the CSA Protocols (Reformat, Volume Viewer, etc.) are restored correctly.

13.) Verify the overall system state content has also restored correctly.

14.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.9 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.
2.) Click the **rat gold** series.
3.) Click SELECT SERIES.
4.) Click on CONFIRM.
5.) Verify the image(s) reconstruct and are displayed.
6.) Click on QUIT.

3.10 FLASH Download

1.) Select the SERVICE DESKTOP.
2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.
3.) Click UPDATE. Ignore any initial errors.
4.) When then Flash Download process is complete, select DISMISS or Close the window.

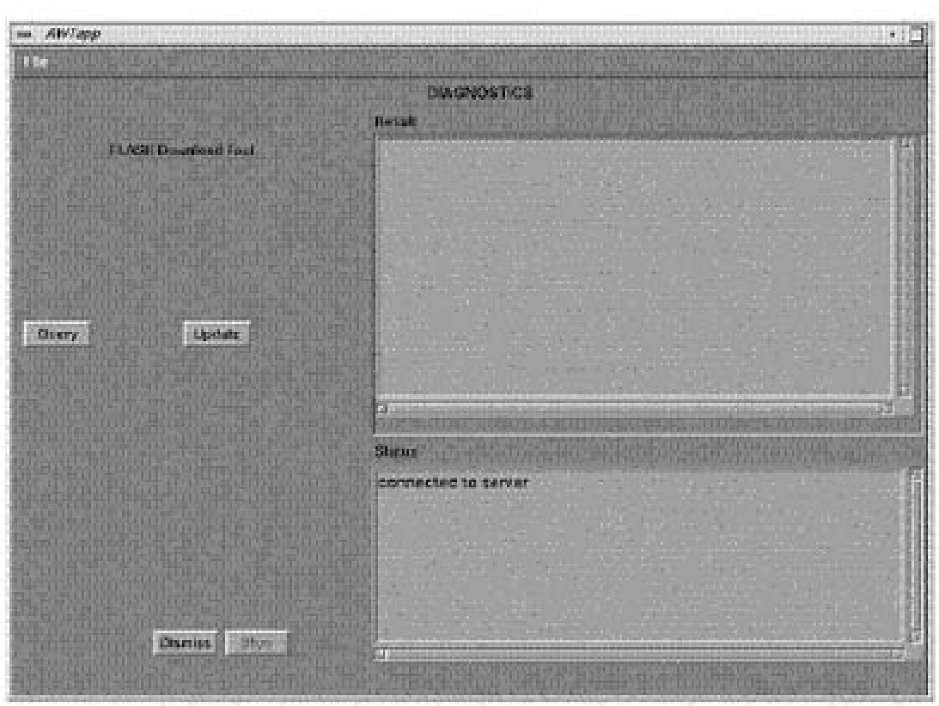


Figure 9-11 Diagnostics screen

5.) If there are any issues with the FLASH Download Tool update, verify the two Console Transceiver DIP switch settings (SW 1 up and SW2 down), and verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

ping SPACE orp ENTER

ping SPACE stc ENTER –OR– ping SPACE tgp ENTER

3.11 Install Options

NOTICE **For systems where CT Perfusion 2 is included in Available Options, perform this Install Options procedure in the English language environment. (The LFC, however, can be performed in other language environment than English.) Otherwise, if you change the language setting after installing CT Perfusion 2, this option will not work; you will not even be able to start the program.**

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 9-12) appears.

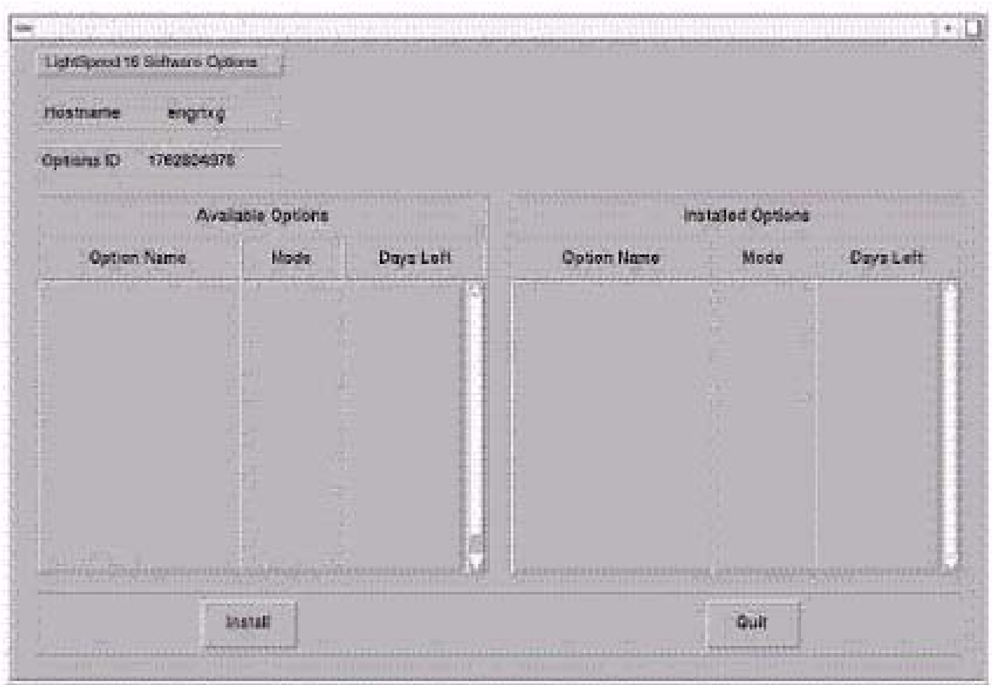


Figure 9-12 Install Options screen

5.) Click INSTALL. A Select Mechanism (Figure 9-13) window opens. Select the mechanism through which you want to install Option Keys.

Figure 9-13 Select Mechanism window

6.) Click PERMANENT. A Select Device (Figure 9-14) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 9-14 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

8.) Click OK. Available options appear on the Software Options screen (Figure 9-15).

Example:
Options screen
with Permanent
& FlexTrial
options.

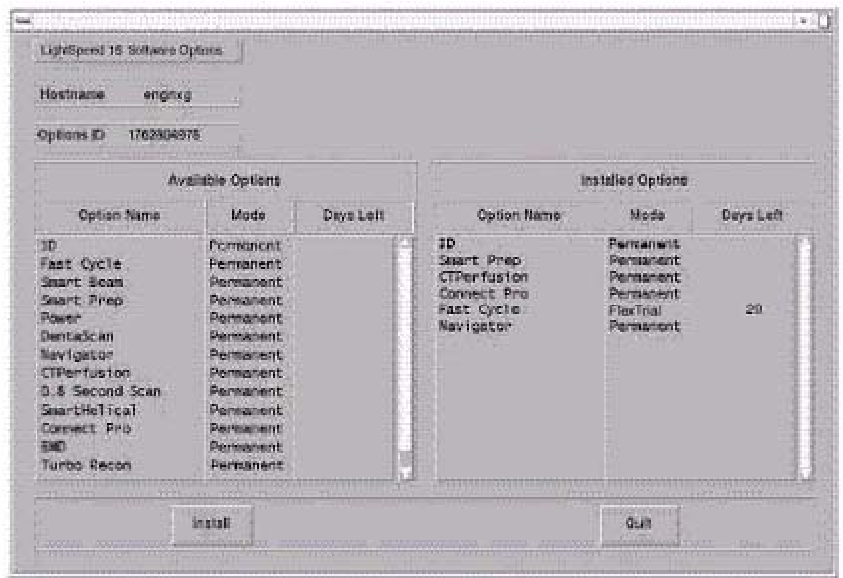


Figure 9-15 Software Options screen

9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

10.) When the process is complete, click QUIT then QUIT to close the window.

3.12 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su SPACE - ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press **ENTER**.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** **ENTER**.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.13 Restart the System

3.13.1 Restarting the System

1.) To restart the system:

a.) Select OK as needed to answer messages. Note any Recon Failures.

b.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 9-16 Shutdown Icon

c.) Select: RESTART then OK to shutdown and Restart the system.

2.) If HIPAA Present SET TO ON --- Admin Screen

When the system comes up the following screen appears if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

a.) Select user: **`admin`**

b.) Password: **`ctAdmin`**

3.13.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.14 System Sanity Scanning

1.) Successfully perform a Scout scan.

2.) Successfully perform a Helical scan.
3.) Successfully perform an Axial scan.

3.15 LightSpeed 5.X (LS^{16}, Ultra, Plus) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type:
  - MDAS_H16_16 for LightSpeed^{16}
  - MDAS8 for LightSpeed Ultra
  - MDAS4 for LightSpeed Plus
- ❑ Tube Type: MX_200MCT
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: GANTRY HP
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: ***CT## WLSNorm 4008 3.57.255.##***

Chapter 10
LFC Procedures – LightSpeed 5.X (RT) M3

Section 1.0
Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed 5.x RT GRE system and should be used ONLY when loading the applicable software package.

Note: To verify if this chapter applies to your system, refer to Software Compatibility, on page 23.

Section 2.0
Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 3 CDROMs. For software versions specific to each installation, see Section 2.1.2.

1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed 5.X RT M3**:

- **2399296 – LightSpeed / HiSpeed Linux Console Operating System**
  Version – GEMS Linux 1.7.10
- **2396451 – LightSpeed RT Applications Software**
  Version – 501WB.4_H5.2M3
- **2396452 – LightSpeed RT DARC Applications**
  Version –
- **2399297 – Xtream Serial-Over-Lan Service Software**
  Version – 1.0

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

1.) Verify the Peripheral Tower DVD RAM power is applied.
2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
4.) Verify and record specific system hardware configuration.

a.) Open a shell.

b.) Type: **`cat /usr/g/config/INFO`** **ENTER**

c.) Record screen information in Section 3.14. For an example output, see Appendix C.

2.2.2 Software Load Prerequisites

1.) With Apps up, select the SERVICE icon to access the Netscape: Service Desktop.

2.) Select PM, SAVE SYSTEM STATE with Applications up. Save Protocols if applicable.

3.) Record Autovoice Volume control settings (**ALT-F3** by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed 5.X (RT) GRE SW Load Checklist

- ❑ Save System State (be sure save any customer reformat files that might exist).
- ❑ Verify and record specific system hardware configuration.
  Open a shell and type: **`cat /usr/g/config/INFO`** **ENTER**
- ❑ Install the Console Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ❑ Select OK by pressing the **ENTER** key and immediately remove the OS CD.
- ❑ Install the LightSpeed Applications SW on the HP Host Computer – **`Yes`** to autorun.
- ❑ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ❑ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ❑ Install the OS on the DARC Node, then type as required.
- ❑ Remove OS CDROM from DARC Node.
- ❑ Install the DARC Applications SW on the HP Host PC, type as required.
- ❑ Remove the DARC Applications SW from the HP Host PC.
- ❑ Install service software as applicable; reboot.
- ❑ Select CANCEL to stop Applications Software from coming up or OK as required.
- ❑ Open a shell and type: **`st`** **ENTER**.
- ❑ Perform the Retro Recon Sanity Check.
- ❑ Restore System State.
- ❑ Perform the Flash Download Update.
- ❑ Install the Options.
- ❑ Reboot the system.
- ❑ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: **`#bigguy`**
- ctuser: **`4$apps`**

3.1.3 Other General Information

Refer to Appendix E for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network

- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State

1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
2.) Select: SERVICE DESKTOP.
3.) Select: PM.
4.) Select: SYSTEM STATE.
5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If applicable, click OK when the following message appears: `System State Media Status: Please insert a DVD or MOD into the drive and press Save again`.
8.) When completed select DISMISS.
9.) Save all reformat / recon protocols.
10.) Close the Service Desktop window in the upper left corner of the screen.
11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

1.) Verify the System State was saved.

2.) Remove the GRE Operator Console's front cover.

3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

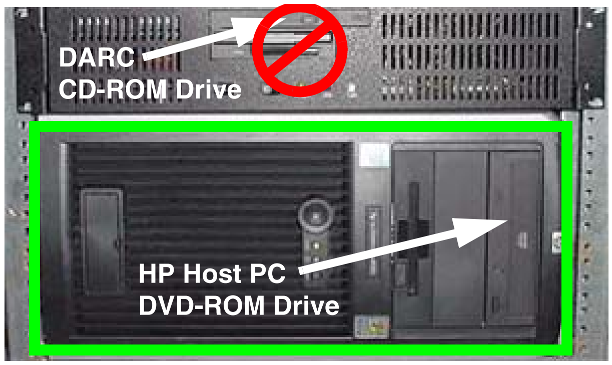


Figure 10-1 HP Host PC DVD-ROM Drive Location

4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power **OFF** the console.

Figure 10-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

5.) Power **ON** the Console or Select: RESTART then OK to shut down and restart the system.

As the computer restarts, the booting process messages appear.

After the booting process completes, the `boot:` prompt appears.

6.) At the boot prompt, type the following depending on system type:
Type: **GEMS2** **ENTER** *(for English Keyboard only)*
or: **iGEMS2** **ENTER** *(international command only)*

Note: The Console Operating System load takes approximately 10 minutes. If the `boot:` prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **GEMS** will result in both scan and display being on one monitor and require a LFC.

Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.

7.) After the OS is loaded, a red button appears and displays OK. Press **ENTER**, and then immediately remove the OS CD (within 10 seconds) when it ejects.

8.) The HP Host Computer reboots. **Do not insert the LightSpeed RT Applications Software CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROM's onto the HP Host Computer.

WARNING **FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM DAS HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND DCB CIRCUIT BOARD REPLACEMENT. (THIS FAILURE WILL OCCUR AFTER THE FLASH DOWNLOAD PROCEDURE FLASHES THE FIRMWARE.)**

Note: For platform-specific software versions, see Section 2.1.2.

1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the **LightSpeed RT Applications Software** CDROM into the DVD-ROM drive on the HP Host PC.

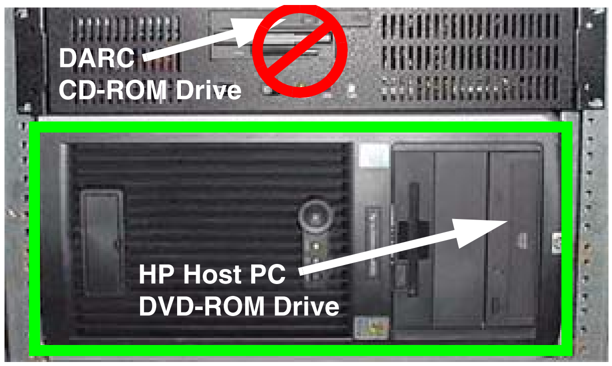


Figure 10-3 HP Host PC DVD-ROM Drive Location

2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

3.) Click YES. After approximately one minute, an Installation Utility window appears.

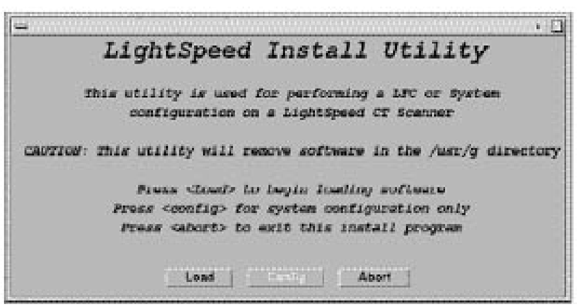


Figure 10-4 Lightspeed Install Utility window

4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.

5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

6.) Click the SYSTEM button.

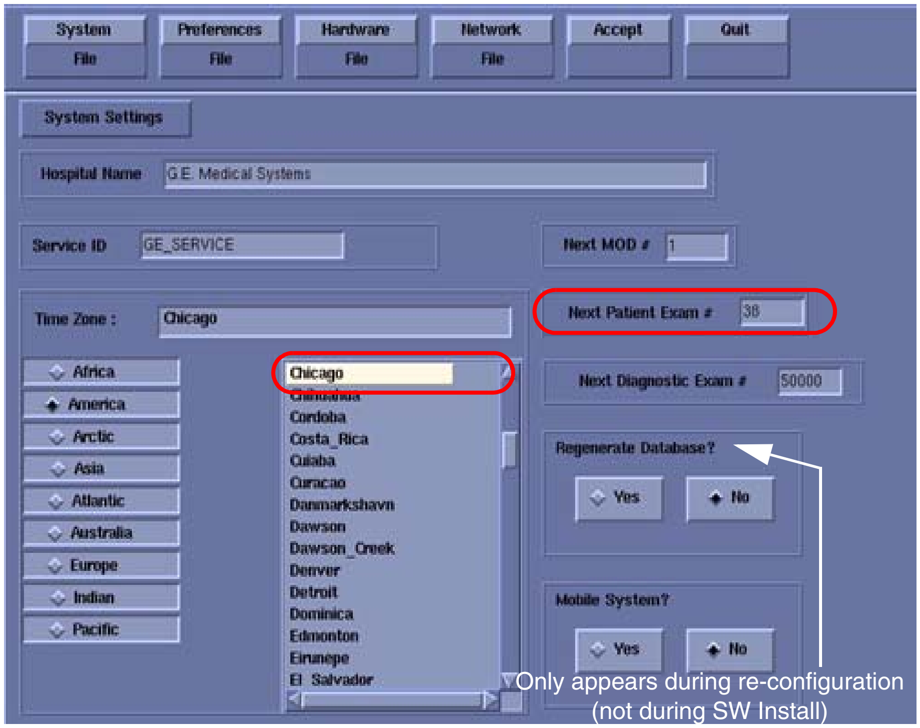


Figure 10-5 System Settings screen

7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = **1**.

8.) Verify that the proper **Time Zone** is selected (example: Chicago).

9.) Click the PREFERENCES button.

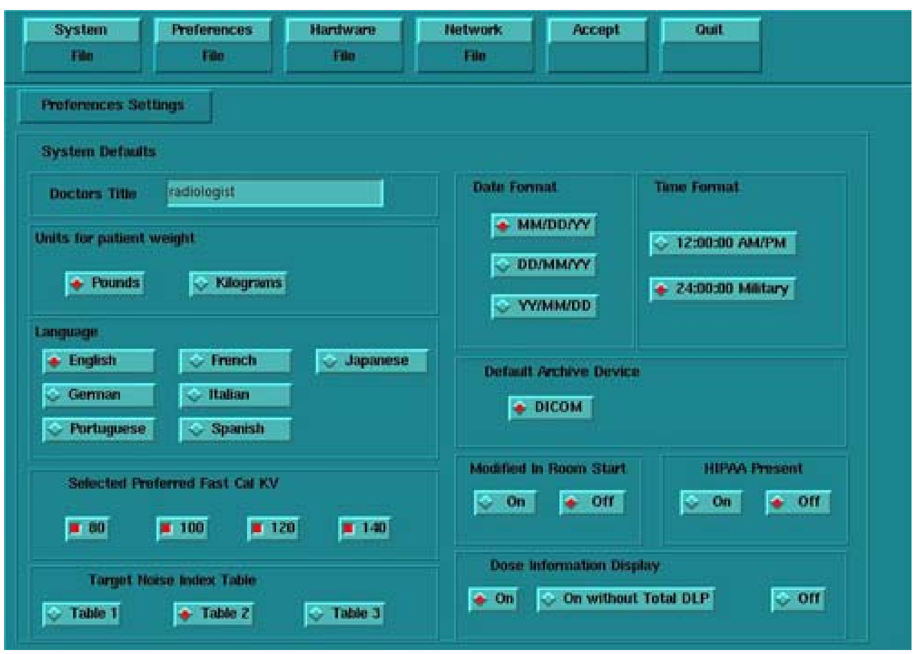


Figure 10-6 Preferences screen

10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).

11.) Select **Language**: (ENGLISH or OTHER).

12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.

13.) Select **Target Noise Index Table**: default is TABLE 2.

14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).

15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).

16.) Default **Archive Device**: add your device here (Example: DICOM.)

17.) Verify **Modified in Room Start** is set to OFF.

18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.

19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:

a.) Select ON (full CTDiw Display)

b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or

c.) Select OFF (no CTDIw Display)

Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.

20.) Click the HARDWARE button.

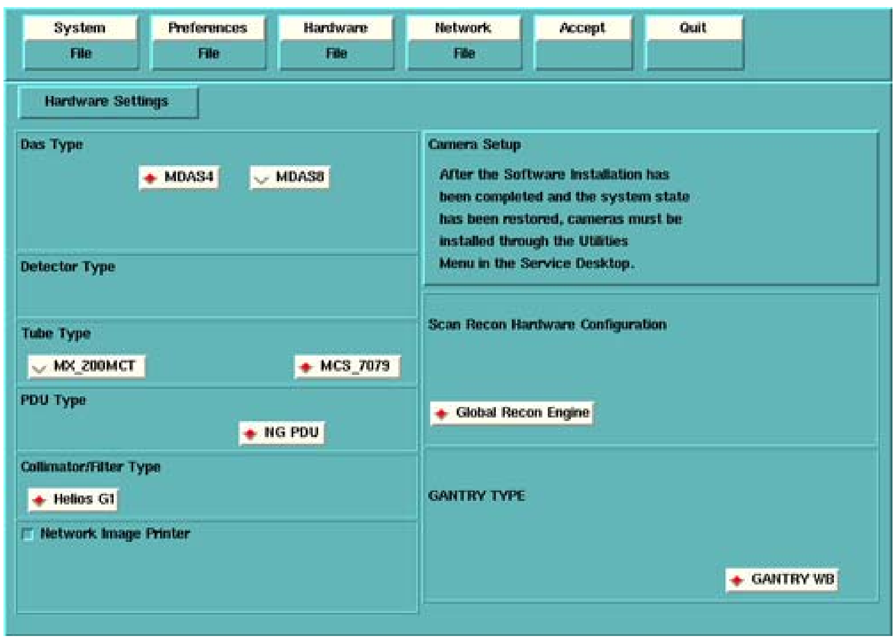


Figure 10-7 RT Hardware Settings screen

21.) Select the proper **DAS Type**: MDAS4.

22.) Select the proper **Tube Type**: MCS_7079.

23.) Select the proper **PDU Type**: NG PDU.

24.) Select the proper **Collimator/Filter Type**: HELIOS G1.

25.) Select the proper **Scan Recon Hardware Configuration**: GLOBAL RECON ENGINE.

26.) Select the proper **GANTRY TYPE**: GANTRY WB.

Note: Confirm that the GLOBAL RECON ENGINE has been selected. If it has not, Software Load will not be successful.

27.) Check the **Network Image Printer** selection ONLY if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active,

the software install will fail.

28.) Click the NETWORK button (site will be different).

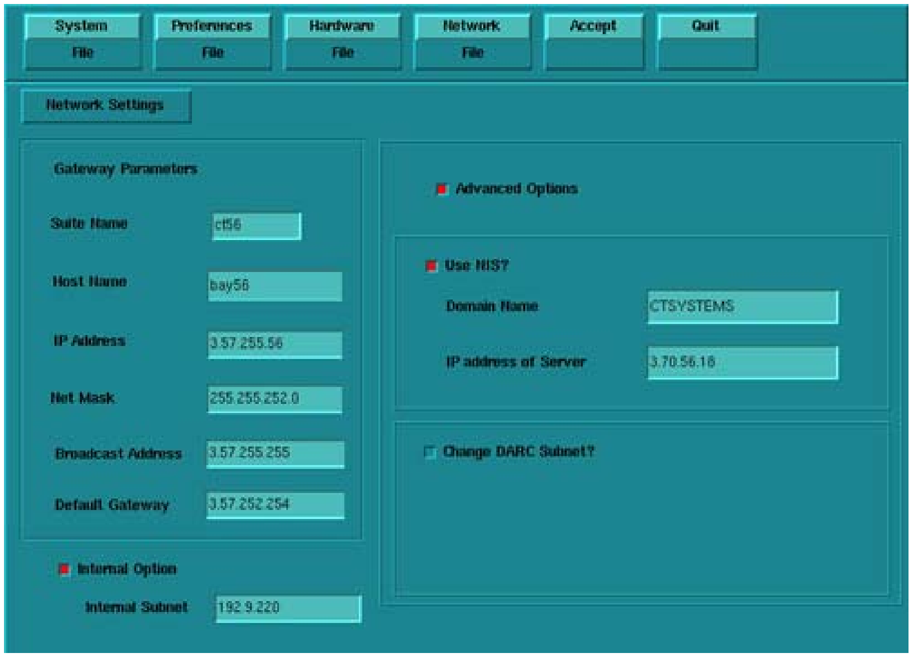


Figure 10-8 Network Settings screen

29.) Select the **Suite Name**: add your name here (Example: ***CT55***).

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.

30.) Select the Host Name: add your host name here (Example: ***BAY55***).

Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:

- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
- MUST NOT exceed 16 Characters.
- MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _

31.) Set the **IP Address**: add your address here (Example: ***3.57.255.55***).

32.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: **169.254.0**

33.) Set the **Netmask**: add your address here (Example: ***255.255.252.0***).

Note: If the **IP Address** is **192.9.220.xxx**, then the **Netmask**: must be set to **255.255.255.252**.

34.) Set the **Broadcast Address**: add your address here (Example: ***3.57.255.255***).

35.) Set the **Default Gateway**: add your address here (Example: ***3.57.252.254***).

Note: If the site does not have a Gateway, enter the value **1.1.1.1**. This field MUST NOT be blank!

36.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**.

37.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.

38.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: ***CTSYSTEMS***.)

39.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: ***3.70.56.18***).

40.) Click the **ACCEPT** button at the right corner of the User Interface. The following message appears: `Please make sure the CT Application SW is in the drive - the system will be rebooted.` Ignore this message.

Note: Do not remove the LightSpeed RT Applications Software CDROM from the HP Host Computer until instructed to do so.

41.) As the system reboots, the following Winterm window message appears: `Starting LFW procedure (~13 minutes to load).` A shell Installation screen appears and the application software starts loading.

42.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.

43.) After approximately 3 minutes, a pink pop up window appears, stating `CT Software Auto-Start Disabled`. In this box select: OK.

44.) Press the button on the DVD ROM drive to remove the LightSpeed RT Applications Software CDROM from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

1.) Open a Unix Shell.

2.) Type the following to verify hardware configuration:

`ctuser:` **`cat space /usr/g/config/INFO`** **ENTER**

Verify DAS type (MDAS4), Tube type (17 for MCS_7079), PDU type (NG PDU), Gantry type (wb), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see Section 3.3.1).

3.) Type the following:

`ctuser:` **`su`** **SPACE** **`-`** **ENTER**

`password:` **`#bigguy`** **ENTER**

`root:` **`cd`** **SPACE** **`/usr/g/scripts`** **ENTER**

`root:` **`./start_darcOS`** **ENTER**

An Attention window appears. See Figure 10-9.

If the `./start_darcOS` script does not work, verify the Global Recon Engine has been selected (see General Information).

Figure 10-9 Insert OS CD Pop-Up Window

4.) Follow the instructions in the Attention window and then insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DARC Node drive.

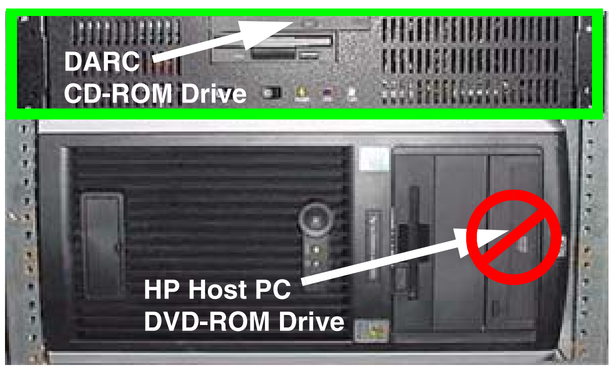


Figure 10-10 DARC CD-ROM Drive Location

5.) Click on **OK**.

A sh window displays initial information, repetitive dpccli> commands, then the message below:

```
The OS load on the DARC Node is started. This takes about 9-11 minutes.
##.## % done.
```

Note: The DARC OS Load will stop prematurely ONLY if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

```
The OS load is complete. Please remove the OS disk from the DARC disk
drive. DARC is rebooting. Wait for the DARC to boot up…
```

7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CDdoes not eject, then press the DARC Node front panel RESET Switch . Manually eject the CD after approximately 2 minutes. See Figure D-8 for location of DARC Reset Switch.

8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See Figure 10-11.

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to Appendix D for additional information on these pop-up windows.

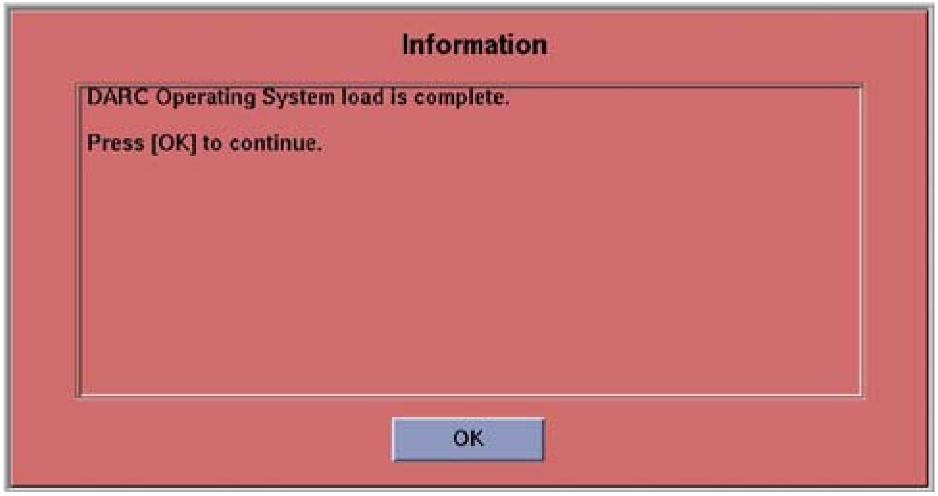


Figure 10-11 DARC OS Load Complete Pop-up Window

9.) Click on **OK**.

10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).

11.) This completes the OS load on the DARC.

12.) Next: load the DARC Applications in the HP Host Computer. See Section 3.3.4.

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully loaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

1.) Insert the **LightSpeed RT DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

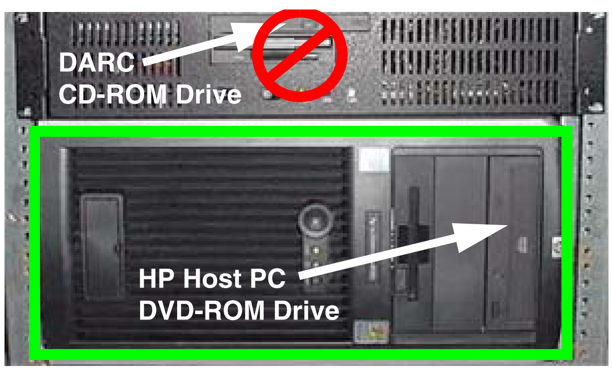


Figure 10-12 HP Host PC DVD-ROM Drive Location

2.) Open a Unix Shell and type the following:

`ctuser:` **`su`** **SPACE** - **ENTER**

`password:` **`#bigguy`** **ENTER**

3.) Verify the DARC OS load was successful by logging into the DARC Node:

root: **rsh** SPACE **darc** ENTER

[root@localhost root] (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)

[root@localhost root] **exit** ENTER

4.) If login was successful, continue with DARC Apps load:

root: **cd** SPACE **/** ENTER

root: **mount** SPACE **/mnt/cdrom** ENTER

root: **cd** SPACE **/mnt/cdrom** ENTER

root: **./copy_rpms** ENTER

The following messages appear.

```
Copying darc rpms...
Please wait...
Done...
```

5.) Type the following:

root: **cd** SPACE **/** ENTER

root: **umount** SPACE **/mnt/cdrom** ENTER

root: **eject** ENTER

6.) Remove the LightSpeed RT DARC Applications CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

root: **cd** SPACE **/usr/g/scripts** ENTER

root: **./start_darc_load** ENTER

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Reboot the System

1.) Open a Unix Shell.
2.) Type: **su** SPACE **-** ENTER
3.) Enter the root password: **#bigguy** ENTER
4.) Type: **sync** ENTER
5.) Type: **sync** ENTER
6.) Type: **reboot** ENTER
A message appears: `The system is going down for reboot NOW!`.
7.) After approximately 4–5 minutes a pink pop up window appears with the message:
`CT Software Auto-Start Disabled.`
Click OK.

3.5 Start Up the CT Applications

1.) When the Host Computer is up, open a Unix shell (window).
2.) Type: **st** ENTER or **startup** ENTER.
3.) A pink pop-up Attention Message appears: `OC initializing. Please wait…`.

Read the Note(s) below.

4.) Click OK as needed to answer message prompts.

5.) If any Recon Selftest Failures are encountered, review the error log.

Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
- If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.6 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.

2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

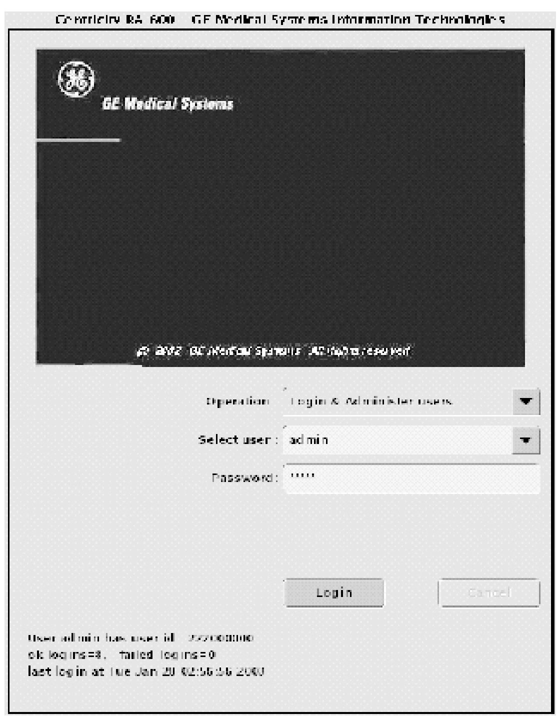


Figure 10-13 Login/Admin screen

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

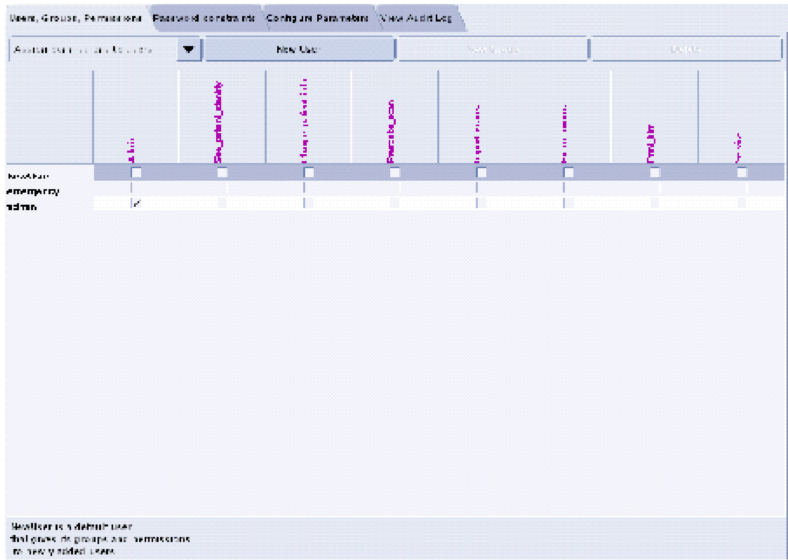


Figure 10-14 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 10-15).

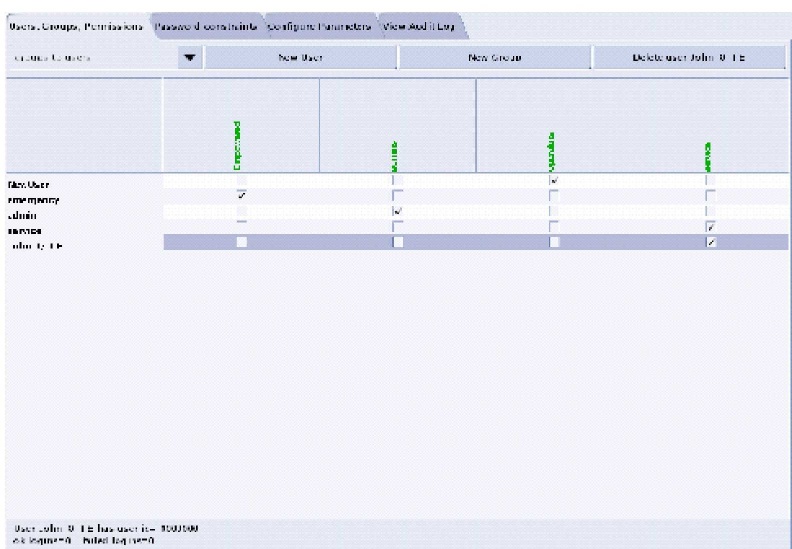


Figure 10-15 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.7 HIPAA Present set to ON — Admin Screen

When the system comes up, the following screen appears if **HIPAA Present** was set to ON during the Load process. Verify the admin screen below has the following 'Select user':

`Select user:` **`admin`**

`Password:` **`ctAdmin`**

The list of users who have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

Figure 10-16 Login/Admin screen

3.8 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State MOD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.

2.) Select: SERVICE DESKTOP.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

3.) Select: PM.

4.) Select: SYSTEM STATE.

5.) Click the ALL selection which highlights the cals, characterization, etc.

6.) Make certain to click RESTORE.

7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.

8.) When completed, select DISMISS or close the window.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.9 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

1.) Click on the RETRO RECON selection on the left Monitor.

2.) Click the **rat gold** series.

3.) Click SELECT SERIES.

4.) Click on CONFIRM.

5.) Verify the image(s) reconstruct and are displayed.

6.) Click on QUIT.

3.10 FLASH Download

1.) Select the SERVICE DESKTOP.

2.) Select in sequence: UTILITIES > INSTALL > FLASH DOWNLOAD.

3.) Click UPDATE. Ignore any initial errors.

4.) When then Flash Download process is complete, select DISMISS or Close the window.

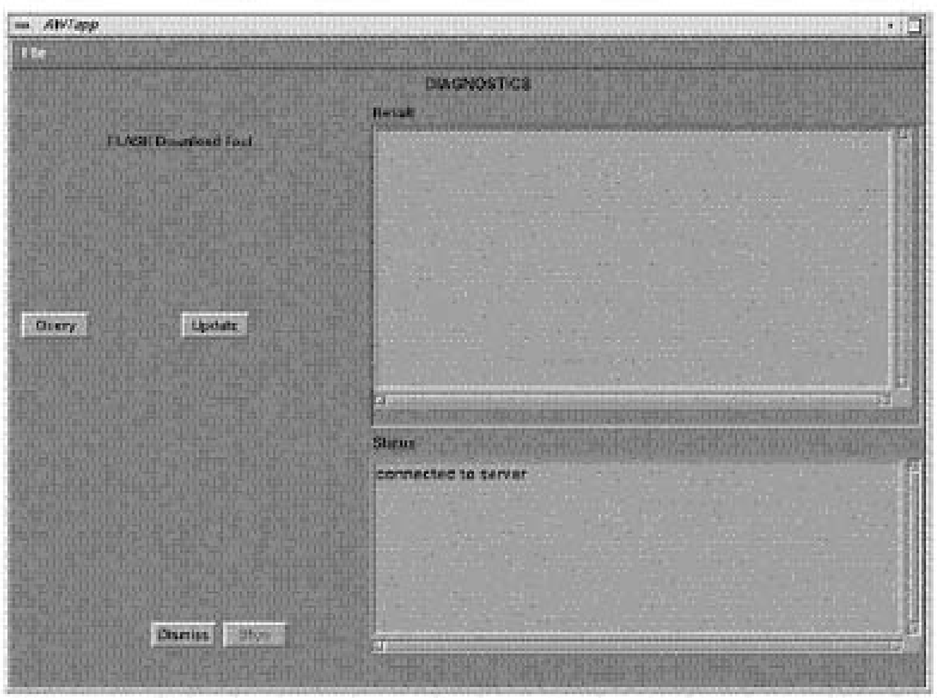


Figure 10-17 Diagnostics screen

5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (**CTRL-C** and close the shell when finished pinging):

`ping` SPACE **`orp`** ENTER

`ping` SPACE **`stc`** ENTER **–OR–** **`ping`** SPACE **`tgp`** ENTER

3.11 Install Options

1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
2.) With the Applications up, select the SERVICE DESKTOP.
3.) Select UTILITIES.
4.) Select INSTALL OPTIONS. A blank Software Options screen (Figure 10-18) appears.

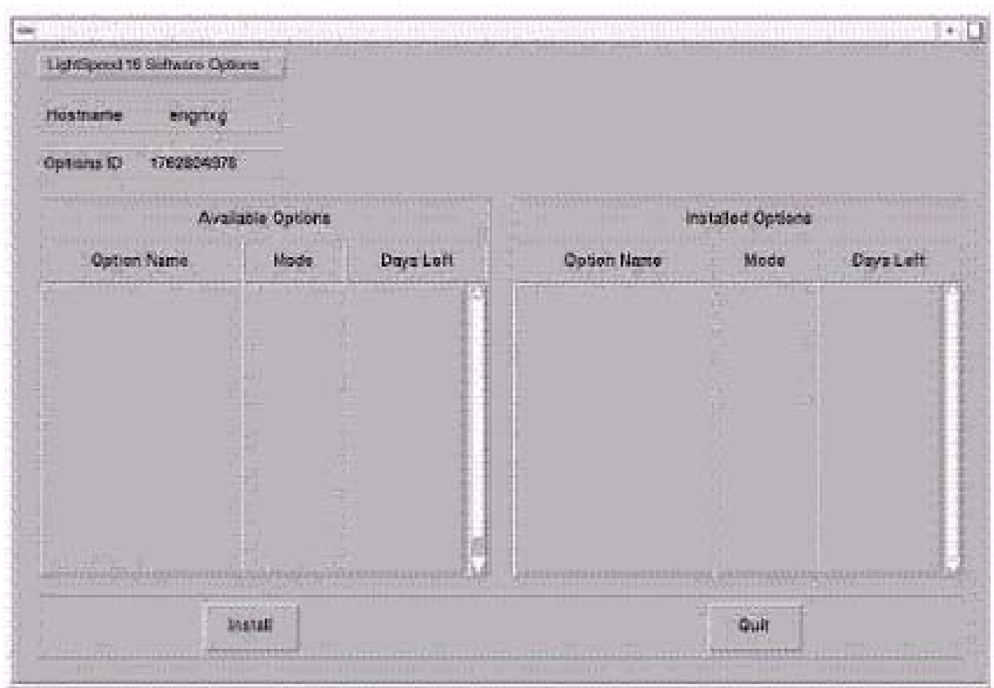


Figure 10-18 Install Options screen

5.) Click INSTALL. A Select Mechanism (Figure 10-19) window opens. Select the mechanism through which you want to install Option Keys.

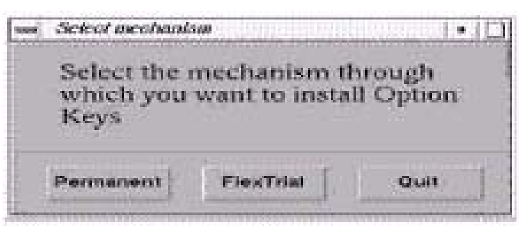


Figure 10-19 Select Mechanism window

6.) Click PERMANENT. A Select Device (Figure 10-20) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

Figure 10-20 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in Figure 10-20 is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

8.) Click OK. Available options appear on the Software Options screen (Figure 10-21).

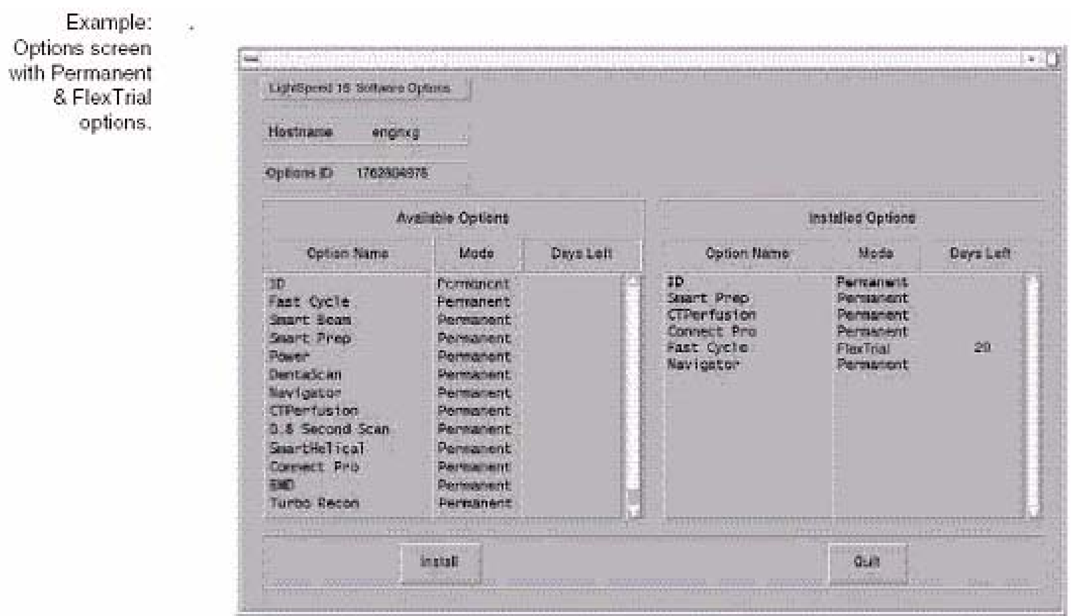


Figure 10-21 Software Options screen

9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.

10.) When the process is complete, click QUIT then QUIT to close the window.

3.12 Set Time and Date

Note: *If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.*

With Application Software down, enter the time and date if needed.

1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.

2.) Open a Unix shell.

Type: **su** **SPACE** **-** **ENTER**.

Type the root password: **#bigguy** and press **ENTER**

3.) Type: **`setdate`** ***`mmddhhmmyyyy`*** (e.g.: **`setdate 050409002003`**) and press ENTER.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. *Where:*

mm is month (01-12)

dd is day (01-31)

hh is hour (00-23)

mm is minutes (00-59)

yyyy is year (1980-2030)

4.) The **`setdate`** program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.

5.) To exit the root session, Type: **`exit`** ENTER.

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.13 Restart the System

3.13.1 Restarting the System

To restart the system:

1.) Open a Unix shell and type: **`reboot`** ENTER

2.) Select OK as needed to answer messages. Note any Recon Failures.

3.) Select the red SHUTDOWN ICON. A set of options appears.

Figure 10-22 Shutdown Icon

4.) Select: RESTART then OK to shutdown and Restart the system.

3.13.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see Section 3.7.

3.14 LightSpeed 5.X (RT) GRE Console Information Sheet

- ❑ Site ID #: ____________ Console Serial Number ________________
- ❑ DAS Type: MDAS4
- ❑ Tube Type: MCS_7079
- ❑ PDU Type: NG PDU
- ❑ KV Values: 80 100 120 140
- ❑ Collimator/Filter Type: Helios G1
- ❑ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ❑ GANTRY TYPE: GANTRY WB
- ❑ Target Noise Index Table: Table 2
- ❑ Dose Info Display: On
- ❑ Modified in Room Start: Off
- ❑ HIPAA Present: Off
- ❑ Default Archive Device: DICOM
- ❑ Suite Name: ___________________
- ❑ Host Name: ___________________
- ❑ IP Address: ____ . ____ . ____ . ____
- ❑ Netmask: ____ . ____ . ____ . ____
- ❑ Broadcast Address: ____ . ____ . ____ . ____
- ❑ Default Gateway: ____ . ____ . ____ . ____
- ❑ Printer: _______________________
- ❑ Printer IP: ____ . ____ . ____ . ____
- ❑ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ❑ Advanced Option: this item is checked
- ❑ Use NIS? Enter Domain Name: ____________________
  IP Address of Internal Server: ____ . ____ . ____ . ____
- ❑ ConnectPRO Option Info: ***`CT##   WLSNorm   4008   3.57.255.##`***
- ❑ Installed Hpower ME6 PATCH A

Appendix A System State Conversion

Section 1.0 Introduction

This procedure describes a system state conversion from IRIX to LINUX.

Section 2.0 Procedure

Do the following steps before powering down the Irix console and installing the GRE console.

1.) Save system state to an MOD on the Irix console before installing the new GRE/Linux console.

2.) Write down all of the system INFO information on the reconfig screens, including the network information. During the Load from Cold on the new console, you cannot restore the INFO file during the load.

3.) If the console has Connect Pro installed, write down the information when you run `installhisris` so it can be entered on the new console when installing the Connect Pro option.

4.) Power down the Irix console.

   a.) Install the new GRE/Linux console. You will be using your existing SCIM/Keyboard, monitors, and mouse unless you received a new mouse. You will receive a new trackball that must be used on the new console. Your old trackball will no longer work. Install the new tower that has a DVD Ram drive installed and an MOD drive installed. When you power up the GRE/Linux console, make sure the power to the tower has been turned on (Switch on front of the tower) and the new trackball is installed. It plugs in to a cable near the connections for the keyboard and mouse. The extension cable for the trackball is already connected to the USB port at the rear of the HP computer.

   b.) With the front cover off, note their are two places where you will be installing a CD during the LFC. The HP Host computer CD drive and the DARC CD drive. The OS and applications are loaded using the host computer CD. During the DARC load the OS is inserted in the DARC CD. The DARC application CD is loaded using the host computer CD.

5.) Start the LFC for the GRE/Linux console. When it asks if you want to restore the `INFO` file during the load, select NO.

6.) Complete the OS and Applications load and then reboot when it asks you to.

7.) Start up applications.

8.) When applications are up, select the Service Desktop and open a unix shell. Become **`root`** and load Class C & M. Close the shell.

9.) Open a new unix shell. You must be ctuser.

   a.) Type **`whoami`** ENTER. The system should respond *`ctuser`*.

   b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.

c.) Execute the following commands as `ctuser`.

`cd /usr/g/bin` ENTER

`sysstategui -MOD -IRX2LNX` ENTER

d.) When the system state GUI comes up, select ALL and then select RESTORE.

e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.

10.) In the same unix shell opened in step 9, execute the following command as ***`ctuser`***:

`gettubeusagedata` ENTER.

Note: If the above command is not found on the system due to older version software, then type the following set of commands as ***`ctuser`***:

a.) **`mountMOD -I2L`** ENTER

b.) **`mv /usr/g/service/.bb  /usr/g/service/bb.lfc`** ENTER

c.) **`mkdir /usr/g/service/.bb`** ENTER

d.) **`rm -rf /tmp/*bb`** ENTER

e.) **`cp /MOD/service_mod_data/system_state/*bb  /usr/g/service/.bb/`** ENTER

f.) **`/usr/g/bin/SwapTubeData`** ENTER

g.) **`/bin/cp -f  /tmp/*bb /usr/g/service/.bb/`** ENTER

11.) Remove the MOD from the drive.

12.) Put the option DVD into the DVD drive and install the options on the system.

13.) Reboot the system.

14.) Complete a flash download.

You should now be ready to scan.

Complete an Insite checkout. The system is a different model type, so you will have to have the insite center perform a checkout.

After insite has been checked out, save system state to DVD.

Appendix B
Hardware Parameters Selection

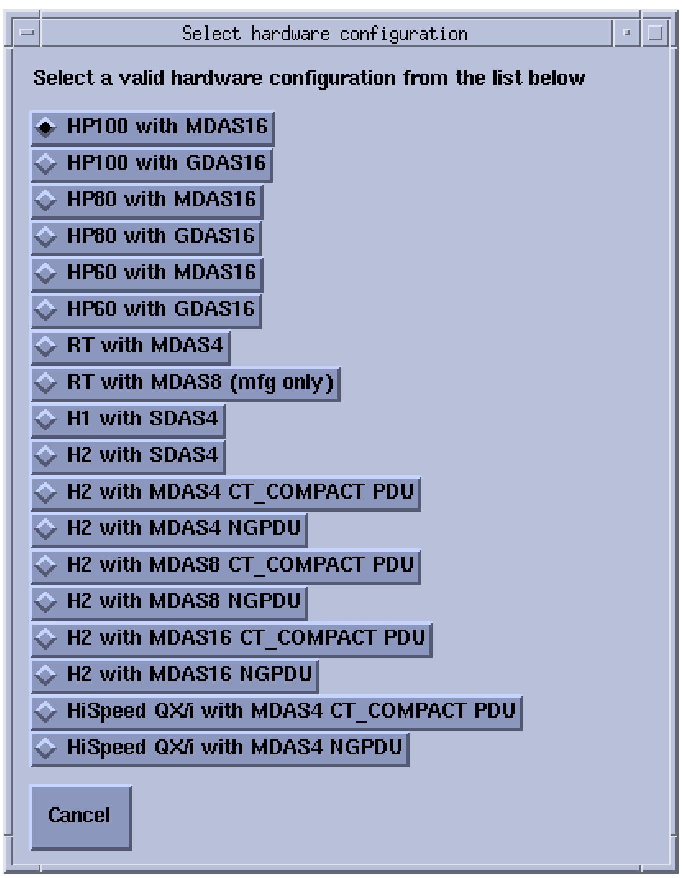
Section 1.0
Introduction

This Appendix applies only to systems using the Chapter 1 - "LFC Procedure – GRE Common Applications Version 04MW13.X, 04MW44.6 & 04MW44.9" software load procedure.

Section 2.0
Procedure

This procedure explains how to correct the the hardware configuration if the DAS Type, Tube Type, PDU Type, or Gantry Type information on Hardware screen is not correct.

1.) On Hardware screen, click the HARDWARE PARAMETERS SELECTION button. This displays the hardware selection popup.



2.) Select the correct system configuration from the displayed list (this closes the hardware selection popup).

Appendix C
Example config/INFO Output

Section 1.0
Sample Output

This section contains an example system hardware configuration INFO file output that is displayed when you type the following in a shell: **`cat /usr/g/config/INFO`** **ENTER**

For some parameters, the values for your system may differ from those in the example.

```
setenv SERVICE_ID GE_SERVICE
setenv HOSPITAL_NAME                          ________________
setenv MENUTZ "America"                       ________________
setenv SUNTZ "US/Central"                     ________________
setenv TZ "CST6CDT US"                        ________________
setenv TZTEXT "Chicago"                       ________________
setenv REGENERATE_DATABASE no
setenv NEXT_MOD_NUMBER 1
setenv NEXT_DIAGNOSTIC_EXAM_NUMBER 50000
setenv GATEWAY_HOSTNAME                       ________________
setenv GATEWAY_IP                             ________________
setenv GATEWAY_NETMASK                        ________________
setenv GATEWAY_BROADCAST                      ________________
setenv GATEWAY_DEFAULT                        ________________
setenv TUBETYPE 15 (Performix) / 16 (Hercules) / 17 (OEM) ______________
setenv PDUTYPE CT_Compact / NG PDU            ________________
setenv DASMTYPE 2
setenv KV_VALUES "80 100 120 140"             ________________
setenv DEFAULTARCHIVE DICOM
setenv ADVANTAGERW FALSE
setenv CTIDATETIMEFORMAT 4
setenv TIMESEPAR /
setenv WEIGHT lb
setenv DOCTITLE "radiologist"
setenv CAMERATYPE ""
setenv SUITEID                                ________________
setenv FILTERTYPE 2
setenv CAMERA no
setenv DEFFORMAT 12
setenv DOGREYSCALE yes
setenv DOSLIDES no
```

```
setenv DOZOOM no
setenv FILMINTERP sharp
setenv PRINTER_HOSTNAME                                  ________________
setenv PRINTER_IPADDRESS                                 ________________
setenv PRINTER no
setenv O2_PRESENT n
setenv GANTRY_TYPE h2 / hp/ wb                           ________________
setenv LANGUAGE en_US
setenv DOSE y
setenv NETWORK yes
setenv USENIS yes
setenv CHGDARCSUBNET yes
setenv DOMAINNAME                                        ________________
setenv NIS_SERVER                                        ________________
setenv DARC_SUBNET                                       ________________
setenv NEXT_PATIENT_EXAM_NUMBER 1
setenv MOBILESYSTEM no                                   ________________
setenv STREAK y
setenv INROOMSCAN off
setenv AAA n
setenv AUTOMA_INDEX_TABLE_TYPE type2                     ________________
setenv RECONHARDWARE GRE / No O2 Pegasus                 ________________
setenv DASTYPE SDAS or MDAS 4 / 8/ 16                    ________________
setenv DETECTORTYPE Warp3
setenv REFRESH_RATE 76

setenv yp s
setenv ypmaster ${SUITEID}_OC0
setenv HOSTNAME SBC0
setenv ALIAS ${SUITEID}_SBC0
setenv UNSYSID $GATEWAY_HOSTNAME
setenv SERVID $GATEWAY_HOSTNAME
setenv tubeType $TUBETYPE
setenv ACRNEMA xxxx
setenv REGEN no

setenv HRCTTITLE                                         ________________
setenv HRPORTNUM                                         ________________
setenv HRAETITLE                                         ________________
setenv HRIPADDR                                          ________________
{ctuser@hosatname}
```

Appendix D
DARC OS Load Pop-Up Boxes

Section 1.0
Introduction

This section describes pop-up message boxes that may appear during a DARC OS load for Version 1.7.10, and provides additional information on what to do if you get one.

This section applies ONLY to products using DARC/Host OS version 1.7.10.

Section 2.0 Procedure

Locate the appropriate pop-up box in the following subsections and refer to the action to take.

2.1 Error - Cannot connect with DARC

POP-UP MESSAGE

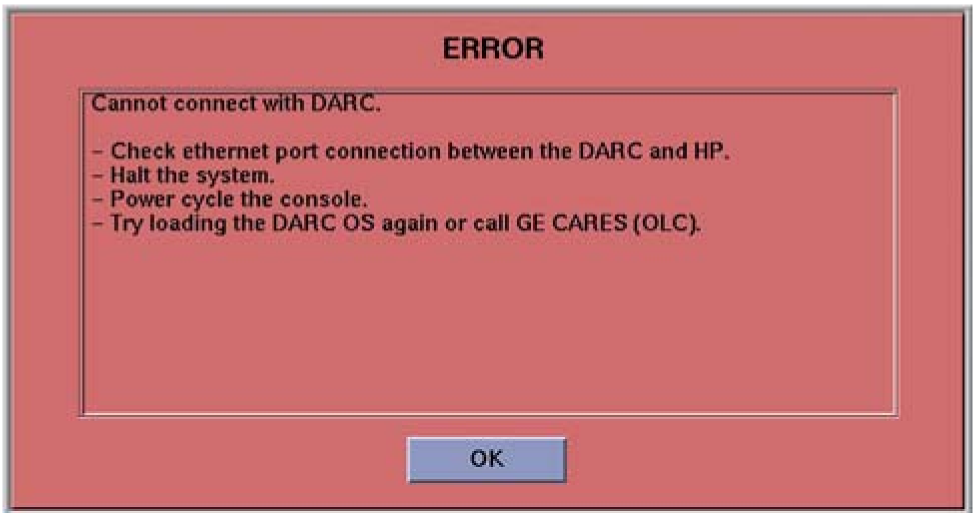


Figure D-1 Cannot Connect With DARC

POP-UP MESSAGE DESCRIPTION

SOL (Serial Over LAN) checks out good but a failure occurs.

ACTION TO TAKE

1.) At the rear of the DARC, verify Ethernet 3 to HP Host PC Ethernet C is installed properly.
2.) Click OK.
3.) Open a Unix Shell.
4.) Type: **`halt`** ENTER.
5.) When console message states power down then Power Off the Operator Console.
6.) Wait 30 seconds and Power On the Operator Console.
7.) Try the DARC OS load again.

2.2 Error - BMC connection on DARC node is not responding

POP-UP MESSAGE

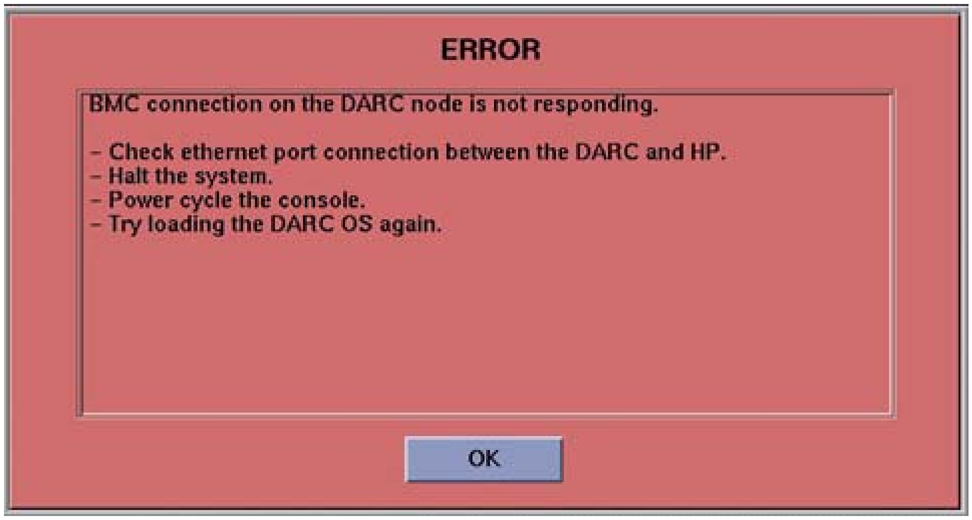


Figure D-2 BMC Connection on DARC Node is Not Responding

POP-UP MESSAGE DESCRIPTION

Cannot connect to the DARC or an internal error in the DARC.

ACTION TO TAKE

1.) At the rear of the DARC, verify Ethernet 3 to HP Host PC Ethernet C is installed properly.
2.) Click OK.
3.) Open a Unix Shell.
4.) Type: `halt` ENTER.
5.) When console message states power down, then Power Off the Operator Console.
6.) Wait 30 seconds and Power On the Operator Console.
7.) Try the DARC Node OS load again.
8.) Try new DARC Node to HP Ethernet cable.
9.) Call OLC before replacing DARC.

2.3 Error - Cannot send boot command to DARC

POP-UP MESSAGE

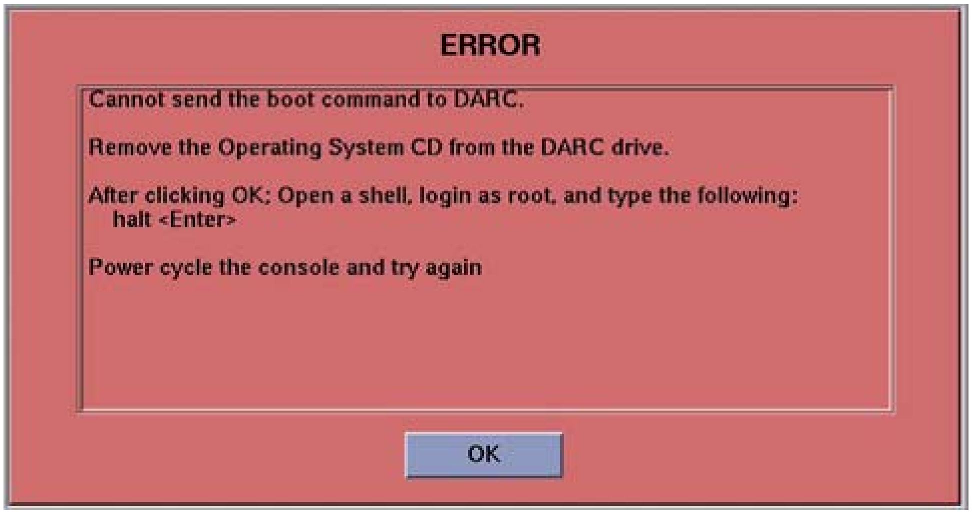


Figure D-3 Cannot Send Boot Command to DARC

POP-UP MESSAGE DESCRIPTION

During initial DARC Operating System load process something goes wrong.

ACTION TO TAKE

1.) Remove OS CD from DARC drive.
2.) Click OK.
3.) Open a Unix Shell.
4.) Type: **`halt`** **ENTER**.
5.) When console message states power down then Power Off the Operator Console.
6.) Wait 30 seconds and Power On the Operator Console.
7.) Try the DARC Node OS load again.

2.4 SOL Service - Run the Serial Over LAN Service CD

POP-UP MESSAGE

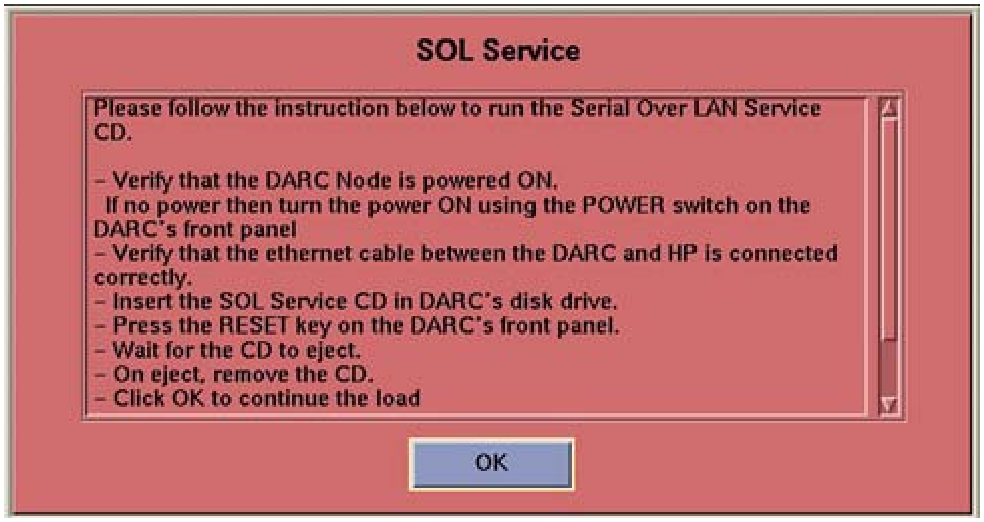


Figure D-4 Run the Serial Over LAN Service

POP-UP MESSAGE DESCRIPTION

DARC Internal BMC chip is faulty or vendor failed to flash the IP Address.

ACTION TO TAKE

Follow instructions in pop-up box using the LightSpeed Xtream Console Serial Over Lan Service CD (P/N 2399297).

2.5 Error - Cannot send a complete signal to DARC

POP-UP MESSAGE

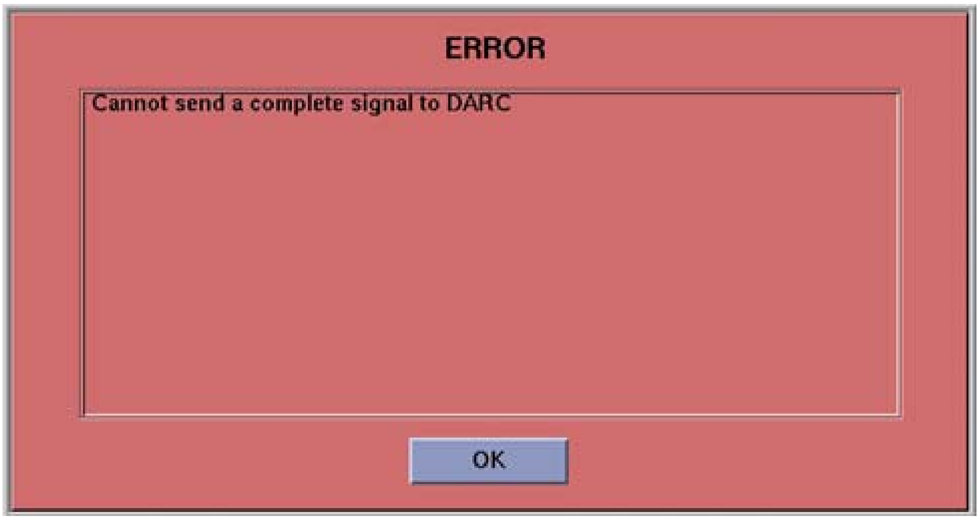


Figure D-5 Cannot send complete signal to DARC

POP-UP MESSAGE DESCRIPTION

This is a Pop-Up that is used in conjunction with the informational pop-ups listed in Section 2.6. It notifies the user an ERROR has occurred.

ACTION TO TAKE

1. Click on OK.
2. Refer to Section 2.6.

2.6 Information - DARC Node loaded OS but complete signal not sent

POP-UP MESSAGE

Figure D-6 DARC Node OS Loaded But No Complete Signal Sent (Tips Ref)

Figure D-7 DARC Node OS Loaded But No Complete Signal Sent

POP-UP MESSAGE DESCRIPTION

Both messages are essentially the same. Ignore the "tips and work around manual" reference. The DARC Operating System was properly loaded properly, however, the DARC "complete" signal was lost or missed within the timeout period.

ACTION TO TAKE

1.) At the DARC front panel, press the RESET Switch. See Figure D-8.

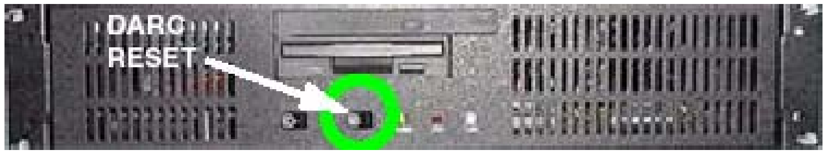


Figure D-8 DARC Reset Switch Location

2.) Eject the CD.
3.) The Pop-Up box in the LFC procedure may pop-up. Click on OK.
4.) Verify the OS was successfully loaded per the LFC procedure:
   a.) Open a Unix Shell.
   b.) `{ctuser@hostname}` **`su`** **SPACE** **`-`** **ENTER**
   c.) Password: **`#bigguy`** **ENTER**
   d.) `[root@hostname]#` **`rsh`** **SPACE** **`darc`** **ENTER**
   `Last login: Thu Oct  9 11:17:35 on ttyS1`
   `You have new mail.`
   e.) `[root@localhost root]` **`exit`** **ENTER** `<-` *The prompt shows the OS was loaded properly.*
   f.) Close the Unix Shell.
5.) Click on OK.
6.) Continue with software install. Load DARC applications on the HP Host Computer next.

2.7 DARC Node - Cable Checkout

1.) Verify the rear power cable is connected, switch is turned on and fan is operating.
2.) Verify front power LED is illuminated.
3.) Verify Ethernet cable is in good condition and connected as follows:
   - Ethernet port C --- DARC Node Ethernet port 3
   - HP (right-card - top Ethernet port)

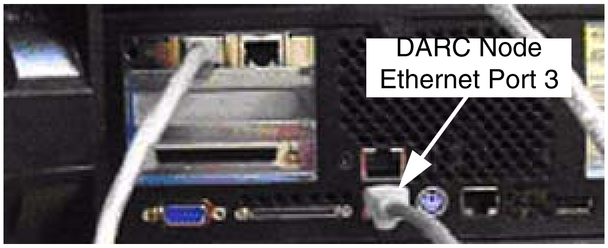


Figure D-9 DARC Node eth3- Bottom Connection Between SCSI and Keyboard

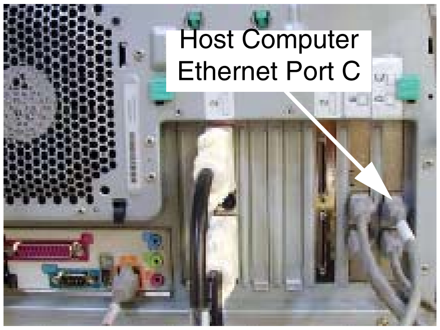


Figure D-10 HP Host Computer Ethernet Port C (See Cable Label)

2.8 DARC Node - Remote Shell

This subsection will verify the DARC Node remote shell can be invoked to determine if the DARC Node is up. The prompt root@localhost lets the user know the DARC OS has been loaded successfully.

Open a Unix Shell and type the following:

```
{ctuser@hostname} su SPACE - ENTER
Password: #bigguy ENTER
[root@hostname]# rsh darc ENTER
Last login: Thu Oct  9 11:17:35 on ####
You have new mail.
[root@localhost root ]# exit ENTER
logout
rlogin: connection closed.
[root@hostname]#
```

2.9 DARC Node --- Serial Over LAN Checkout

Verification of the Serial Over LAN check is almost always the initial test performed on the DARC after power and cable verification. It has been placed after ping and remote shell checkout because they are very quick tests. With SOL Checkout the telnet connection is made and the SOL connection from the HP Host Computer to the DARC Node is then established. This test will verify the DARC Node was booted and is up.

Open a Unix Shell and type the following:

{ctuser@hostname} **su** **SPACE** - **ENTER**

Password: #**bigguy** **ENTER**

[root@hostname]# **telnet localhost 623** **ENTER**

Trying 127.0.0.1...

Connected to localhost. <- *This output indicates HP Host Apps were loaded and the cliservice proxy server is running.*

Escape character is '^]'.

Server: **darc** **ENTER**

Username: **ENTER**

Password: **ENTER**

Login successful <- *This output indicates the SOL connection is established.*

dpccli> **console** **ENTER**

GEMS Linux 1.7.10 <- *This is the Console Operating System Software Revision.*

See you on the darc side of the moon.* <- *This indicates the DARC Node is up.*

darc login: **root** **ENTER** **** See note below. ****

Password: #**bigguy** **ENTER**

Last login: Thu Oct 9 11:11:15 on ttyS1

You have new mail.

[root@darc ~]# **~.**(<- *Enter a "tilde" "period" to terminate the telnet.*

dpccli> **exit** **ENTER**

Connection closed by foreign host.

[root@hostname]#

Note: **If the darc login prompt is not present and localhost.localdomain prompt is displayed, then try reloading the DARC Node OS and DARC Application software per the procedure. The problem could be that the DARC Apps did not load correctly. Always be sure the DARC OS matches the HP Host OS already loaded (example, if the applications software requires Console Operating System CD version 1.7.10, then both the DARC and HP Host must be loaded with this OS version).**

Appendix E
General Information

Section 1.0 Introduction

This Appendix contains the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

Section 2.0 Procedures

2.1 Remote Shell — Check Internal Network

- To become root within a Unix Shell:
  1.) Type: **`su -`** ENTER
  2.) Type the root password: **`#bigguy`** ENTER
- To check the internal network as root, type: **`rsh darc`** ENTER
  1.) Type: **`exit`** to close root@darc shell, or to check IG1, type: **`rsh ig1`** ENTER
  2.) Type: **`exit`** to close root@IG1 shell and **`exit`** to close root@darc shell
  3.) Close the Unix Shell.

2.2 HP Computer CDROM — Mount and Unmount Process

1.) Insert the CD into the HP Host Computer DVD ROM drive
2.) Open up a Unix Shell and type these commands:
   a.) ctuser: **`su -`** ENTER
   b.) password: **`#bigguy`** ENTER
   c.) root: **`mount /mnt/cdrom`** ENTER
3.) The CD has now been mounted. You can see what is on it by typing:
   `ls /mnt/cdrom` ENTER
4.) You must unmount the CD in order to remove it from the drive; do this by typing:
   `umount /mnt/cdrom` ENTER

2.3 Peripheral Tower DVD — Mount and Unmount Process

1.) Insert the DVD into the Peripheral Tower DVD RAM drive
2.) Open up a Unix Shell and type: **`mountDVD`** ENTER
3.) Now the DVD has been mounted. You can see what is on it by typing:
`ls /DVD` ENTER
4.) You must unmount the DVD in order to remove it from the drive; do this by typing:
`unmountDVD` ENTER

2.4 Peripheral Tower DVD — Initializing a DVD

If a DVD needs to be initialized on new media, open a Unix Shell and perform the following:

1.) Type: **`su -`** ENTER
2.) Enter the root password: **`#bigguy`** ENTER
3.) Type: **`mountDVD`** ENTER
4.) Type: **`mkfsDVD`** ENTER; answer **`y`** for yes and close the unix shell.

2.5 Peripheral Tower MOD — Mount and Unmount Process

1.) Insert the MOD into the Peripheral Tower MOD drive
2.) Open up a Unix Shell and type: **`mountMOD`** ENTER
3.) Now the MOD has been mounted. You can see what is on it by typing:
`ls /MOD` ENTER
4.) You must unmount the MOD in order to remove it from the drive; do this by typing:
`umountMOD` ENTER

Note: If MOD cannot be unmounted, select: IMAGE WORKS > ARCHIVE > DETACH.

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006